Learning extrinsic drivers of cell states by niche covariation analysis

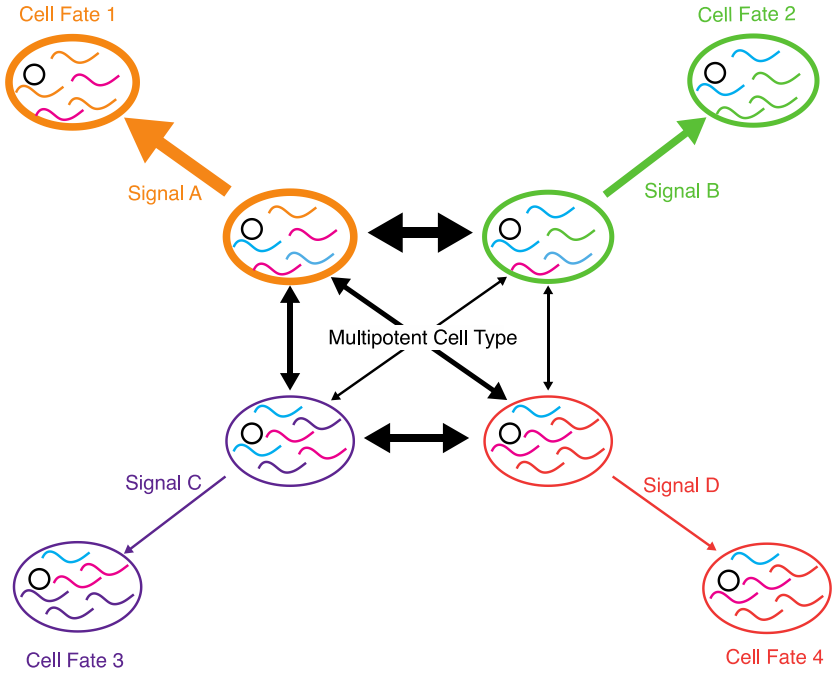
Dominic Grün
MOPITAS
13th November 2025

Julius-Maximilians-
UNIVERSITÄT
WÜRZBURG

CAIDAS

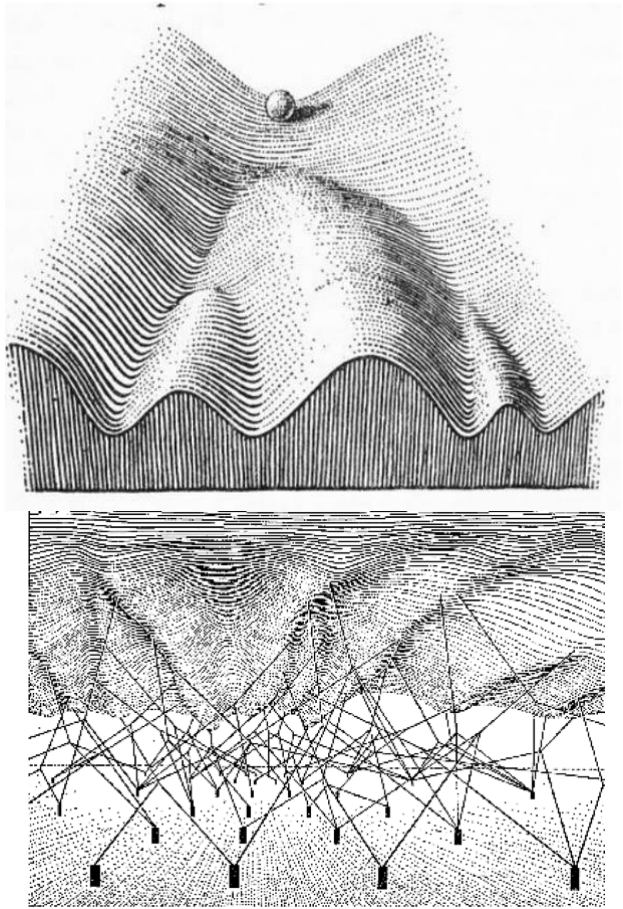
WÜSI SYSTEMIMMUNOLOGIE
WÜRZBURG

Dynamics of cell fate transitions



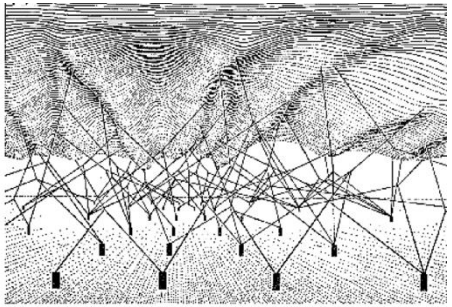
Sagar et al. (2020) Annual Reviews of Biomedical Data Science
Herman et al. (2018) *Nature Methods*
Grün et al. (2016) *Cell Stem Cell*
Grün et al. (2015) *Cell*
Grün et al. (2015) *Nature*

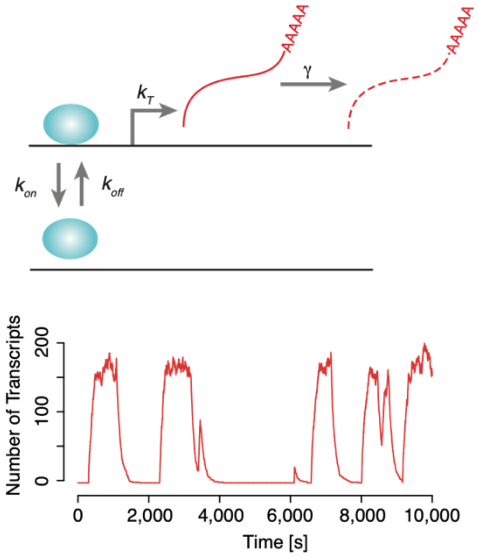
Which forces shape the dynamics of cellular state transitions?

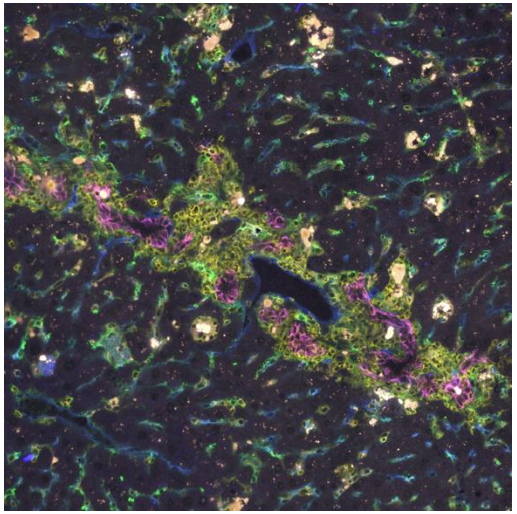


Waddington (1957)

Determinants of cell state variability





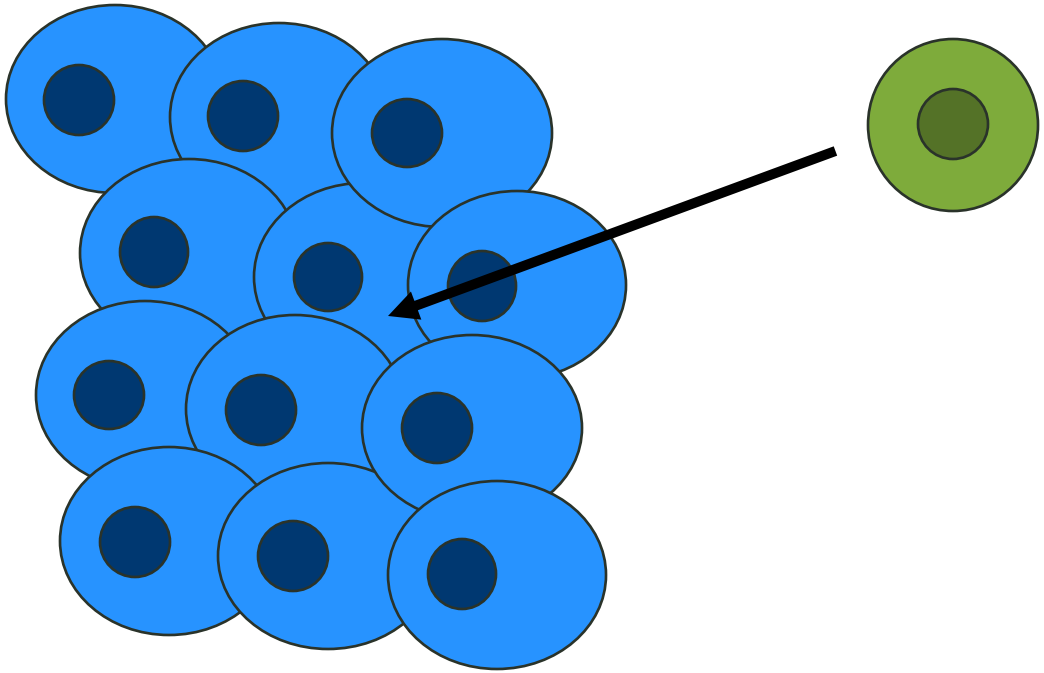


total cell state variability = intrinsic determinants + extrinsic determinants

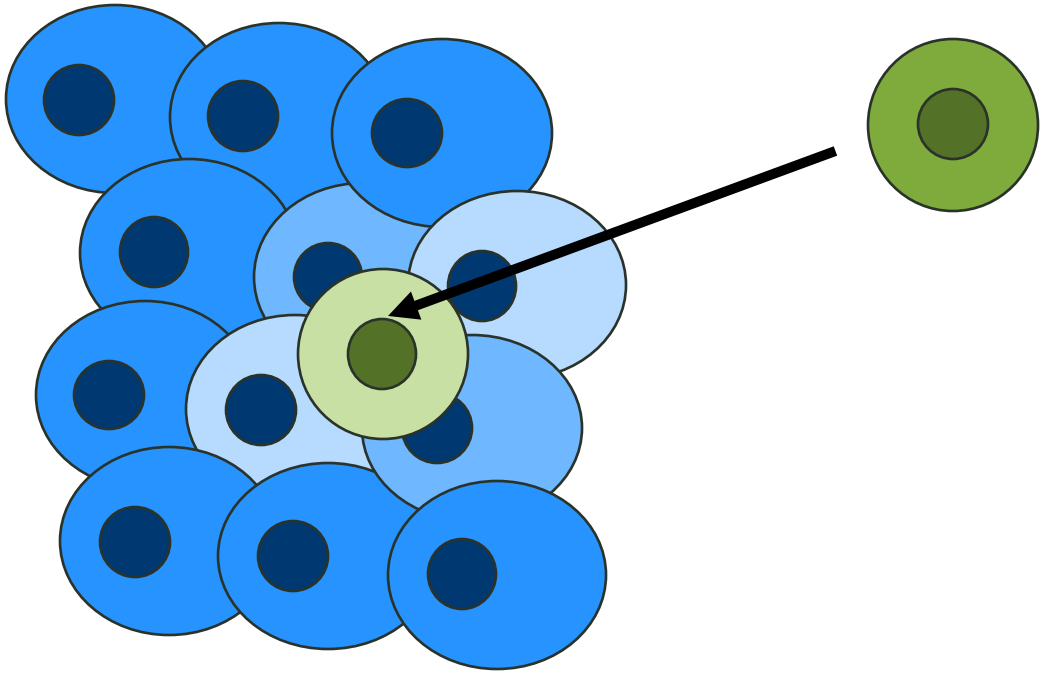
Wang et al. (2024) *Genome Biology*
Rosales-Alvarez et al. (2023) *Genome Biology*
Grün (2020) *Nature Methods*
Grün et al. (2014) *Nature Methods*

Spatial determinants of cell state variability?

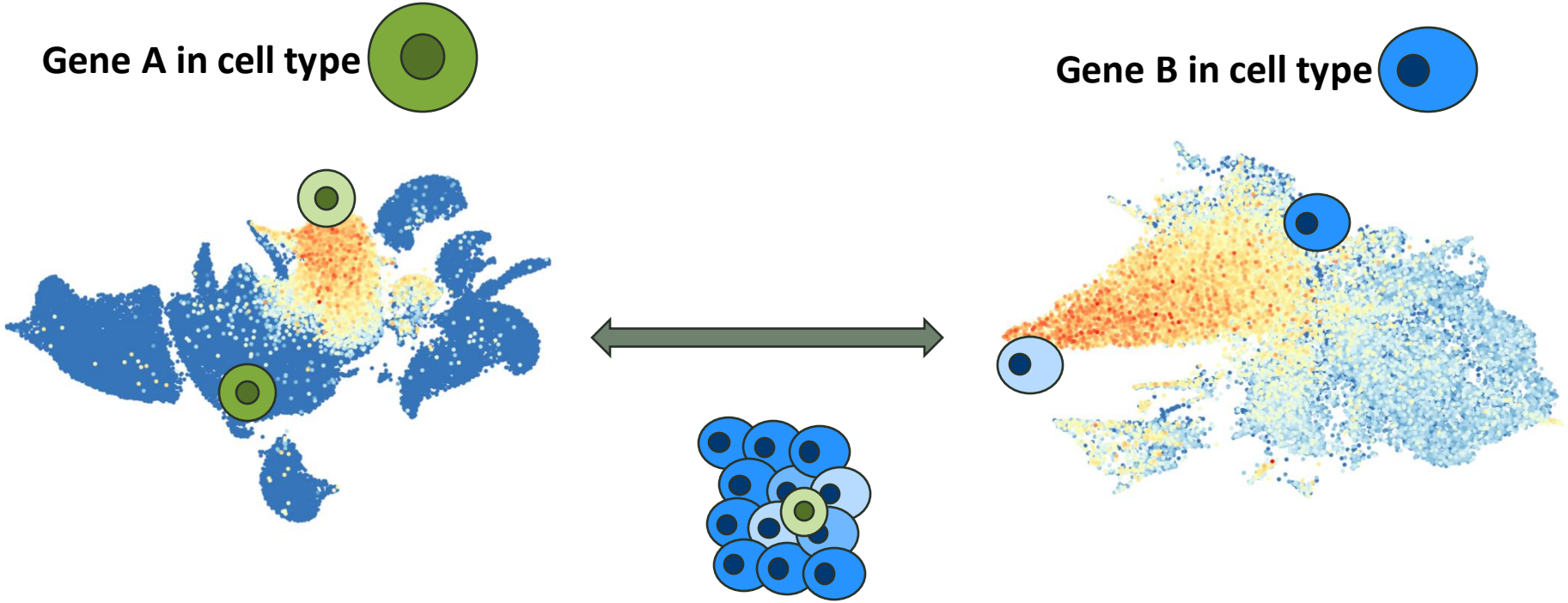
Determinants of cell state variability



Determinants of cell state variability

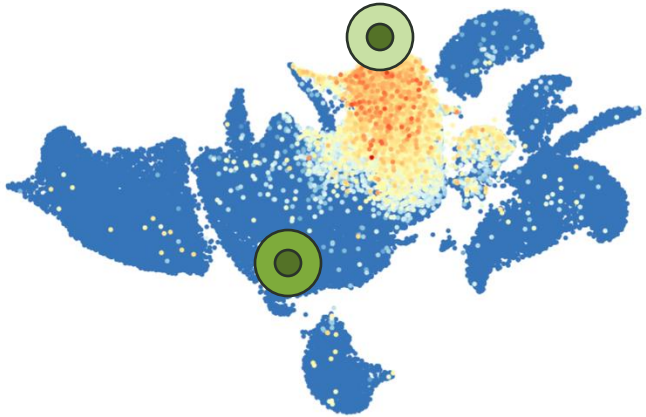


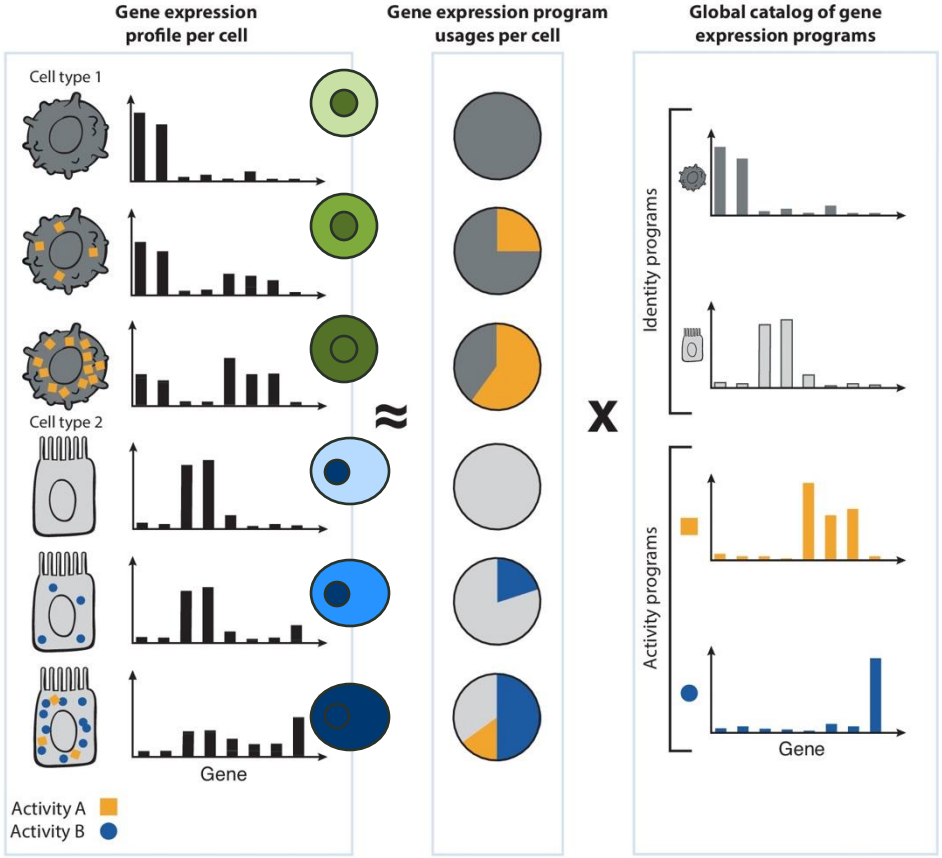
Determinants of cell state variability



Determinants of cell state variability

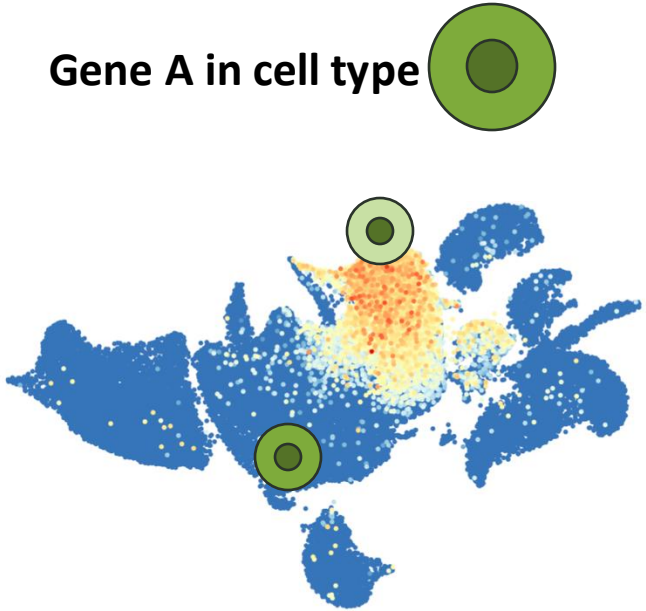
Gene A in cell type



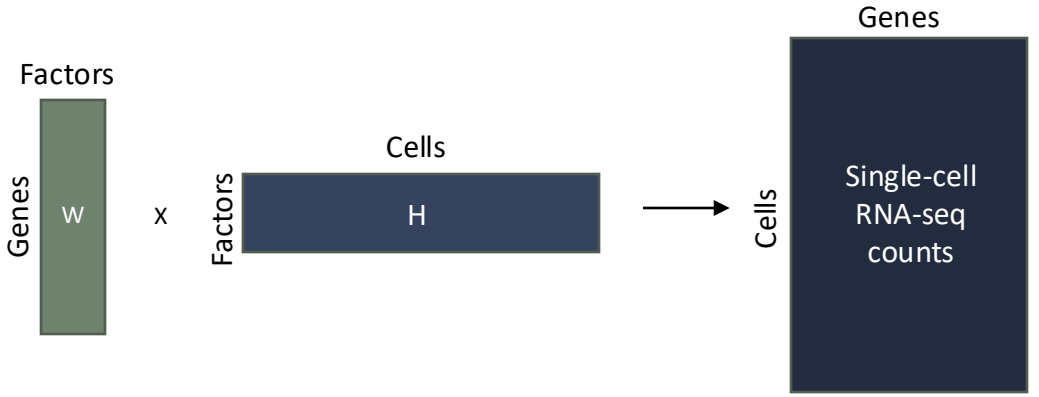


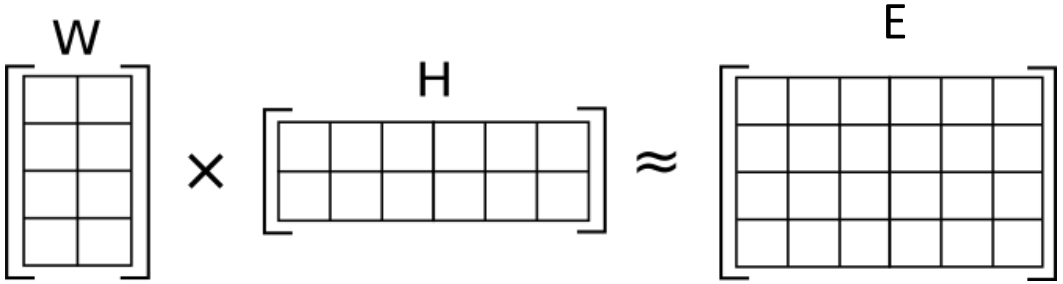
Kotlyar et al. (2019) *eLife*

Inferring latent cell state variables with NMF



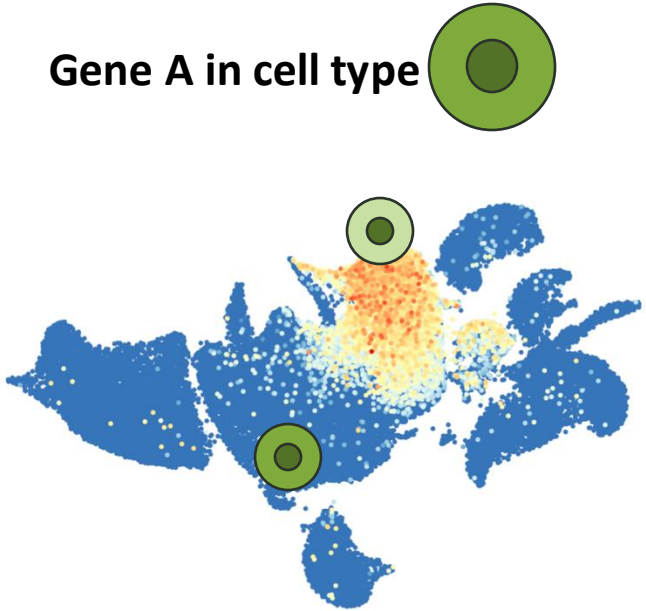
NMF: non-negative matrix factorization



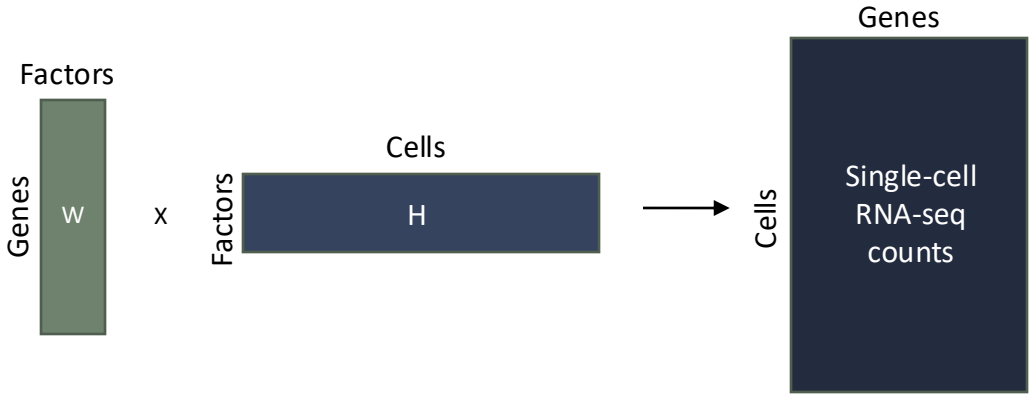


$$\| E - WH \|_F, W \geq 0, H \geq 0.$$

Inferring latent cell state variables with NMF



NMF: non-negative matrix factorization

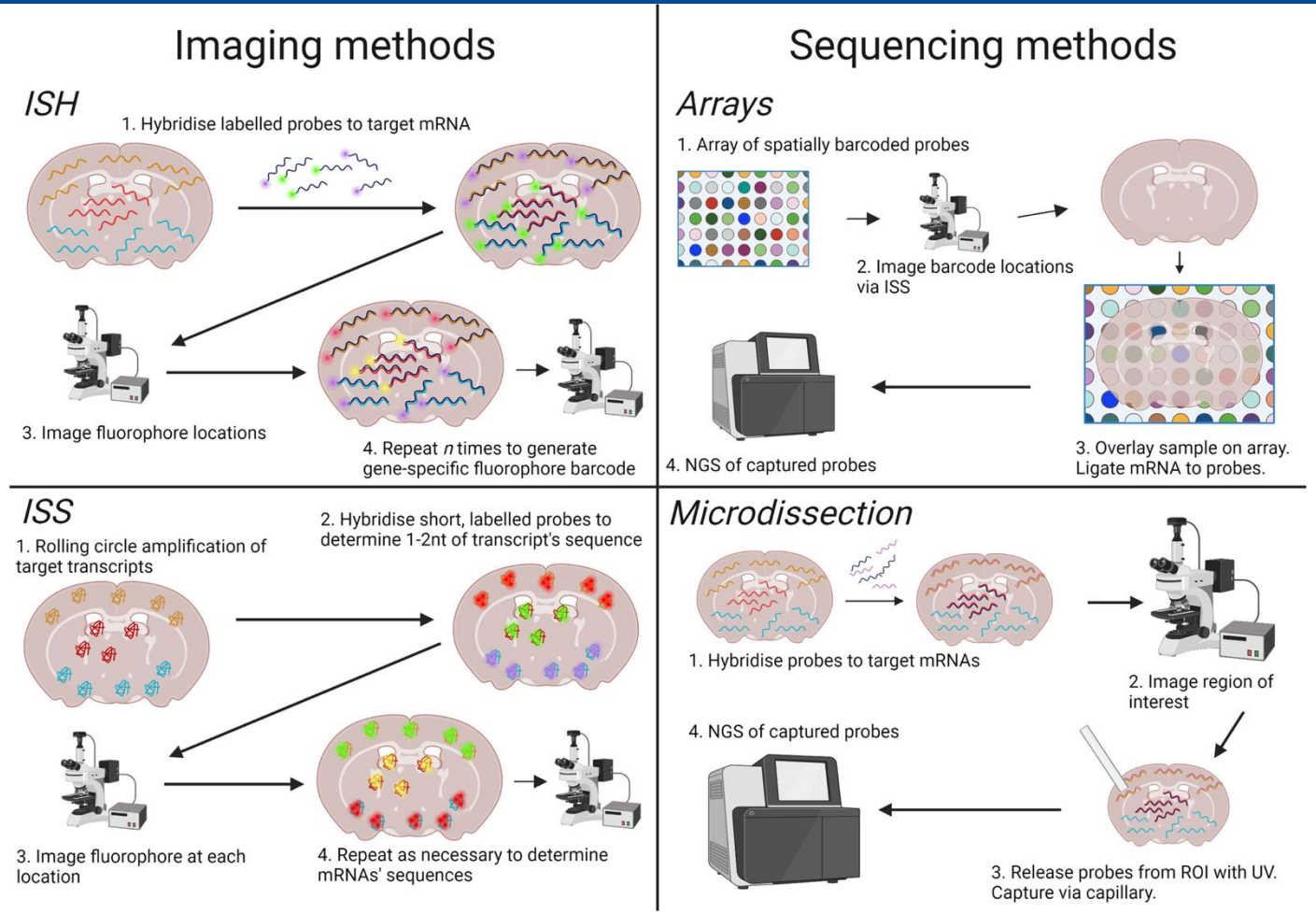


- Single-cell resolution
- High-sensitivity (low noise)
- Transcriptome-wide scale

Experimental strategies for spatial transcriptomics

MERSCOPE
MERFISH
seqFISH
RESOLVE

Xenium
STARmap



Visium
Curio
Stereo-Seq
Seq-Scope

Williams et al. (2022) Genome Medicine

Experimental strategies for spatial transcriptomics

Imaging methods

Pros:

- Single-molecule resolution
- High sensitivity

Cons:

- ~500 genes (more coming)
- (Difficult-to-use)
- High costs

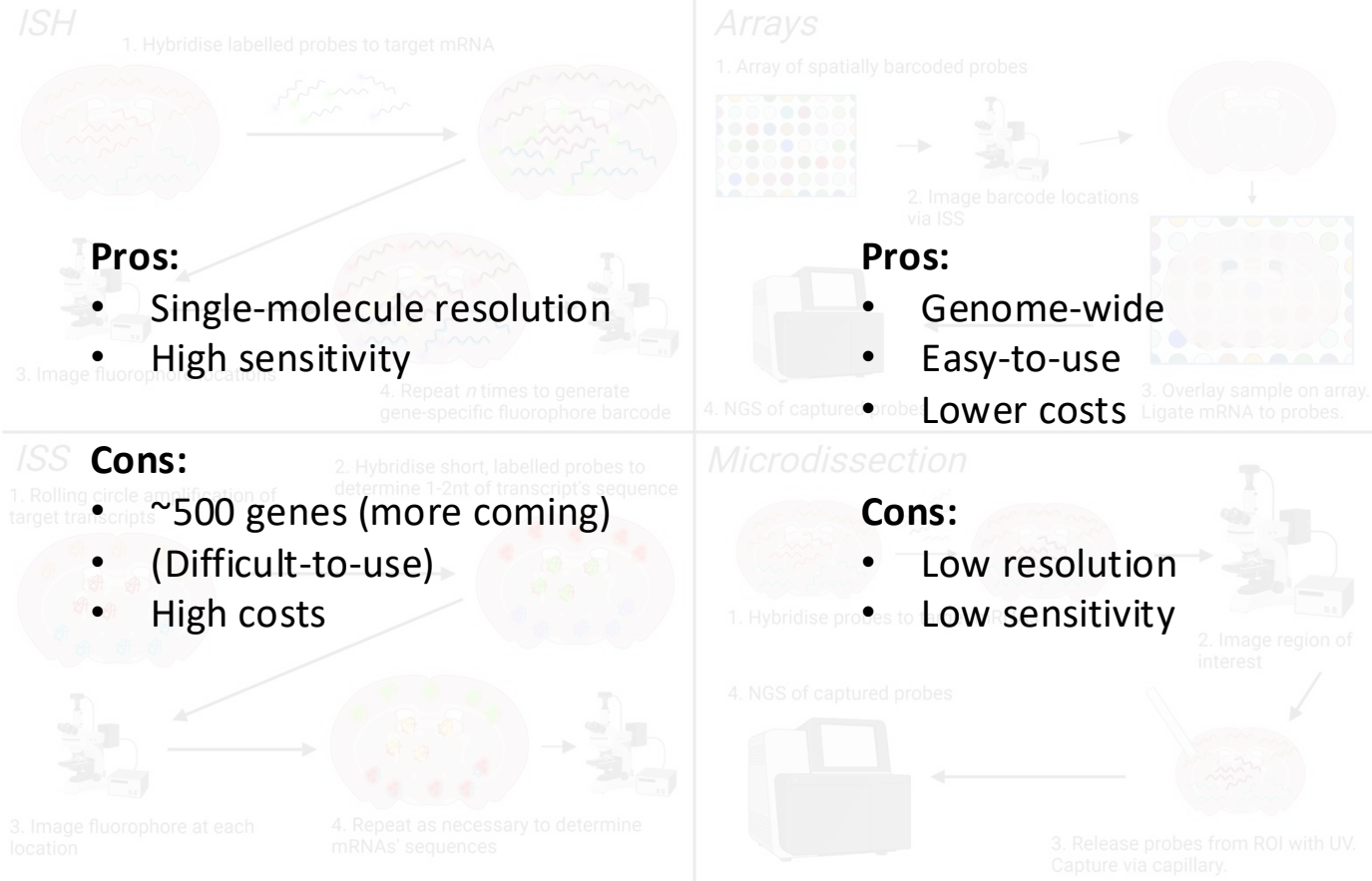
Sequencing methods

Pros:

- Genome-wide
- Easy-to-use
- Lower costs

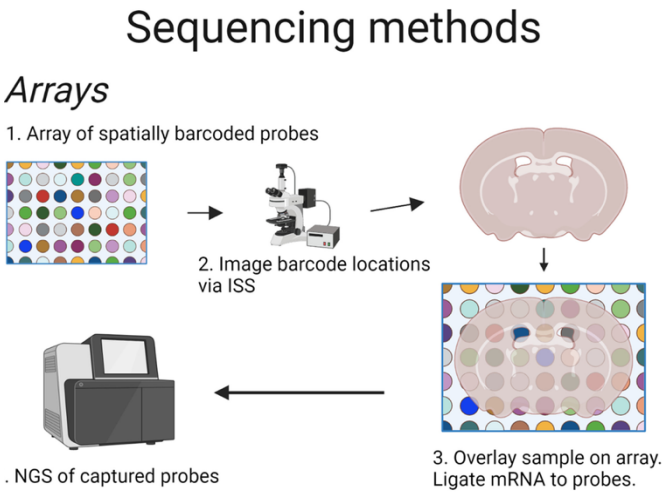
Cons:

- Low resolution
- Low sensitivity



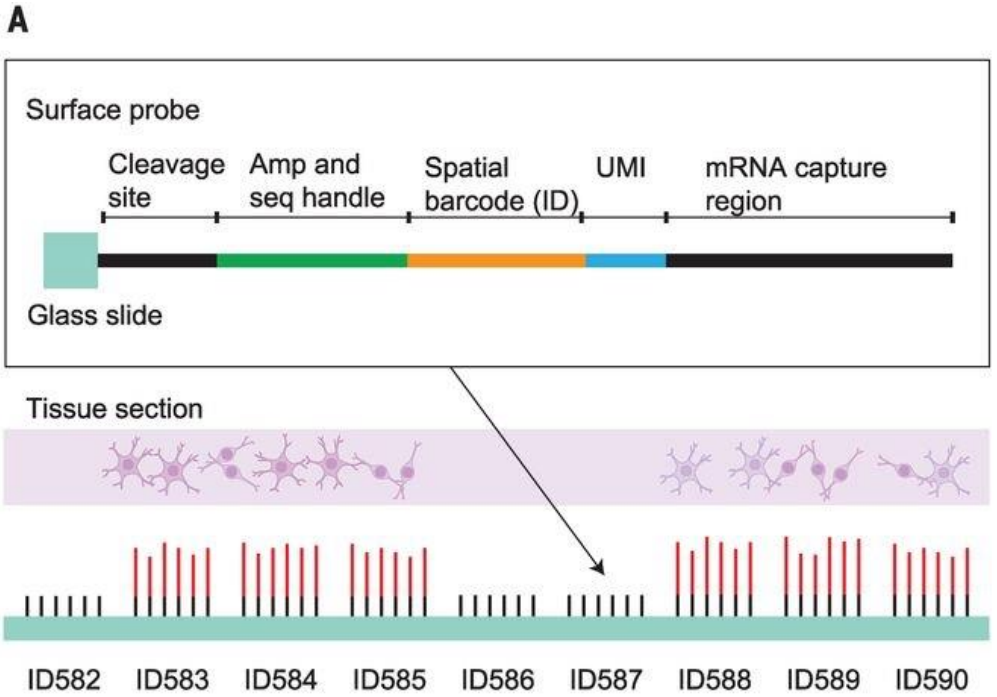
Williams et al. (2022) Genome Medicine

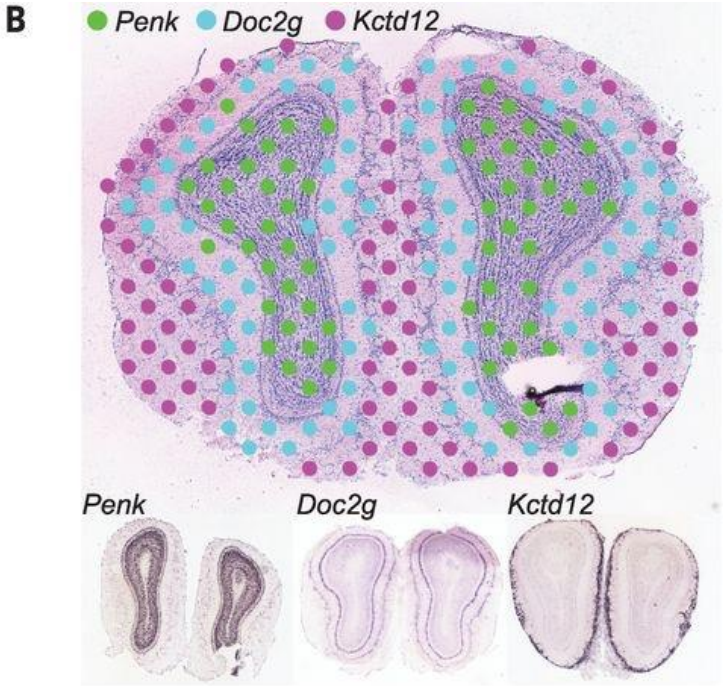
Experimental strategies for spatial transcriptomics



Williams et al. (2022) Genome Medicine

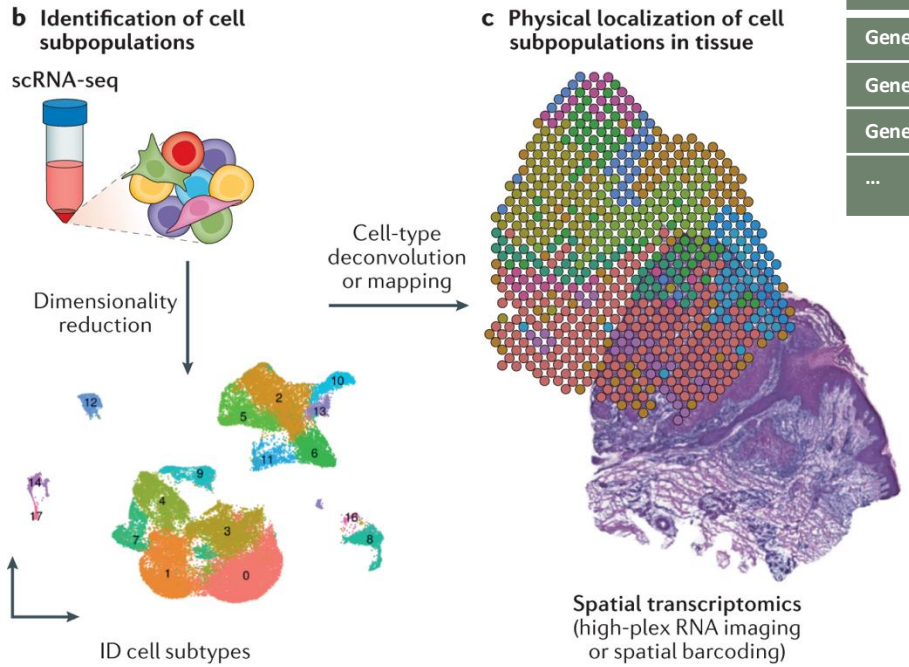
Spatial transcriptomics by sequencing





Joakim Lundeberg and Jonas Frisen: Stahl et al. (2016) *Science*

Integrating scRNA-seq with spatial transcriptomics



scRNA-seq

	Cell 1	Cell 2	Cell 3	...
Gene 1	10	0	0	...
Gene 2	0	2	0	...
Gene 3	1	4	12	...
...	...	...	...	...

clustering

	Cell type 1	Cell type 2	Cell type 3	...
Gene 1	3.7	10.5	2.6	...
Gene 2	6.0	2.0	8.3	...
Gene 3	2.1	7.1	15.4	...
...	...	...	...	...

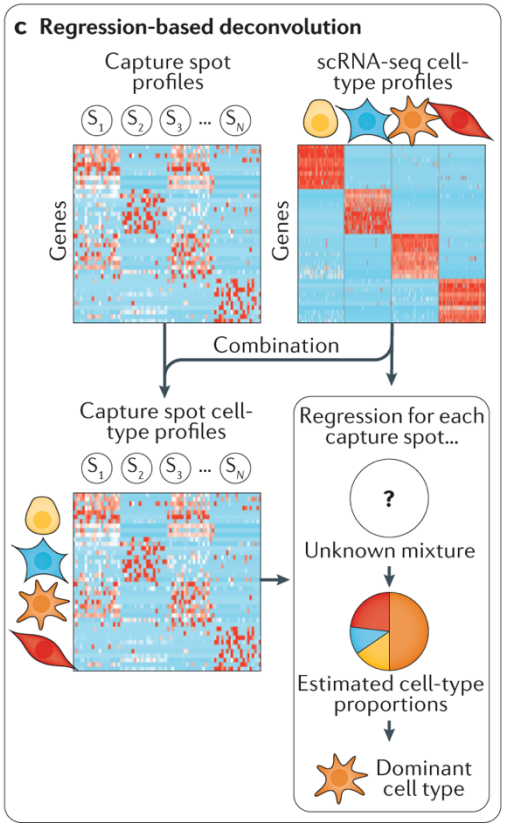
Spatial transcriptomics

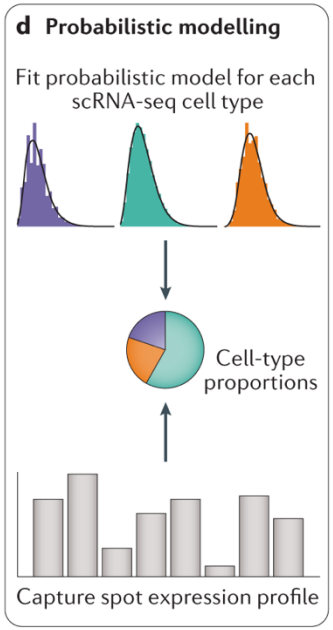
	Spot 1	Spot 2	Spot 3	...
Gene 1	25	41	16	...
Gene 2	80	5	0	...
Gene 3	12	32	12	...
...	...	...	...	...

Spot 1 = 0.1 * Cell type 1 + 0.5 * Cell type 4 + 0.4 * Cell type 10

Longo et al. (2021) *Nature Review Genetics*

Integrating scRNA-seq with spatial transcriptomics





scRNA-seq

	Cell 1	Cell 2	Cell 3	...
Gene 1	10	0	0	...
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...	...	...	...	...

Spatial transcriptomics

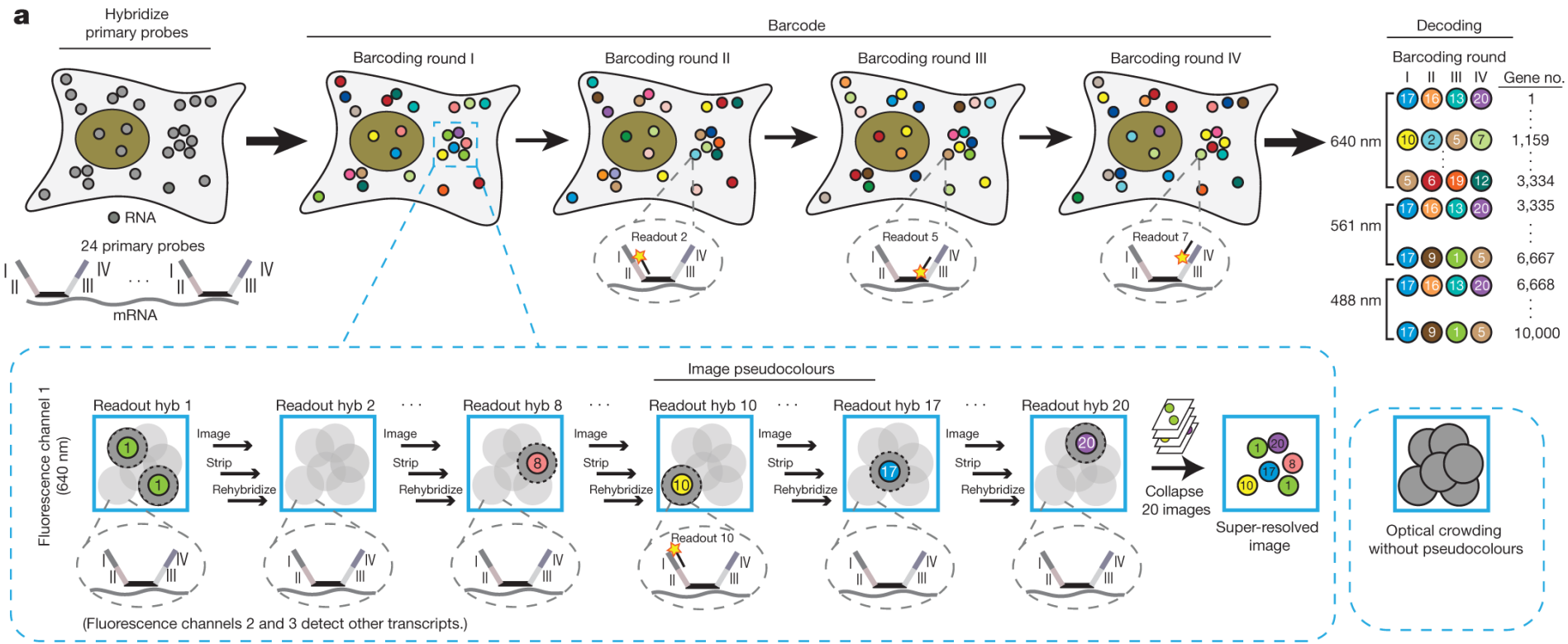
	Spot 1	Spot 2	Spot 3	...
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Spot 1 = 0.1 * Cell type 1 + 0.5 * Cell type 4 + 0.4 * Cell type 10

Longo et al. (2021) *Nature Review Genetics*

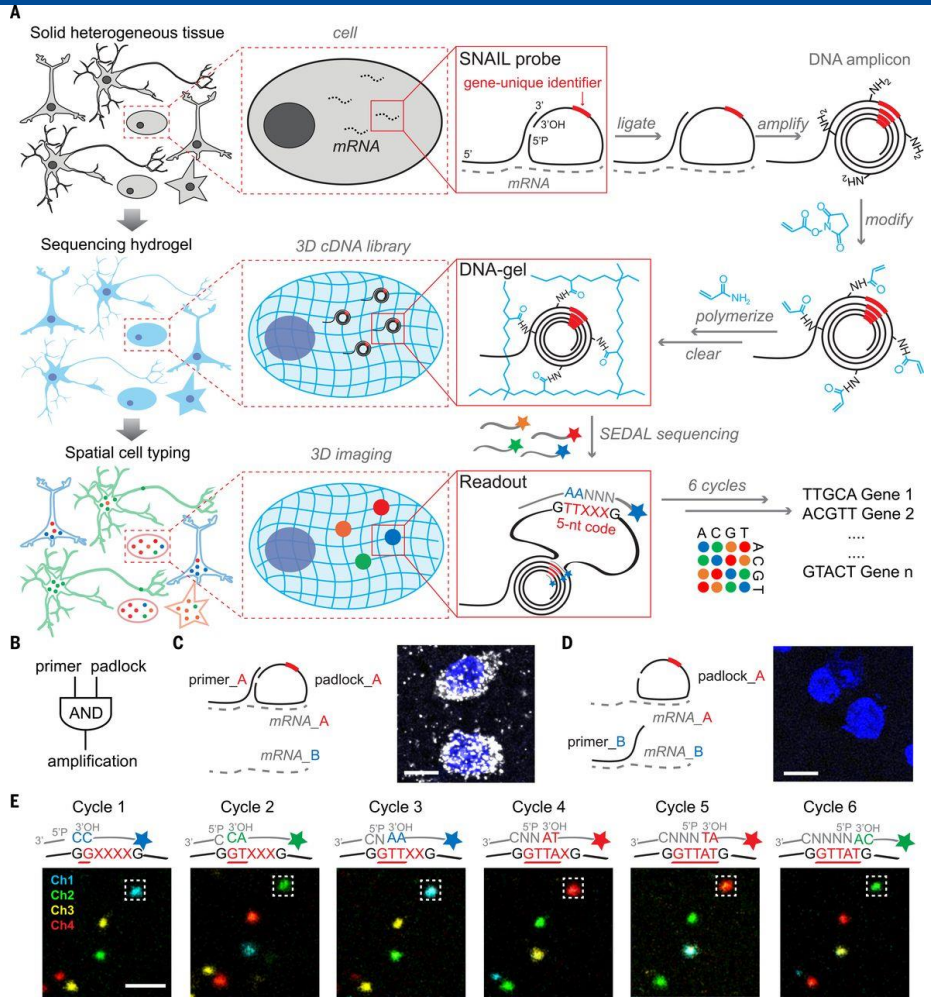
From spots to cells (or even mRNA molecules)…
Imaging-based methods

Single-molecule resolution spatial transcriptomics by multiplexed smFISH



seqFISH+ (Long Cai) : Eng et al. (2019) *Nature*

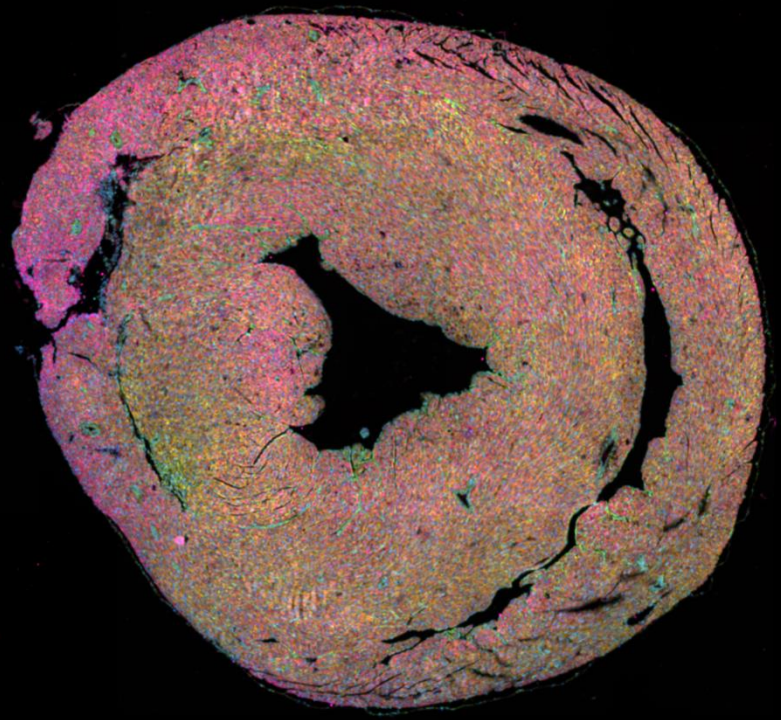
Single-molecule resolution spatial transcriptomics by in situ sequencing



Mats Nilsson:
Larsson et al. (2010) *Nature Methods*

Wang et al. (2018) *Science* (3D)

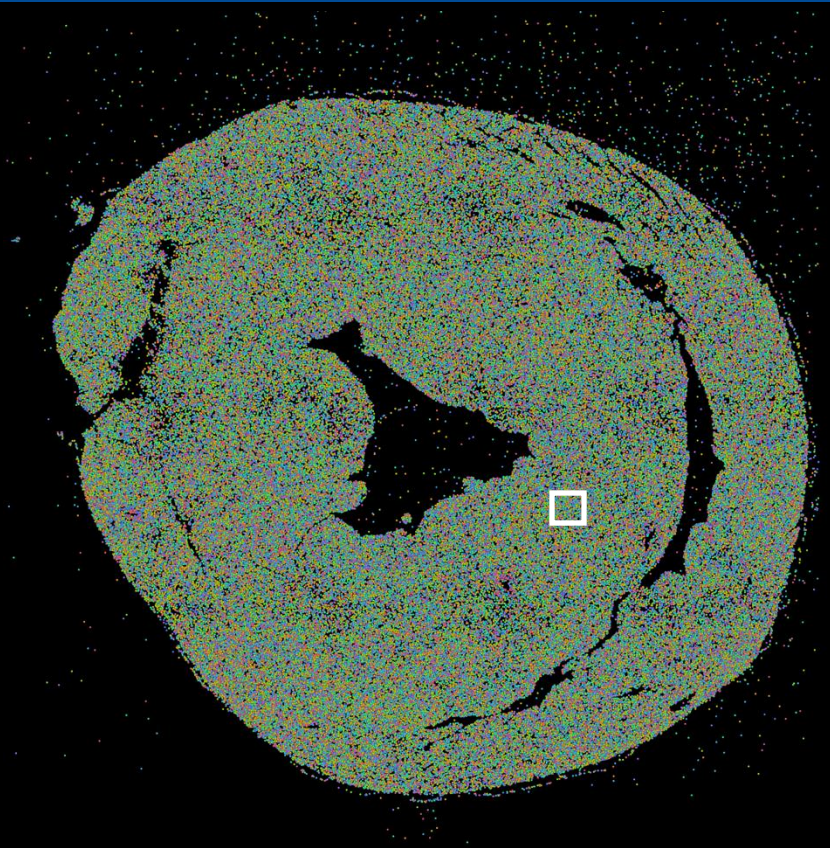
Preliminary Data: Spatial Transcriptomics of the neonatal heart



Cell morpholgy staining:
Intracellular protein
Intracellular RNA
Membrane proteins

10x Xenium on 480 genes

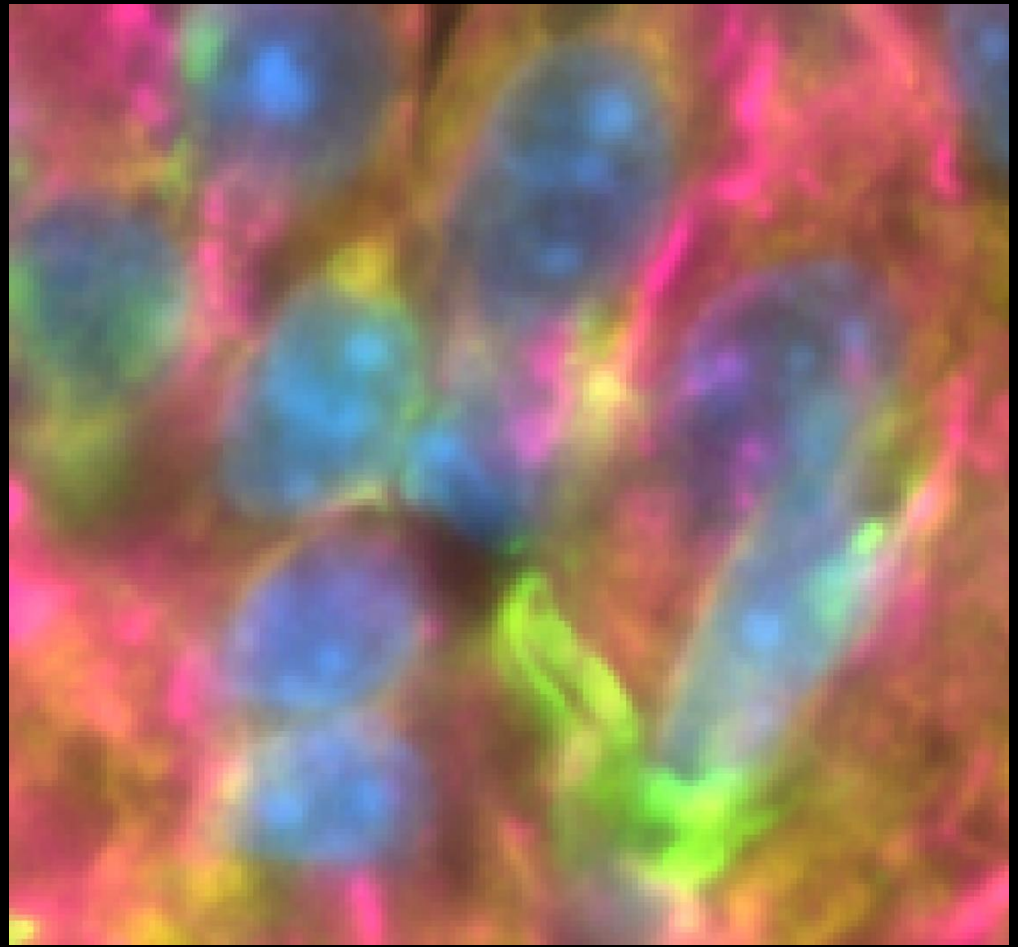
Preliminary Data: Spatial Transcriptomics of the neonatal heart



In situ sequencing of 480 genes

10x Xenium on 480 genes

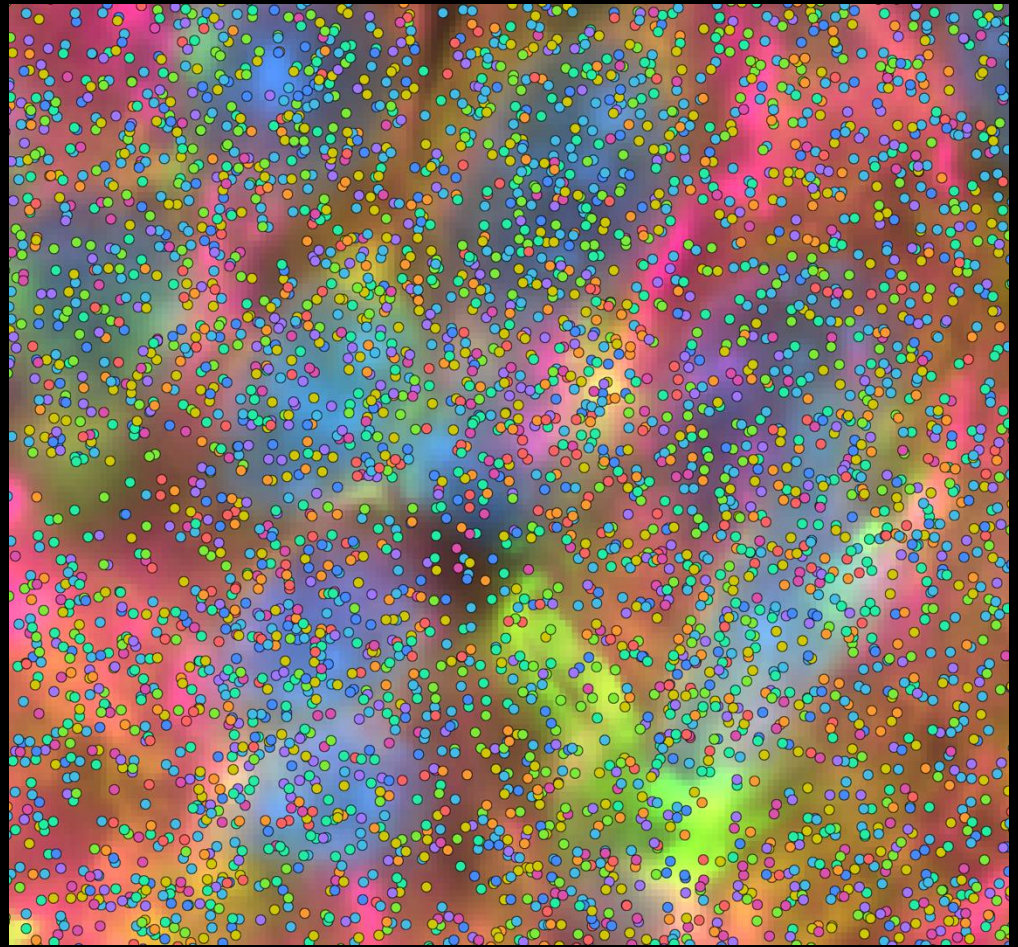
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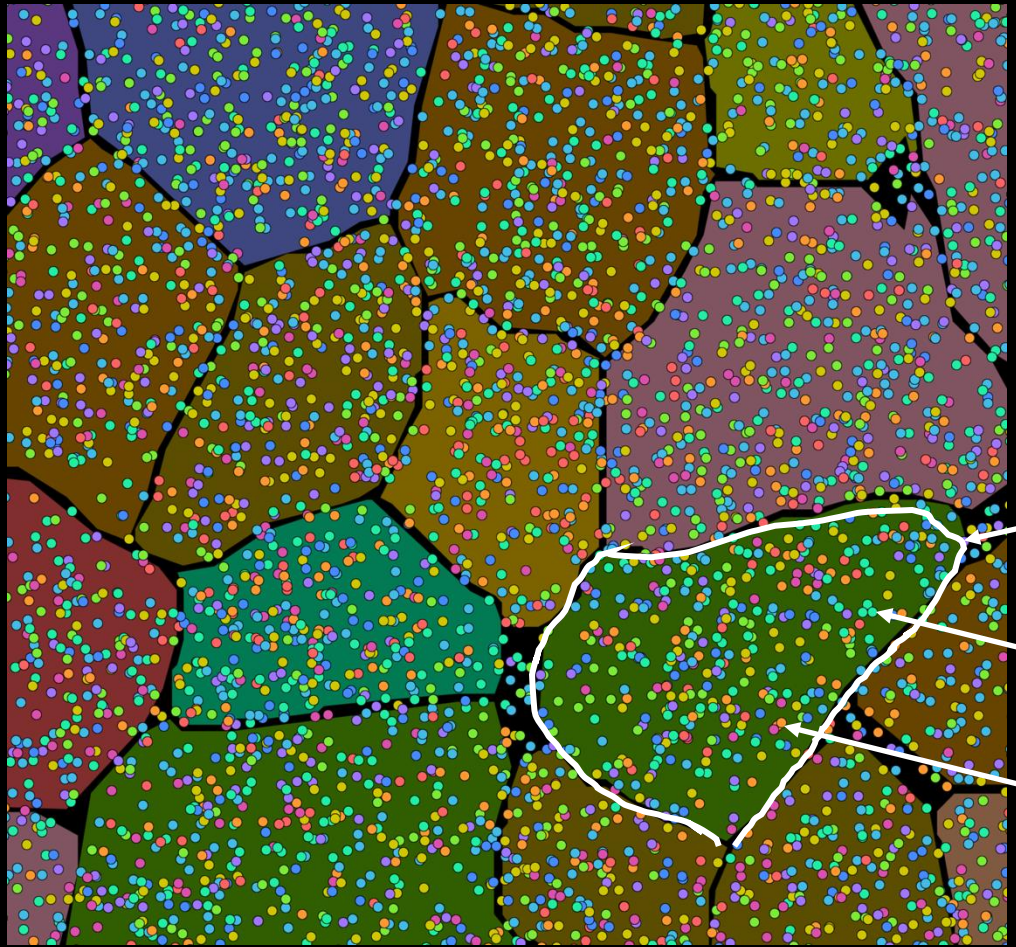
Preliminary Data: Spatial Transcriptomics of the neonatal heart



In situ sequencing of 480 genes

10x Xenium on 480 genes

Preliminary Data: Spatial Transcriptomics of the neonatal heart



Cell morpholgy + Gene expression = spatial cell type map

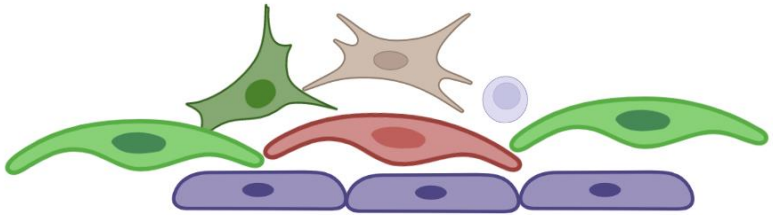
Cell

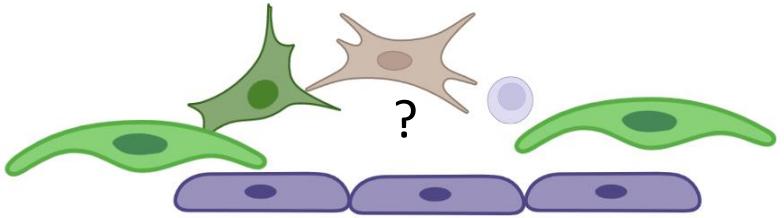
mRNA of gene A

mRNA of gene B

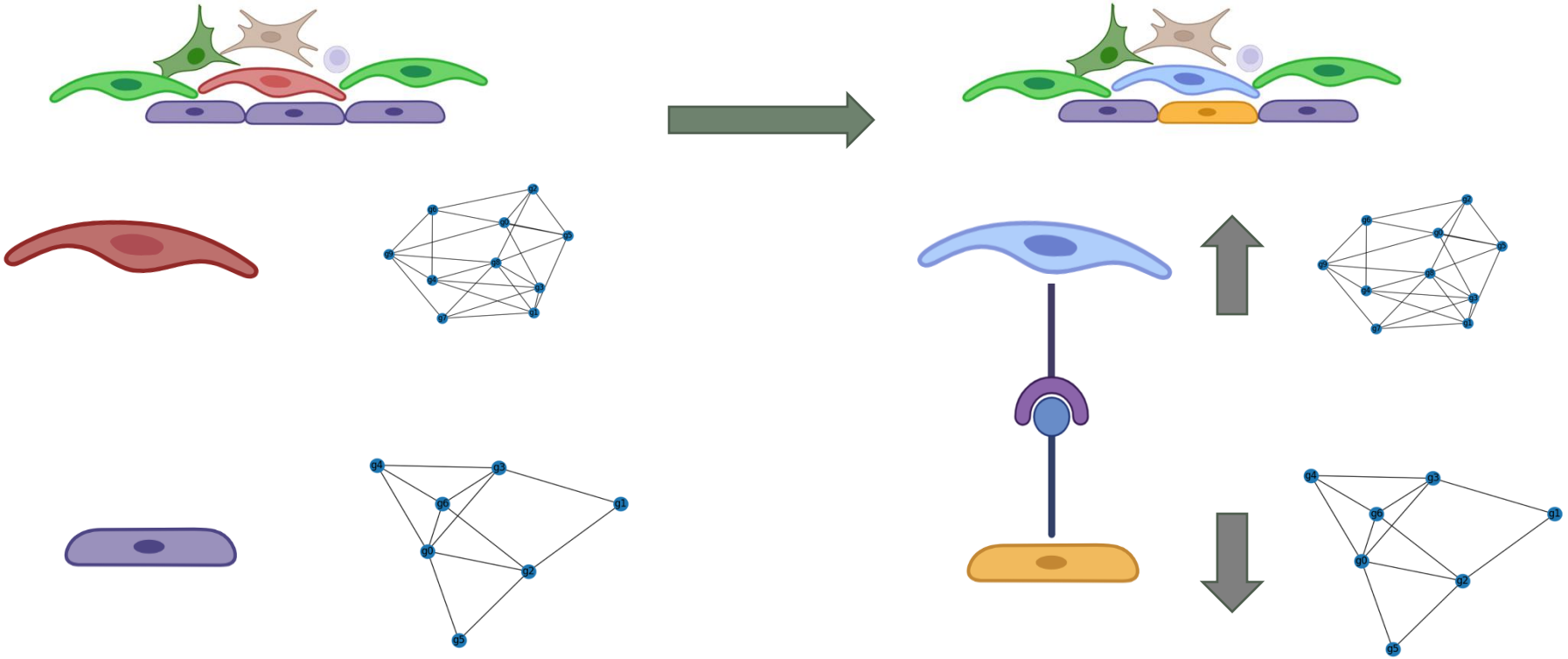
10x Xenium on 480 genes

Question 1: Predictive capacity of the niche?





Question 2: Extrinsic sources of intra-cell type variability?

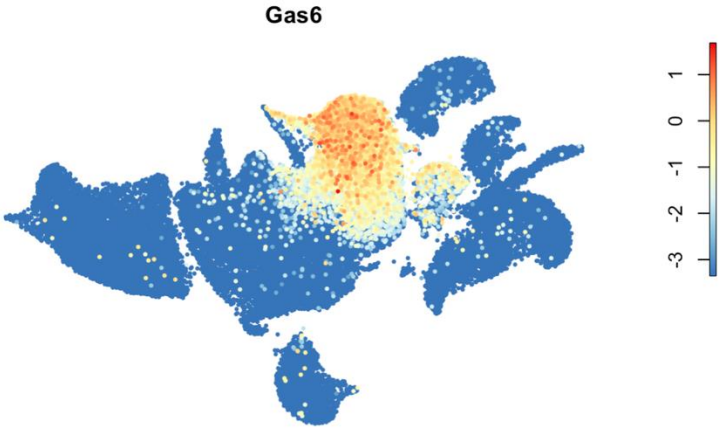


Integration Challenge

single-cell RNA-seq

Genome-wide

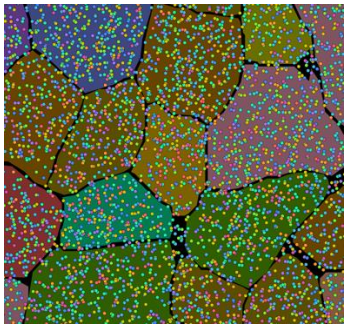
no spatial information



Imaging-based spatial transcriptomics

<500 genes

spatial single-cell (single-molecule) resolution



NiCo: Resolving extrinsic determinants of cell state variability

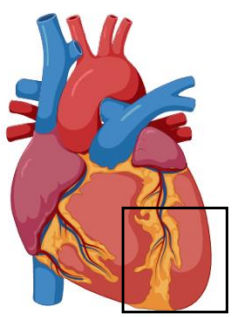


Ankit Agrawal

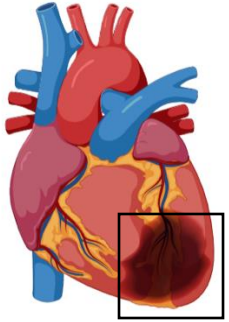
- *Infer cell type identity in spatial data*
- *Identify predictive niche interactions*
- *Identify latent factors capturing intra-cell type variability*
- *Infer covariation of latent factors in co-localized cell types*
- *Predict covarying pathways, gene modules, and ligand-receptor pairs mediating these interactions* ***at cell-type resolution in local niches (as opposed to continuous local domains).***

Agrawal et al. (2024) *Nature Communications*

Cellular crosstalk during lesion formation upon myocardial infarction



Myocardial infarction (MI)



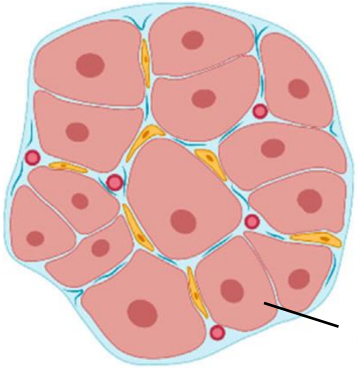
Excessive fibrosis, prolonged inflammation



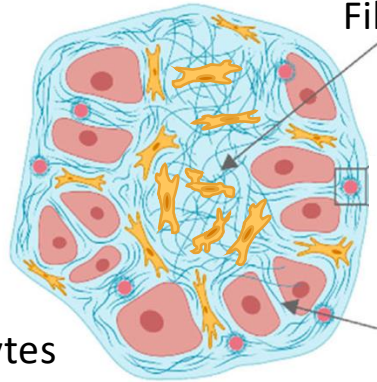
Heart Failure

?

Inflammation + Fibrosis



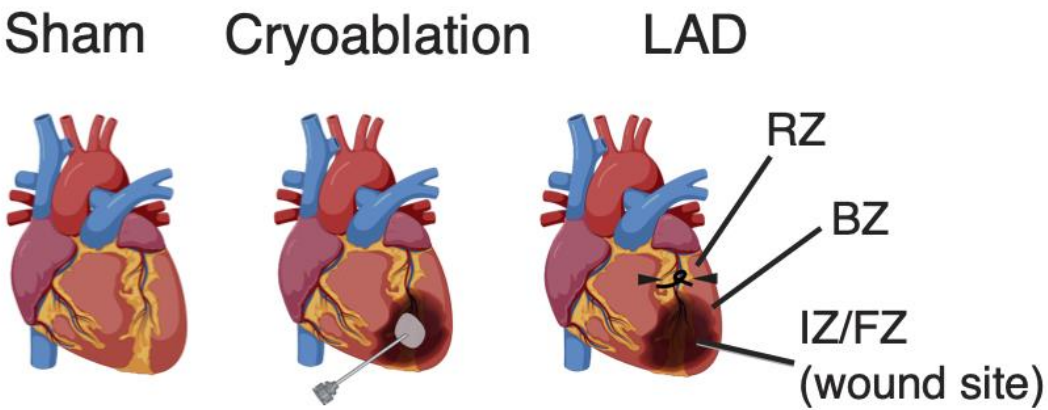
Cardiomyocytes (CM)



Fibrotic scar

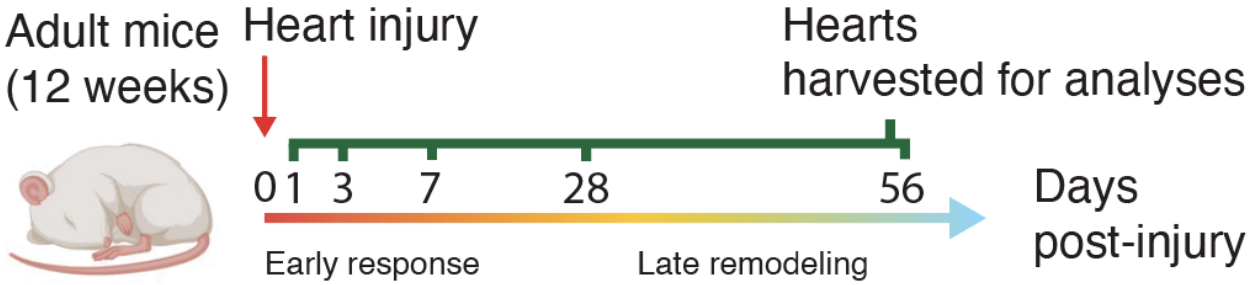
Remaining CM

Spatio-temporal analysis of cardiac scar formation in the mouse model



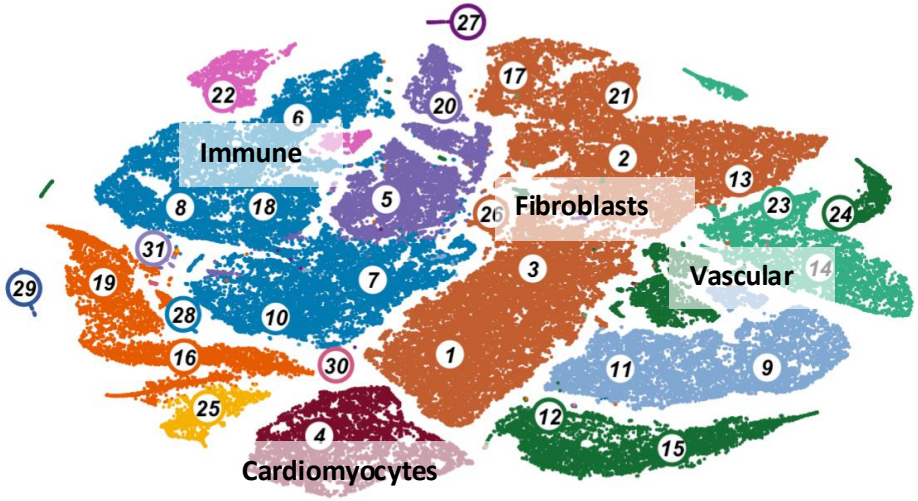


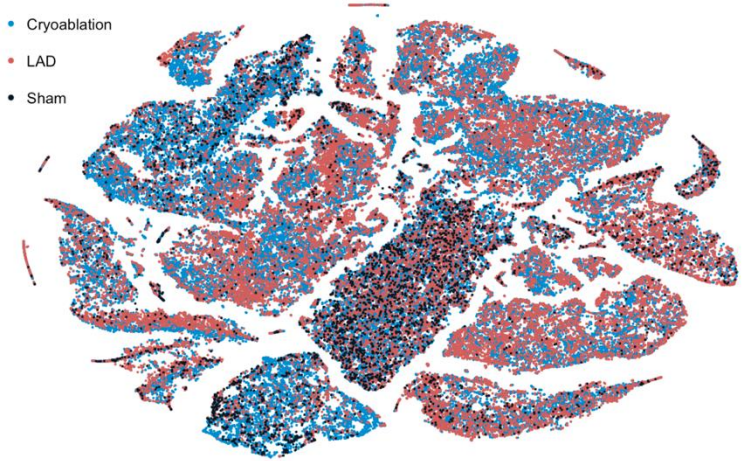
Andy Chan

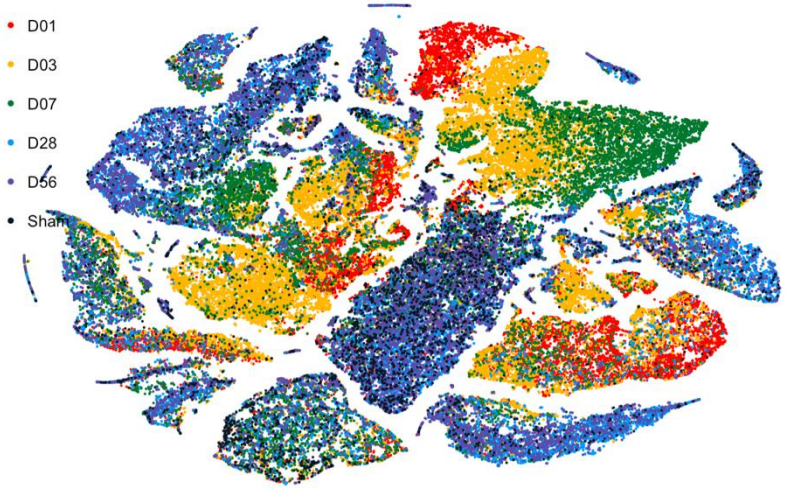


Chan et al. (2025) *Nature Cardiovascular Research*

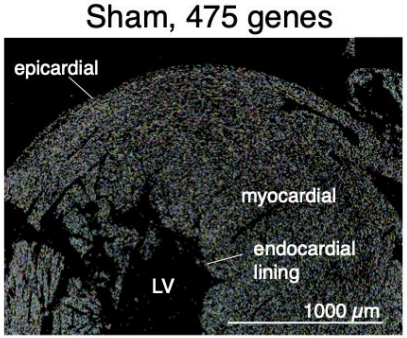
Temporal cell type atlas of the cardiac lesion

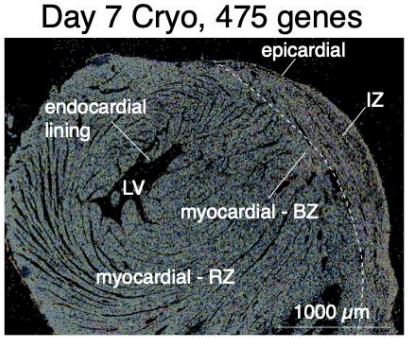


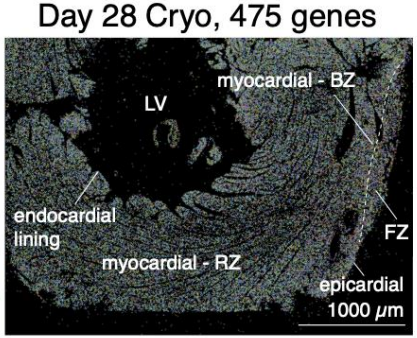


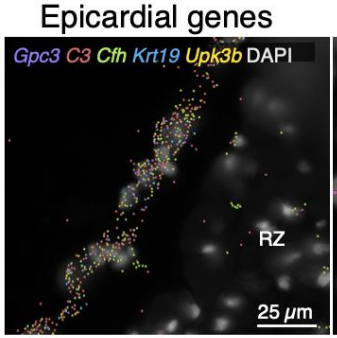


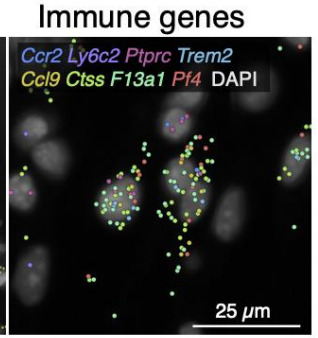
Spatial Transcriptomics of the cardiac lesion

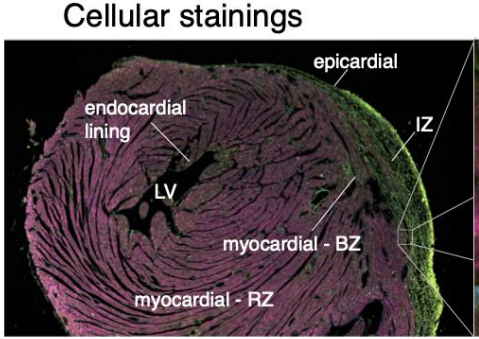


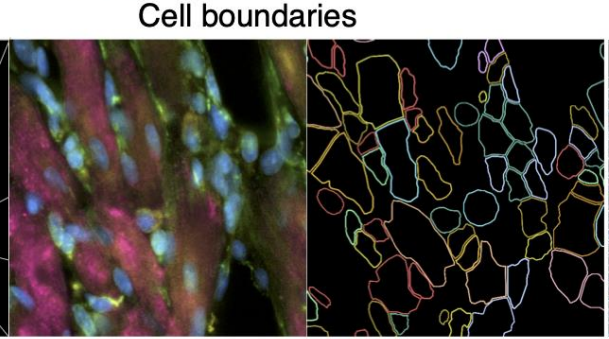


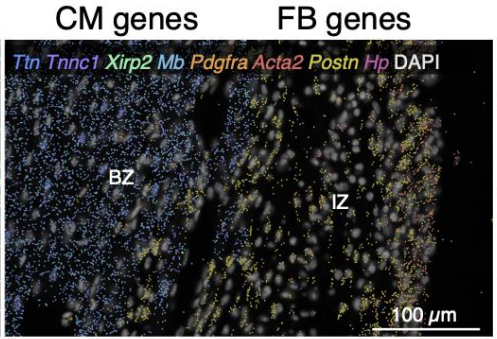


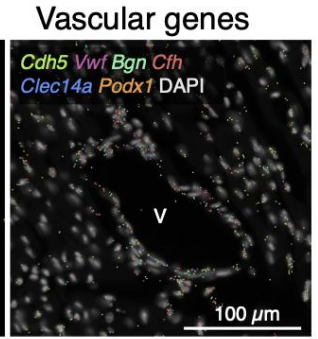




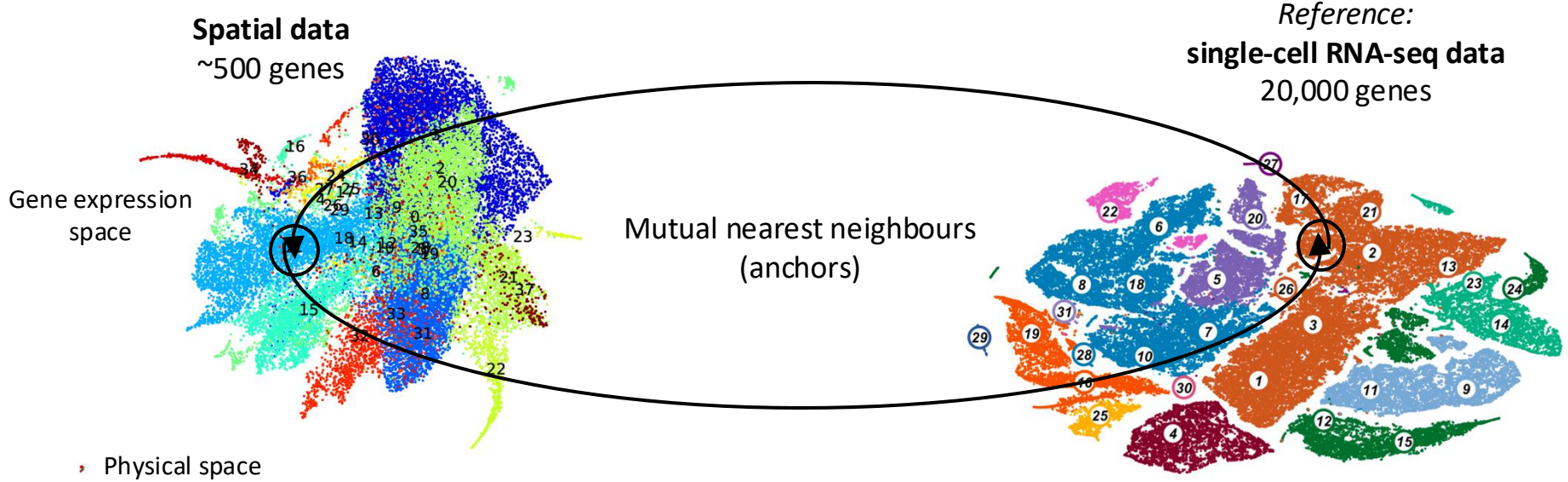


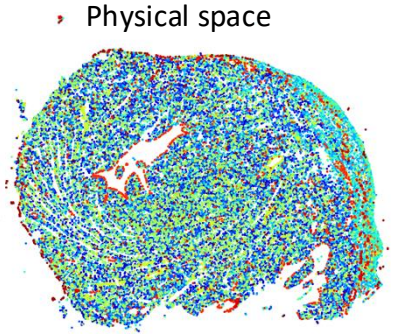






Cell type annotation by label transfer





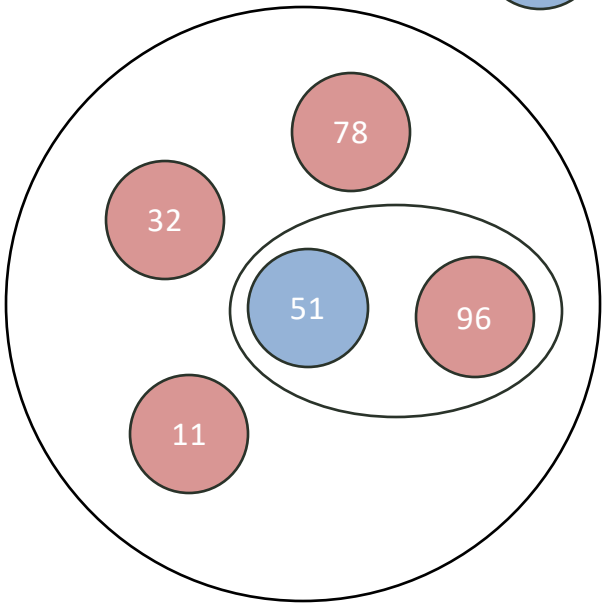
- Elimination of total count dependence by regression (Pearson Residuals, scTransform)
- PCA transformation
- Euclidean distance between reference and query cells
- Mutual nearest-neighbourhoods (anchors)

Mutual nearest neighbours as anchors for label transfer

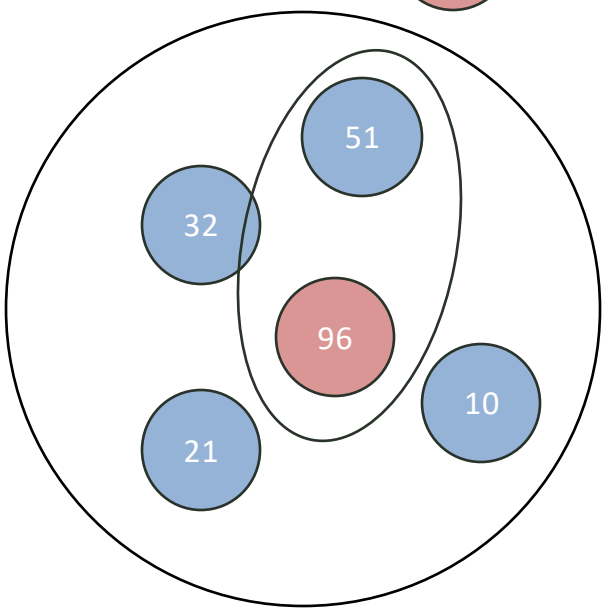
i Query Cells i=1,…,N

j Reference Cells j=1,…,M

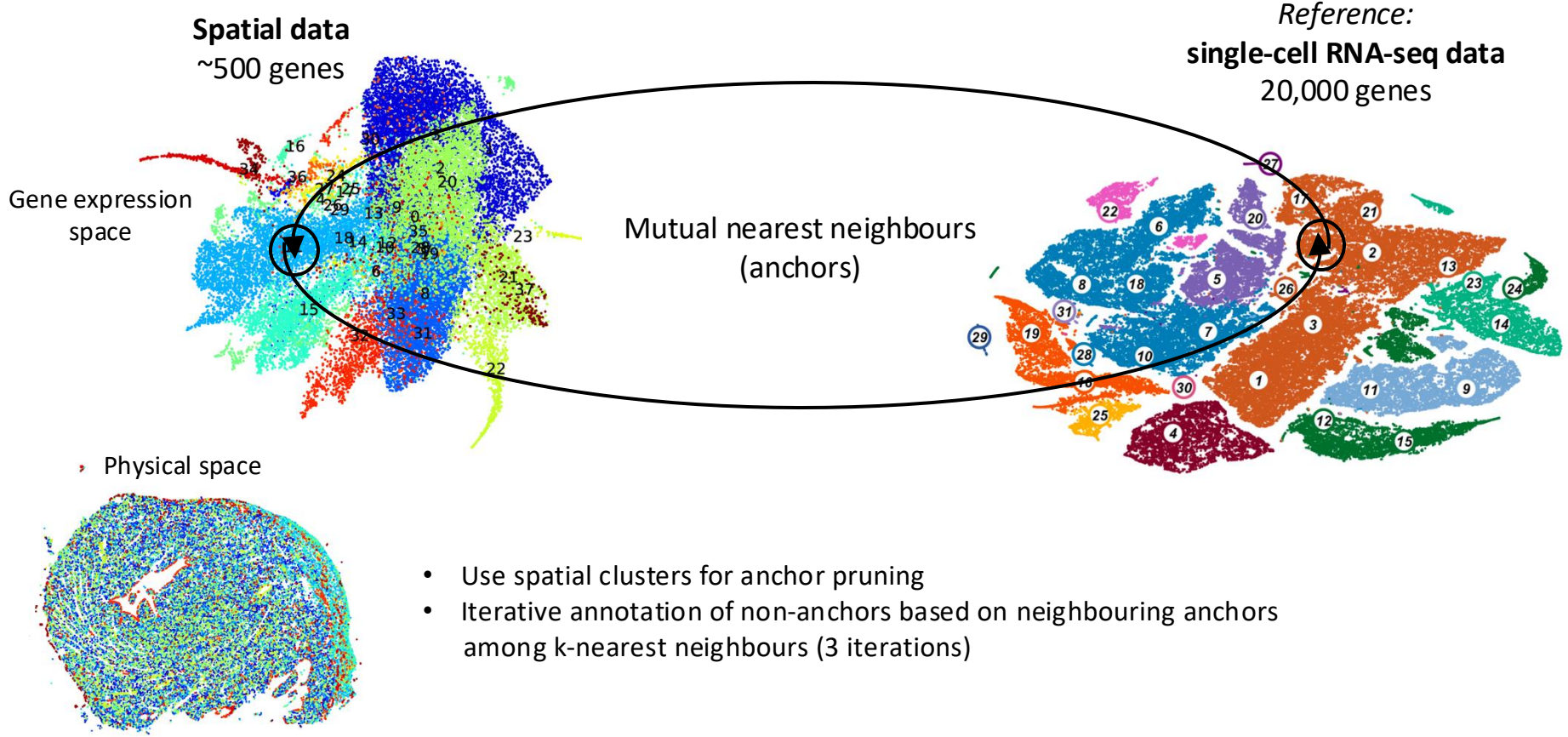
K-nearest reference neighbours of 51



K-nearest query neighbours of 96

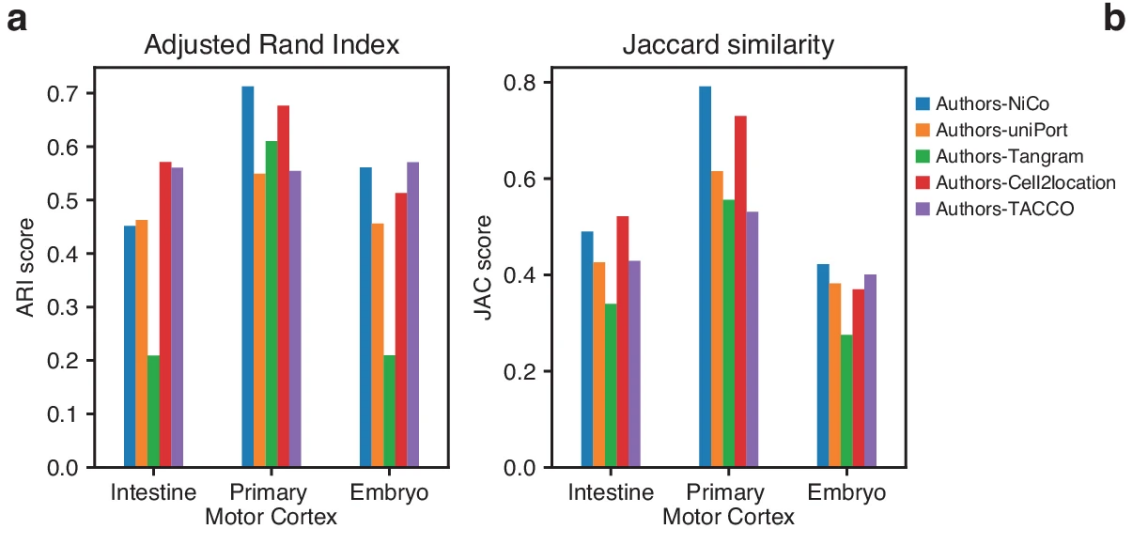


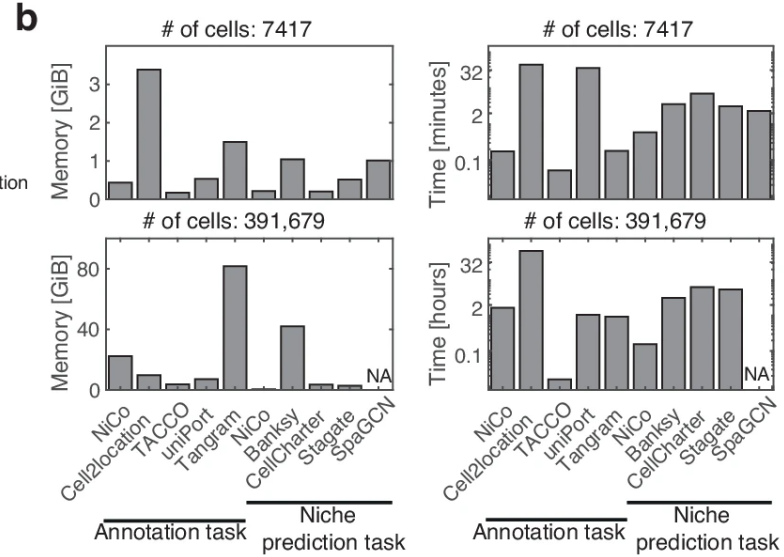
Cell type annotation by label transfer



- Use spatial clusters for anchor pruning
- Iterative annotation of non-anchors based on neighbouring anchors among k-nearest neighbours (3 iterations)

Cell type annotation by label transfer



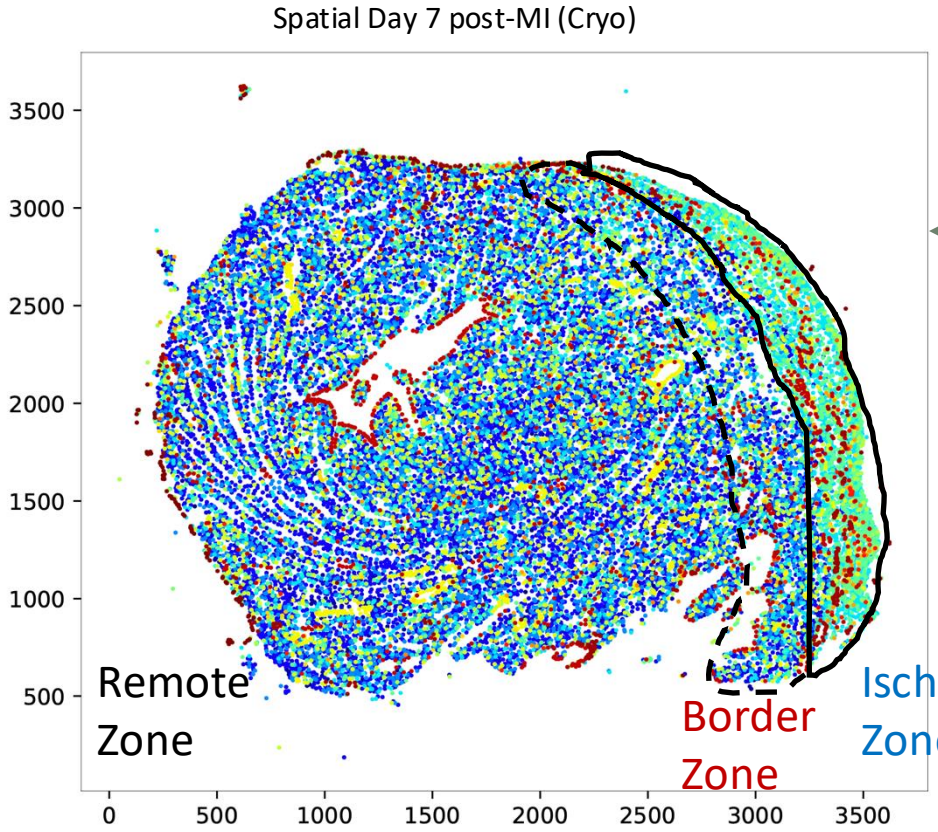


Embryo (MERFISH):
Petukhov et al. (2021) Nature Biotechnology

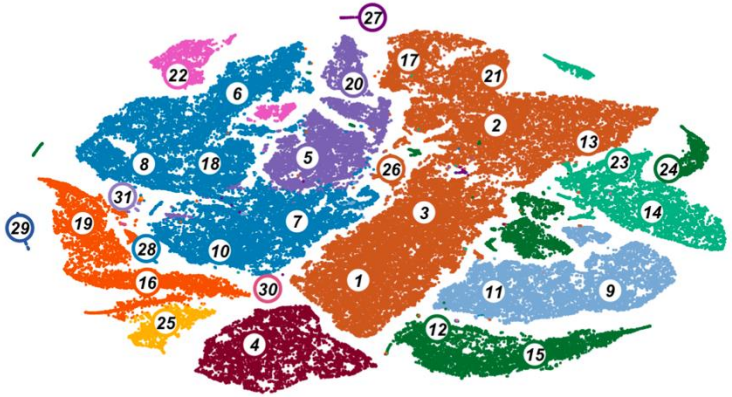
Primary Motor Cortex (MERFISH):
Zhang et al. (2021) Nature

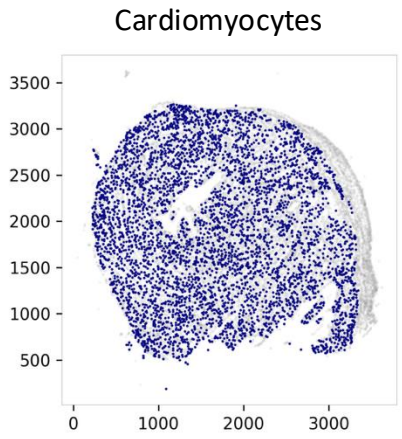
Embryo (seqFISH):
Lohoff et al. (2021) Nature Biotechnology

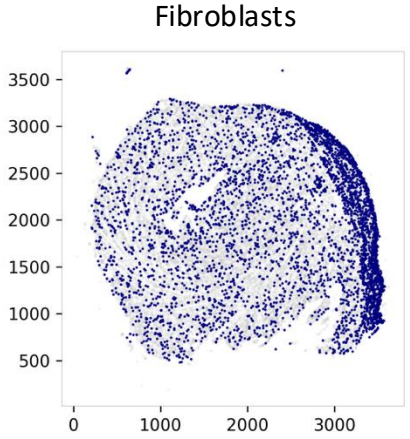
Spatial Transcriptomics of the cardiac lesion: Annotation



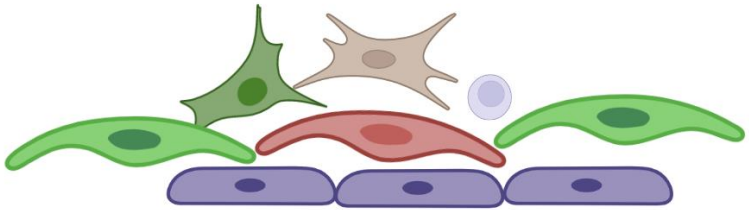
NiCo annotate





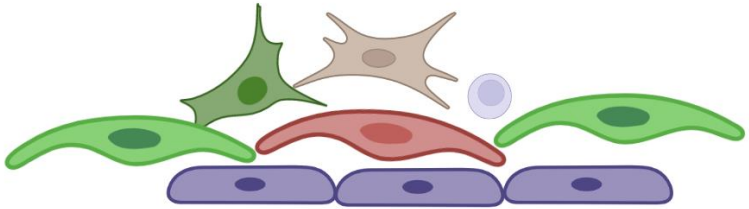


Inference of predictive neighbours by multinomial logistic regression



$$P(\text{cell}) \sim w_{\text{cell}} x_{\text{cell}} + w_{\text{cell}} x_{\text{cell}} + w_{\text{cell}} x_{\text{cell}} + w_{\text{cell}} x_{\text{cell}} + w_{\text{cell}} x_{\text{cell}}$$

Inference of predictive neighbours by multinomial logistic regression

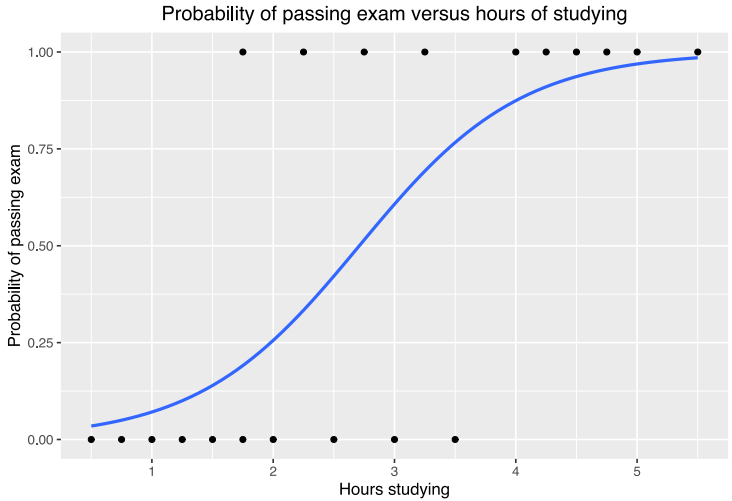


P( ) ~ W X + W X + W X + W X + W X

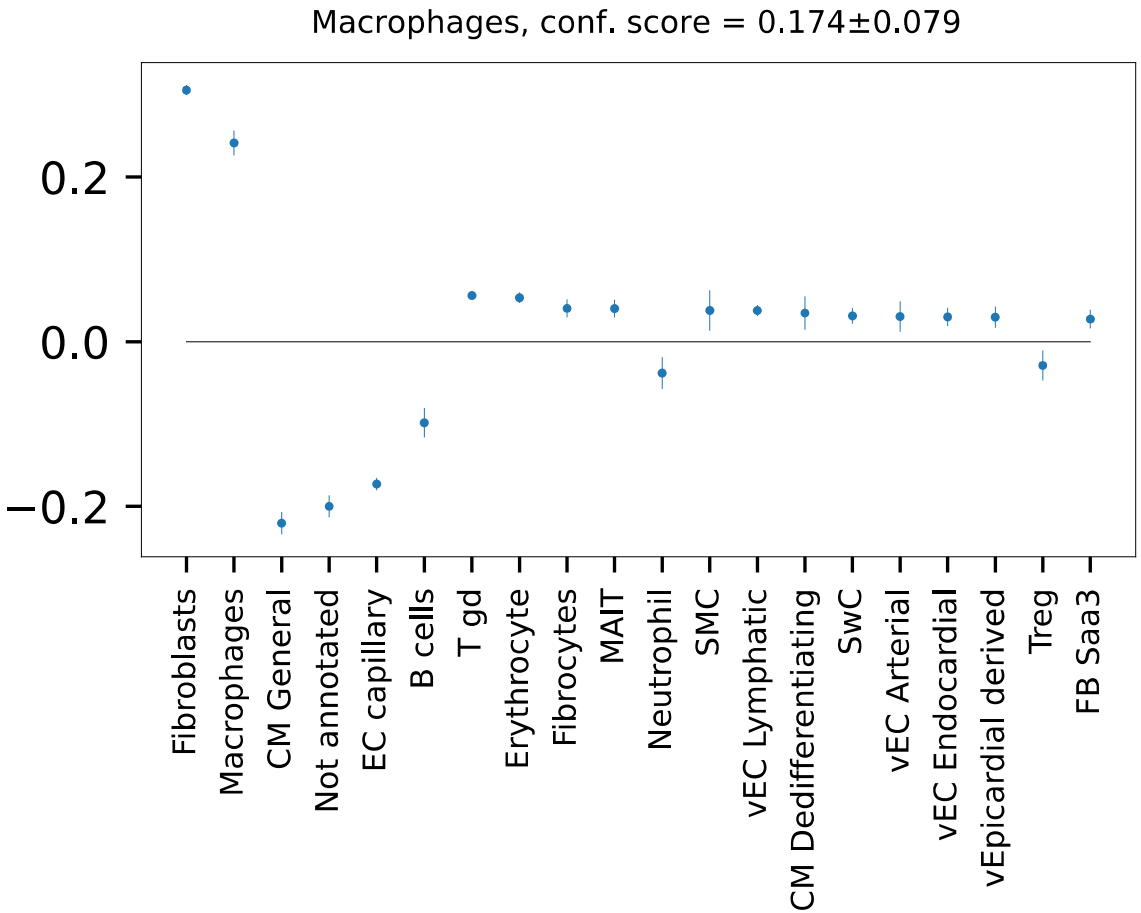
$$P(y_i = k|\mathbf{x}_i; W) = \frac{\exp(\mathbf{w}_k \mathbf{x}_i + w_{0,k})}{\sum_{j=1}^{K} \exp(\mathbf{w}_j \mathbf{x}_i + w_{0,j})}$$

$$X_{ij} = \Lambda_{ij} / \sum_k f_j \cdot \Lambda_{ik}$$

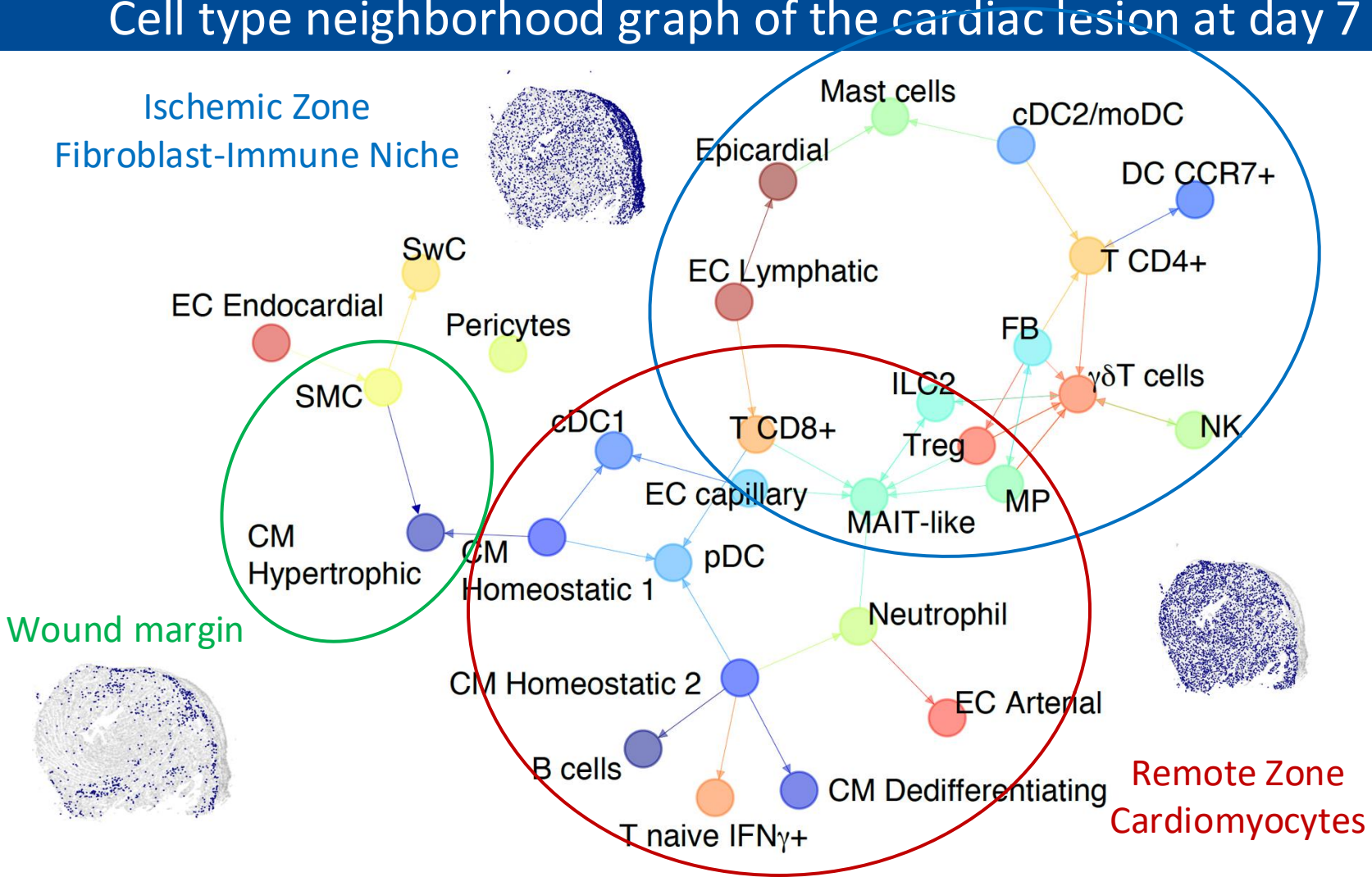
$$p(x) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x)}}$$



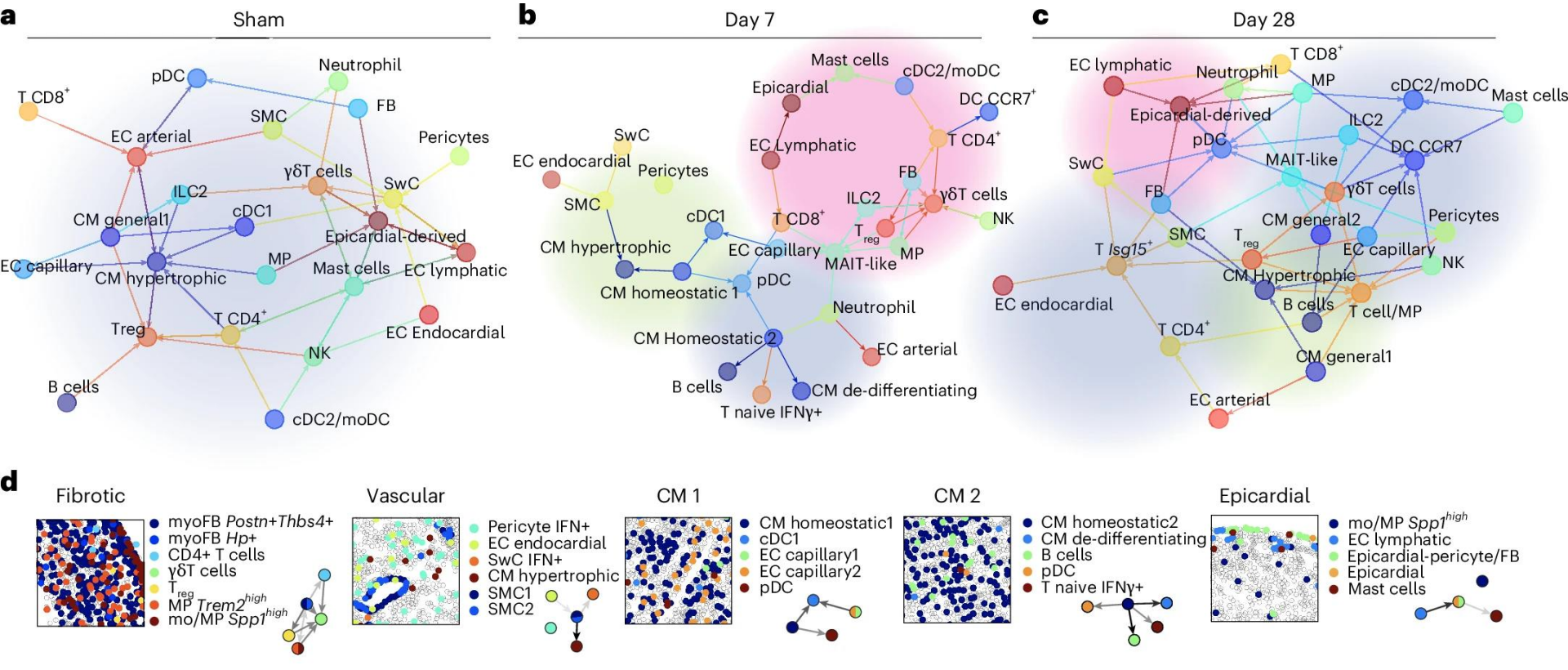
Predictive capacity of cardiac lesion niche



Cell type neighborhood graph of the cardiac lesion at day 7

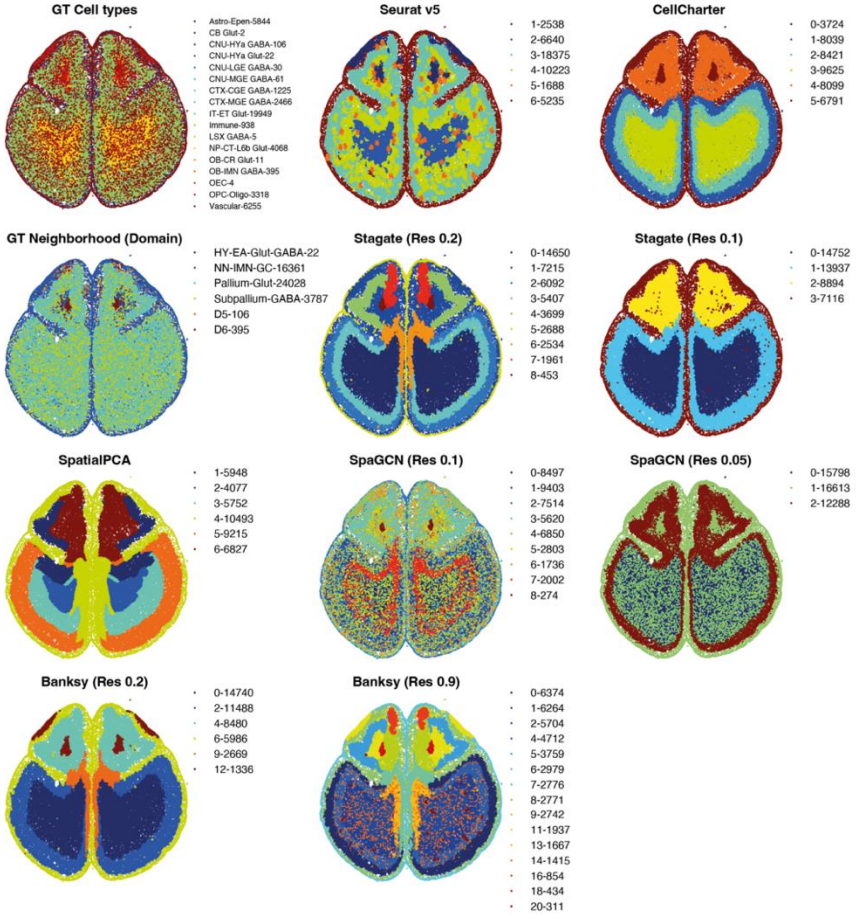


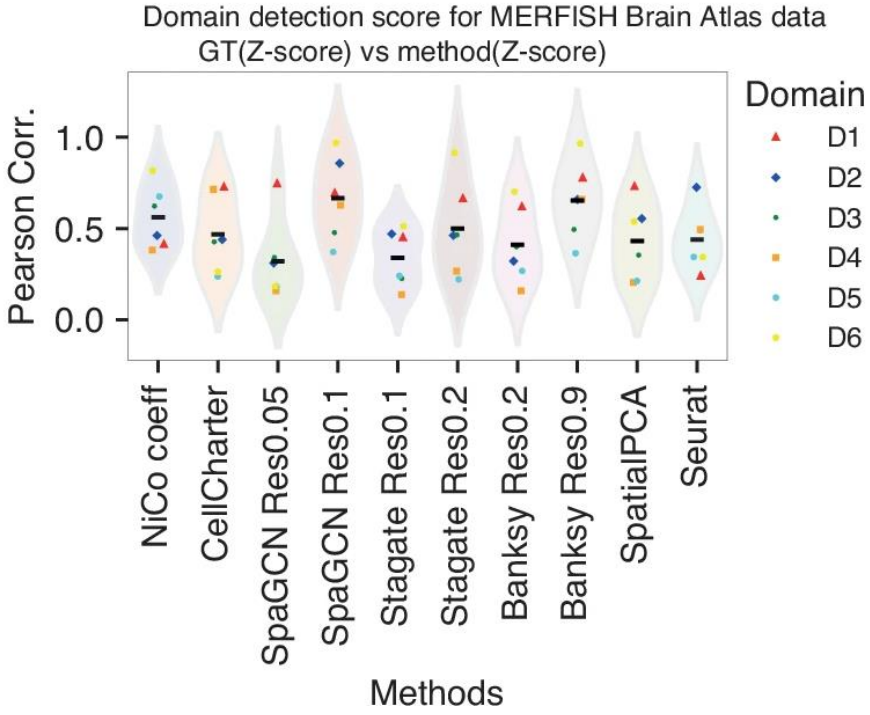
Cell type neighborhood graph dynamics



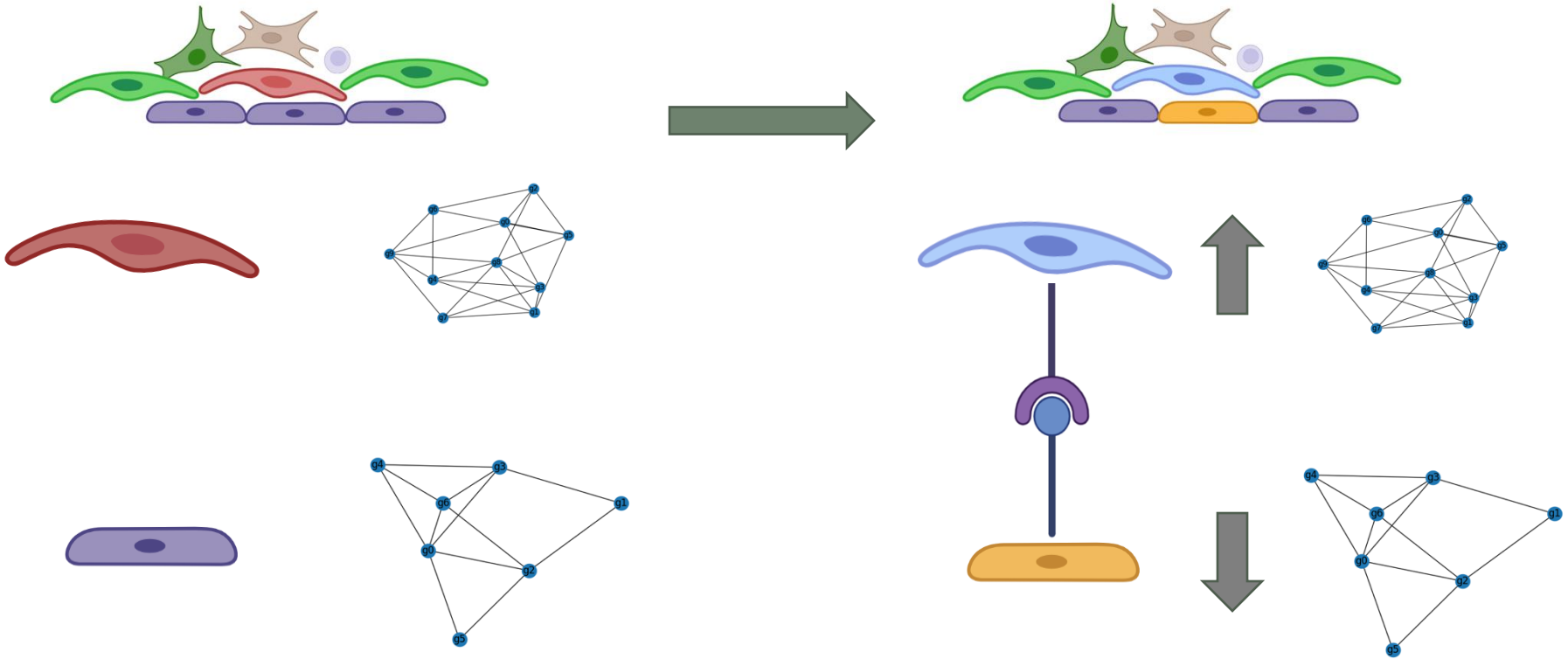
Cell type neighborhood graph dynamics

Allen Brain Atlas MERFISH Data

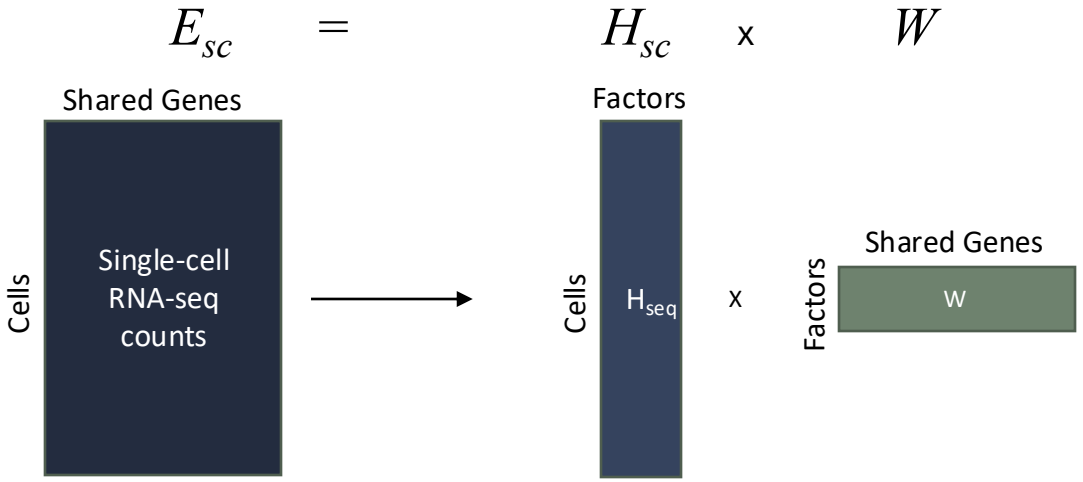




Co-variation of gene expression in neighboring cells



Latent factors of intra-cell type variability

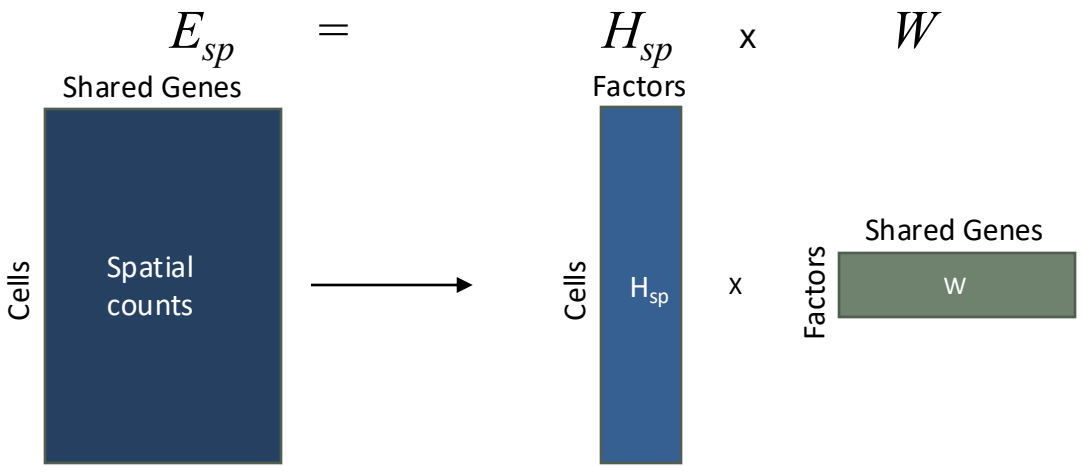


Integrated NMF

$$\text{argmin}\left(||E_{sc} - H_{sc}\,(W + V_{sc})\,||_F^2 + \lambda \|H_{sc} V_{sc}\|_F^2 \right.$$

$$\left. + ||E_{sp} - H_{sp}(W + V_{sp})||_F^2 + \lambda ||H_{sp} V_{sp}||_F^2 \right)$$

or

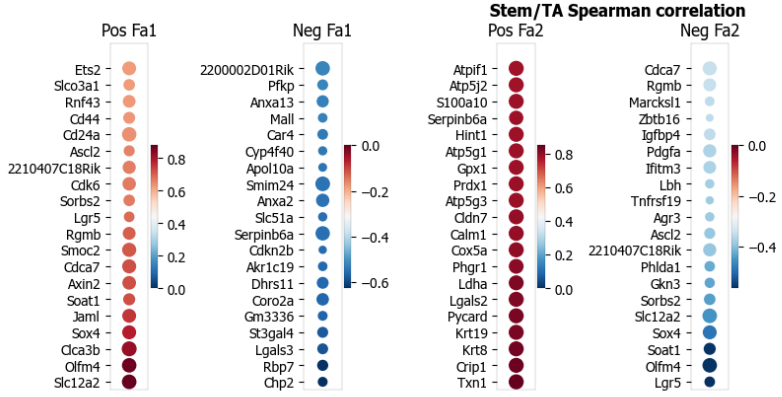


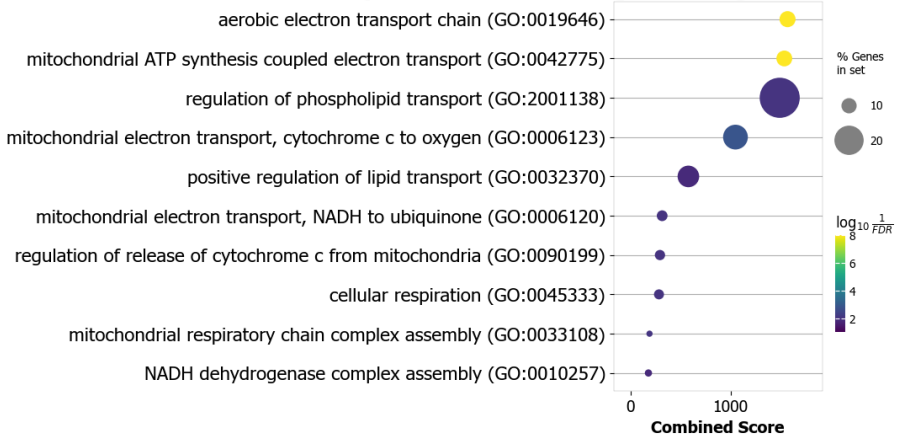
NMF + transfer of factors

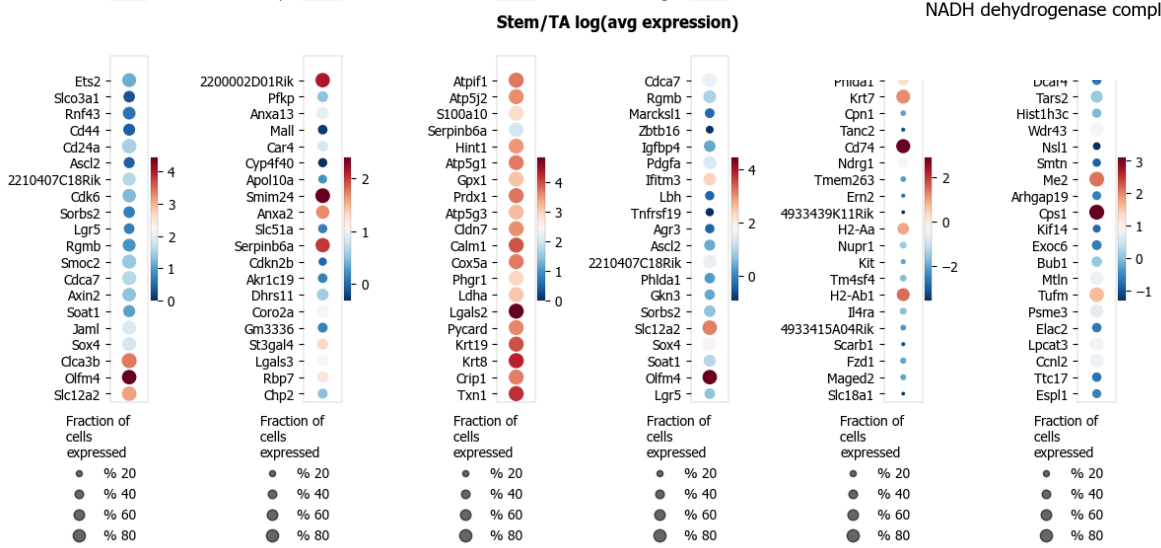
$$H_{sp}[i,j] = \frac{H_{sp}[i,j] * (W^T E_{sp}^T)[i,j]}{\left(W^T W H_{sp}\right)[i,j]}$$

Yang and Michailidis (2016) *Bioinformatics*
Welch et al. (2019) *Cell*

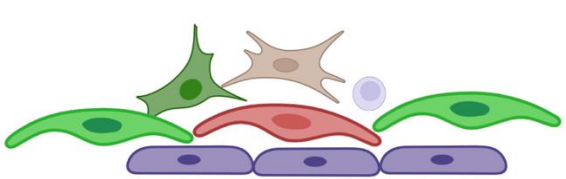
Inference of factor covariation within the niche

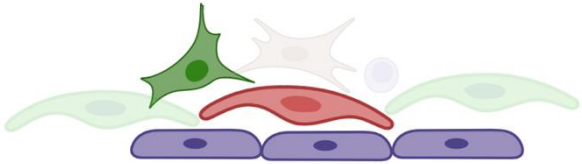


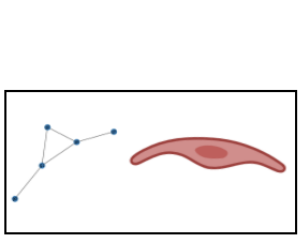




Inference of factor covariation within the niche





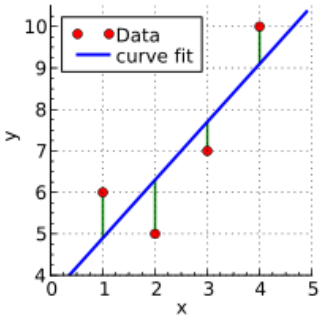


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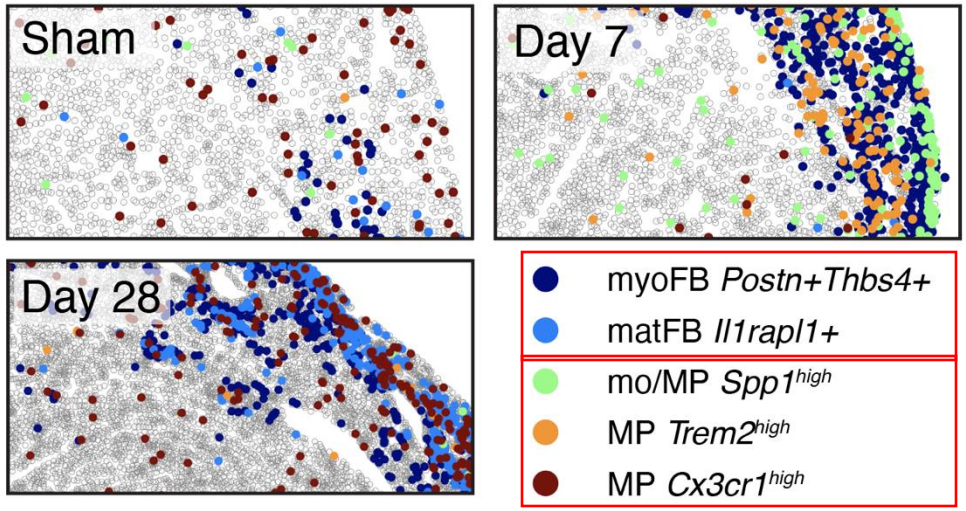
$\beta_{__,1}$ + $\beta_{__,2}$ + $\beta_{__,3}$

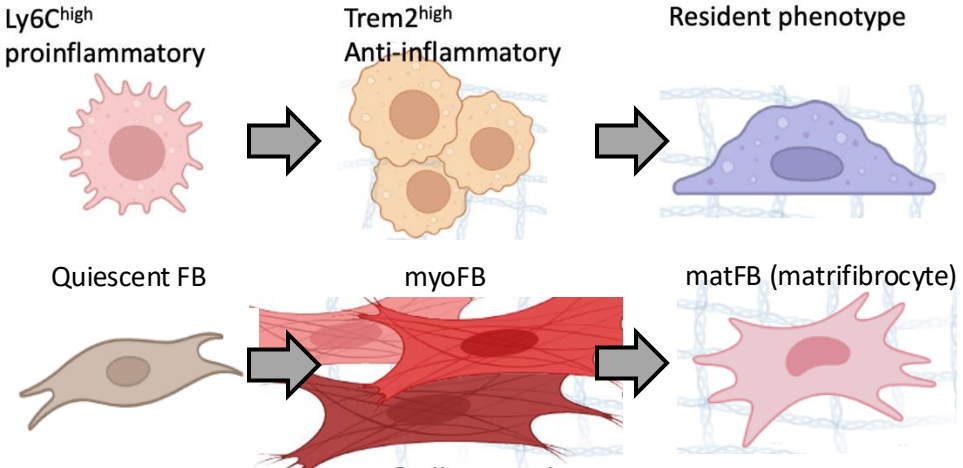
+ $\beta_{__,1}$ + $\beta_{__,2}$ + $\beta_{__,3}$

$$y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip} + \varepsilon_i = \mathbf{x}_i^{\mathsf{T}} \boldsymbol{\beta} + \varepsilon_i, \qquad i = 1, \ldots, n,$$

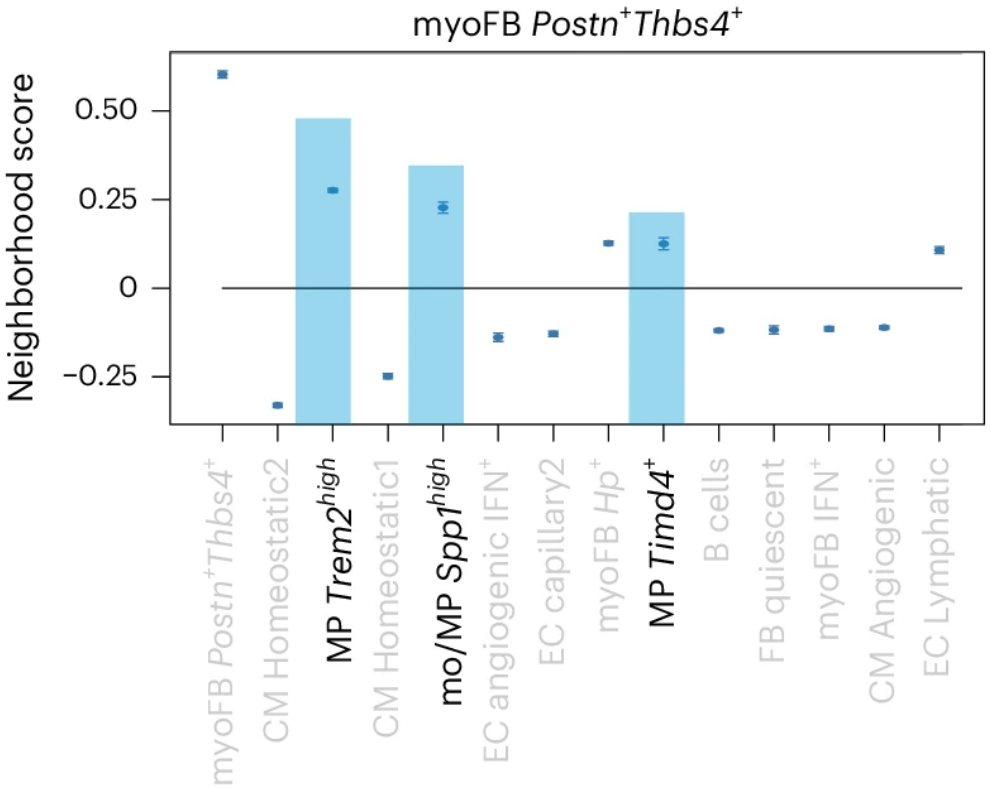


Covariation between cardiac Macrophages and Fibroblasts

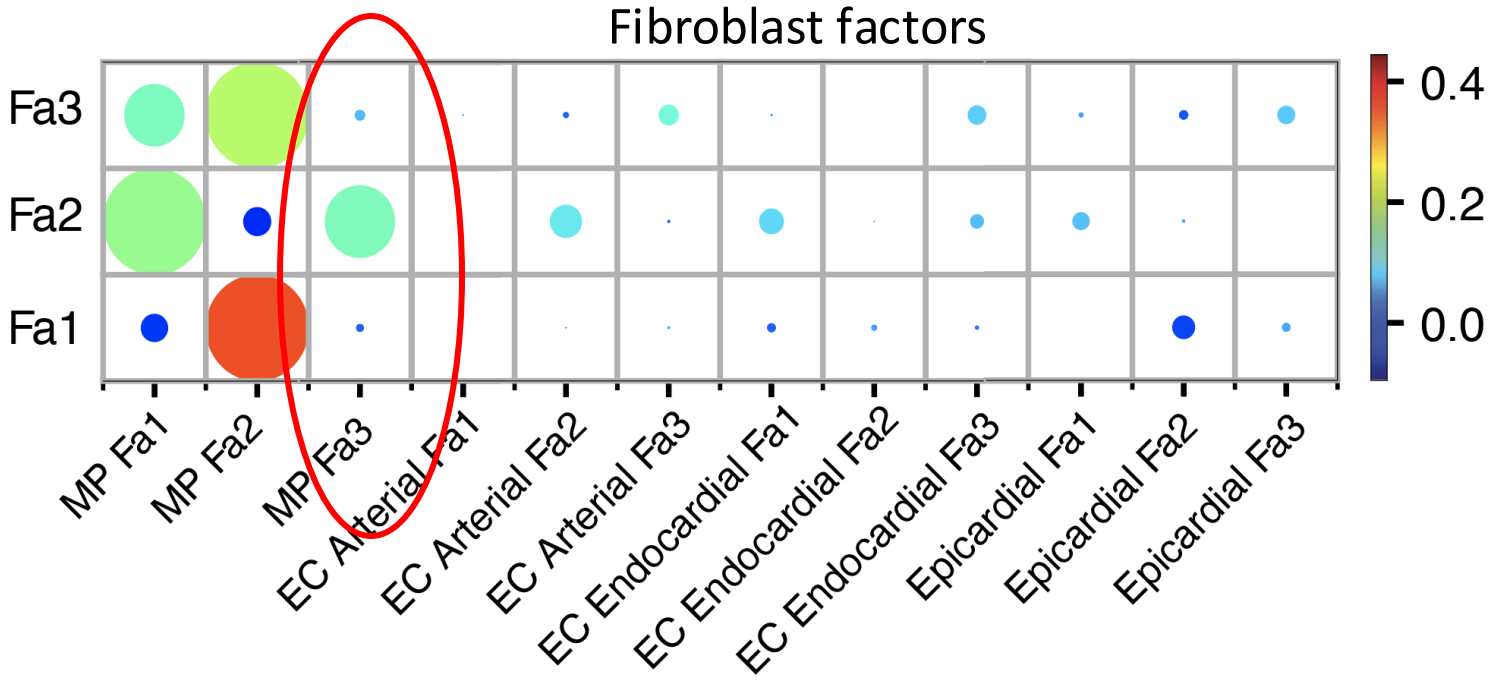




Inference of factor covariation within the niche



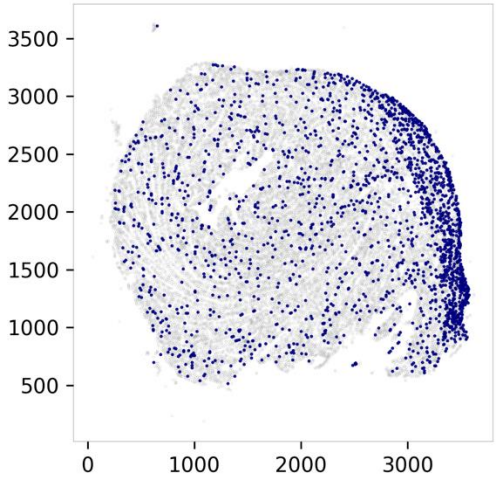
Covariation between cardiac Macrophages and Fibroblasts

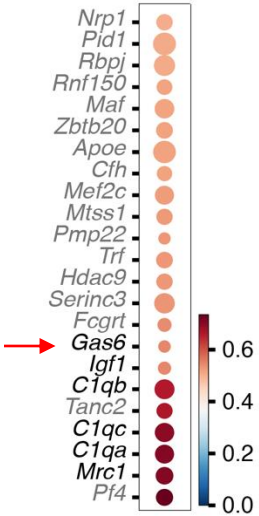


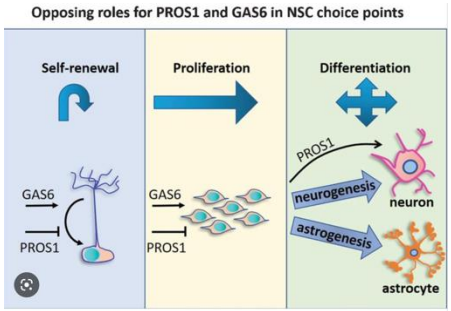
MP: Macrophage

Association of latent factors with full transcriptome

Macrophages (Fa3)

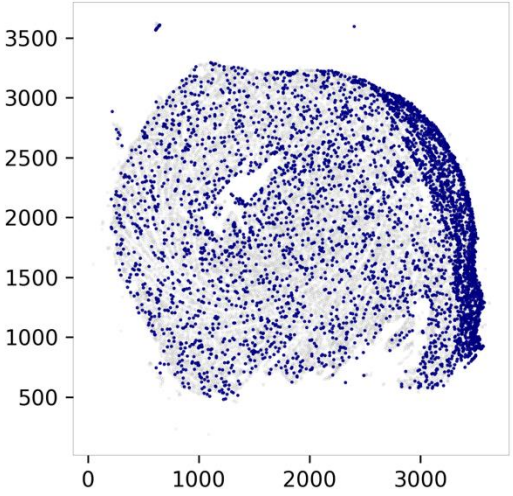


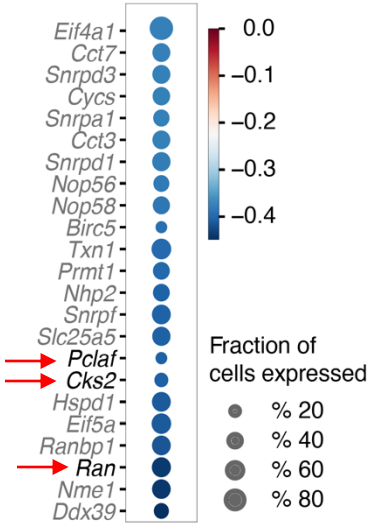


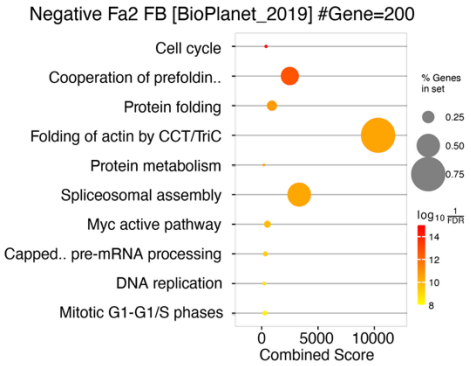


Burstyn-Cohen and Quan (2021) *Brain Plast.*

Fibroblasts (Fa2)

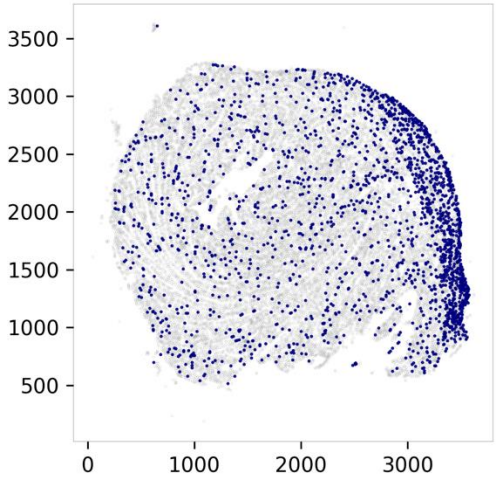


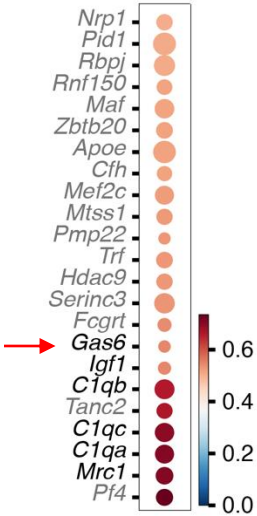




Association of latent factors with full transcriptome

Macrophages (Fa3)



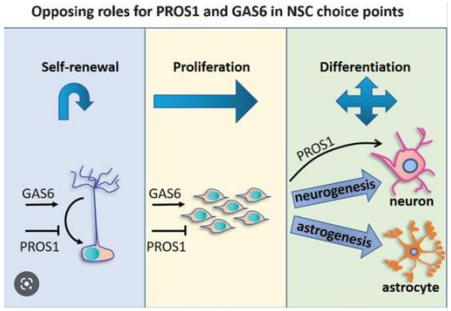


Fraction of cells expressed

% 20

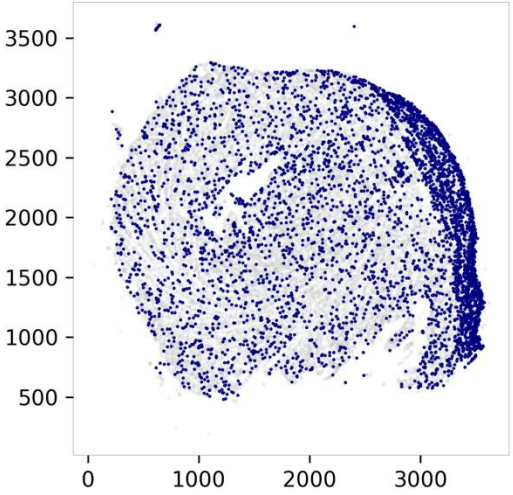
% 40

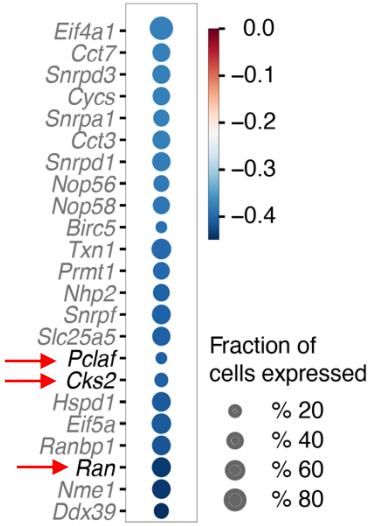
% 60

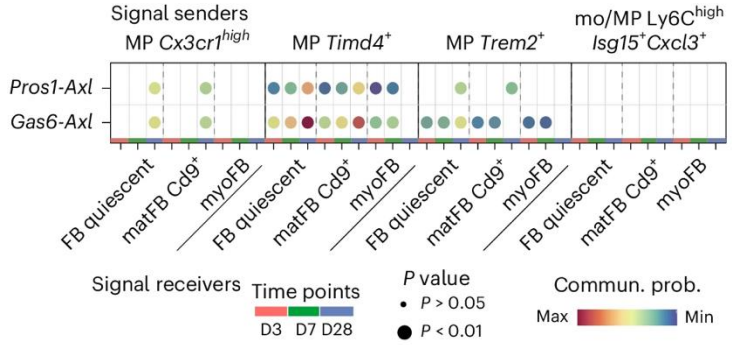


Burstyn-Cohen and Quan (2021) *Brain Plast.*

Fibroblasts (Fa2)

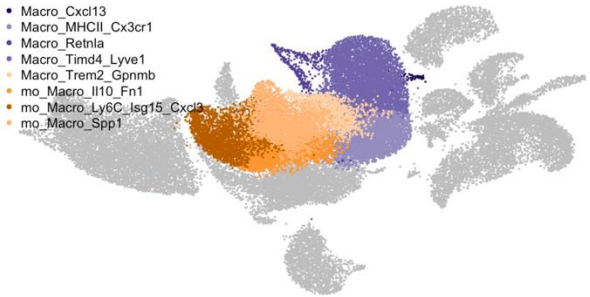




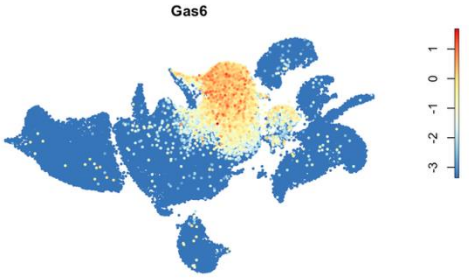


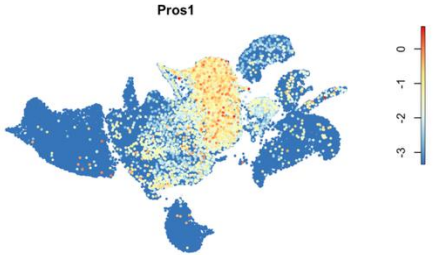
Predicted ligand-receptor interaction controlling covariation

Macrophages

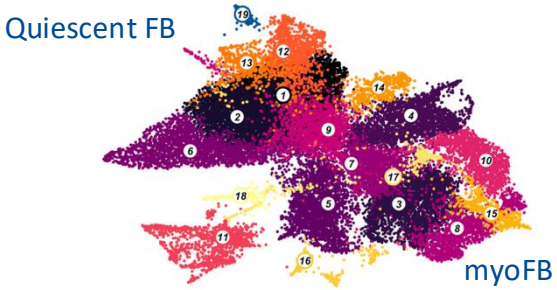


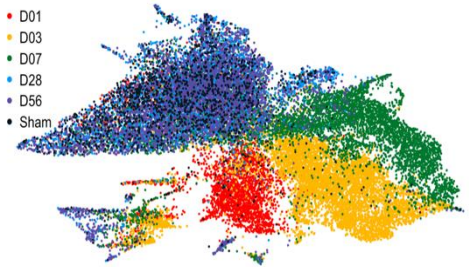


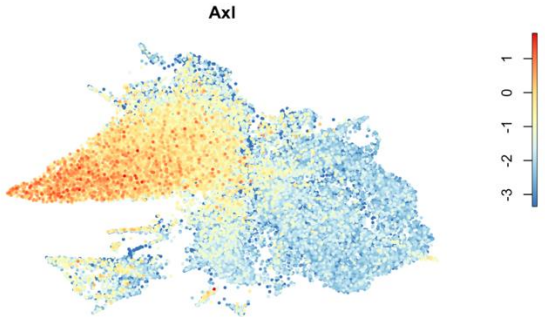




Fibroblasts

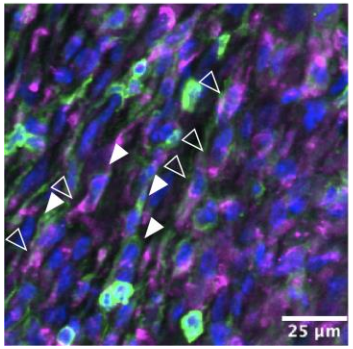


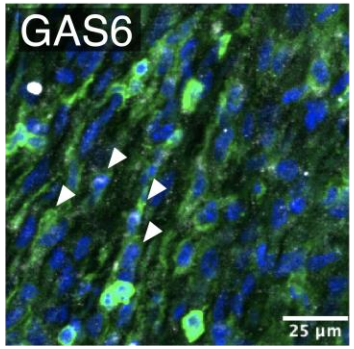


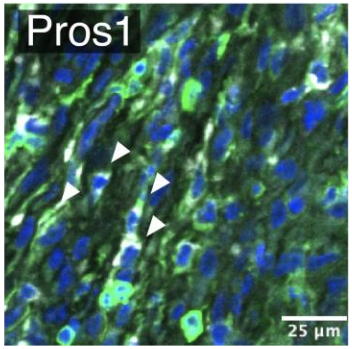


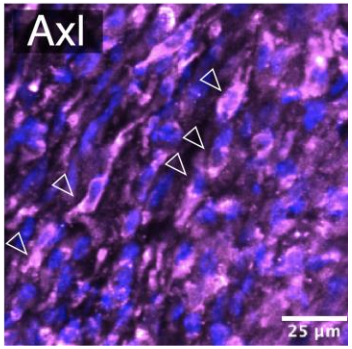
In situ validation and functional in vitro validation

Cardiac slice, Day 7 Cryoablation, IZ CD45 PDGFRA DAPI

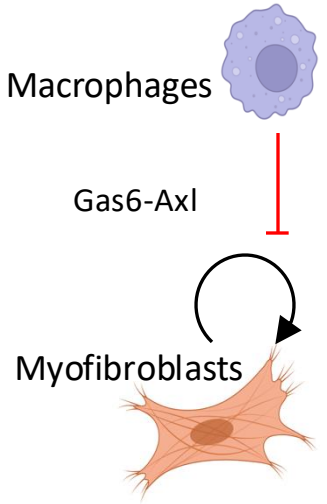




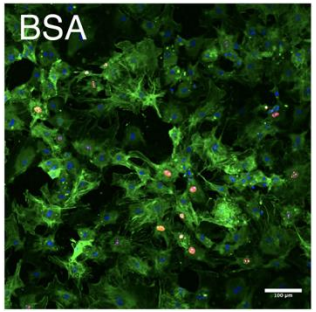


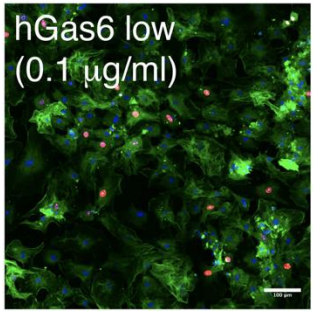


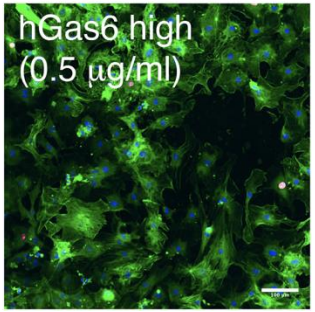
At >7days post MI

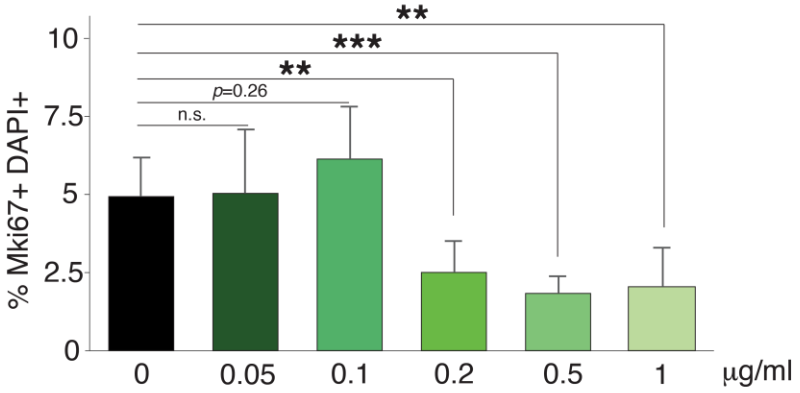


cardiac myoFB culture αSMA Mki67 DAPI









Conserved expression of Gas6-Axl in human MI samples

Article | Published: 10 August 2022

Spatial multi-omic map of human myocardial infarction

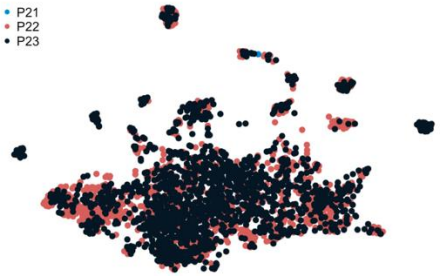
Christoph Kuppe, Ricardo O. Ramirez Flores, Zhijian Li, Sikander Hayat, Rebecca T. Levinson, Xian Liao, Monica T. Hannani, Jovan Tanevski, Florian Wünnemann, James S. Nagai, Maurice Halder, David Schumacher, Sylvia Menzel, Gideon Schäfer, Konrad Hoeft, Mingbo Cheng, Susanne Ziegler, Xiaoting Zhang, Fabian Peisker, Nadine Kaesler, Turgay Saritas, Yaoxian Xu, Astrid Kassner, Jan Gummert, ... Rafael Kramann + Show authors

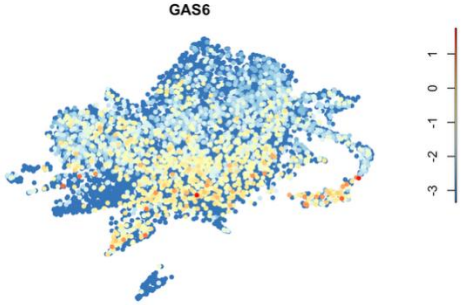
Nature **608**, 766–777 (2022) | Cite this article

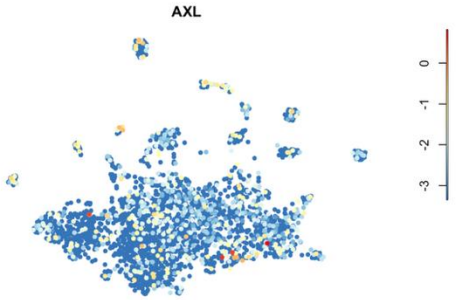
Macrophages



Fibroblasts



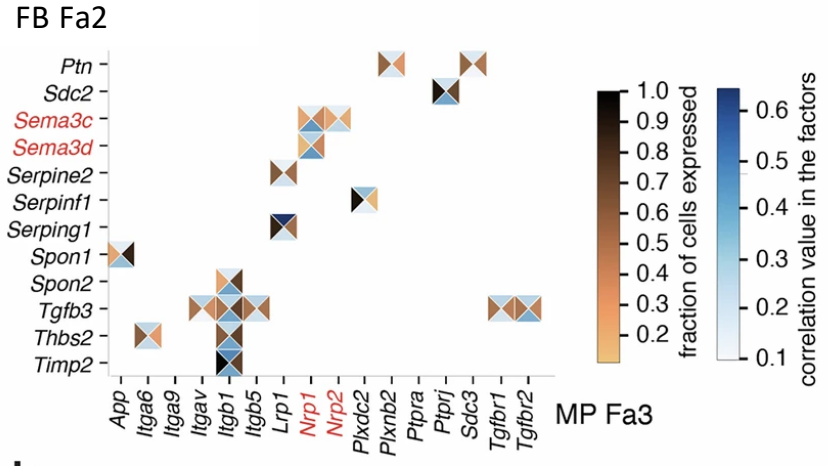




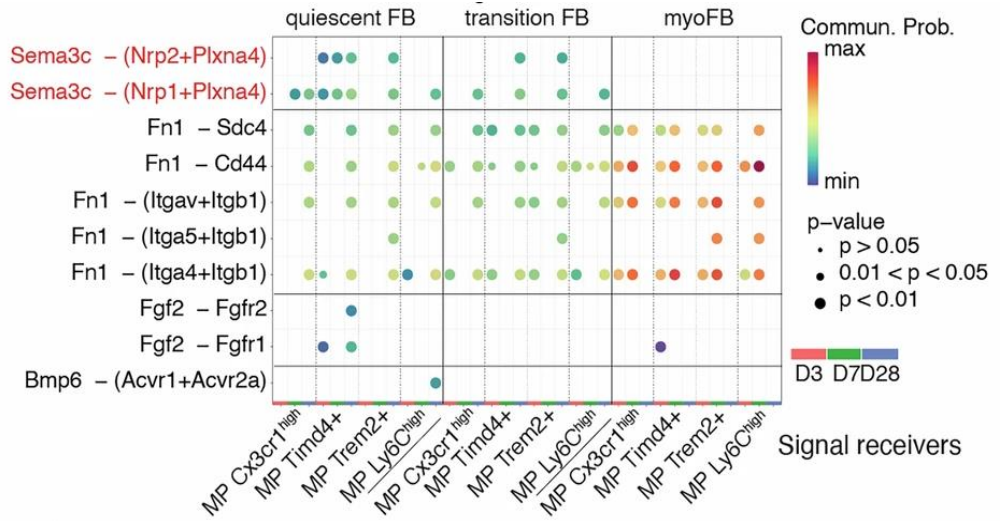
Macrophage GAS6 – Fibroblast AXL potential therapeutic target for limiting scar tissue.

Signaling from FB to MP

Correlation to factors



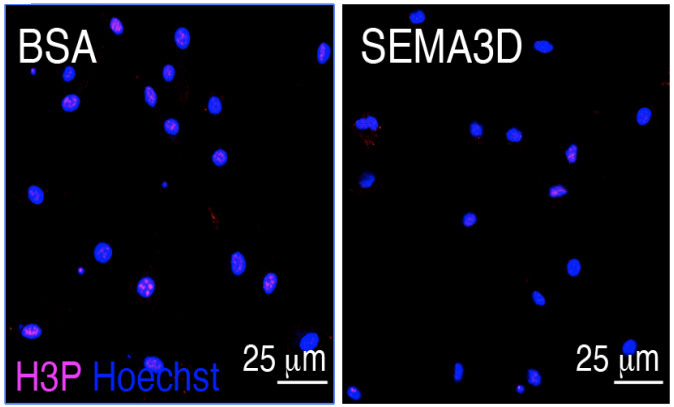
CellChat

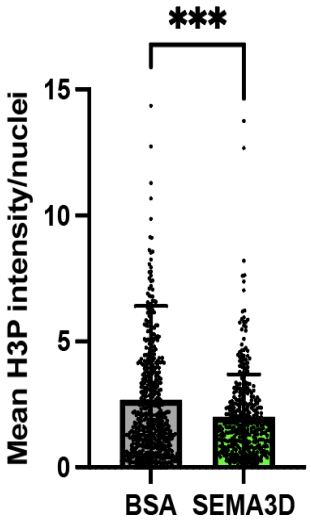


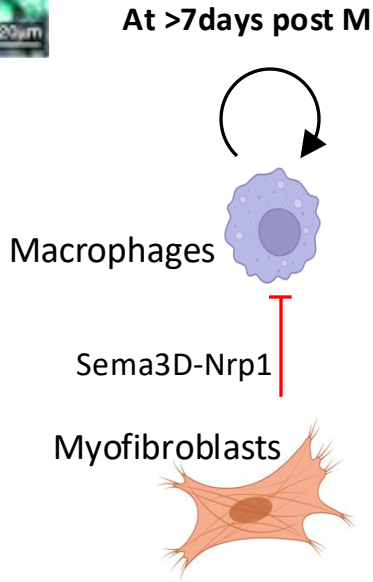
Association of latent factors with full transcriptome



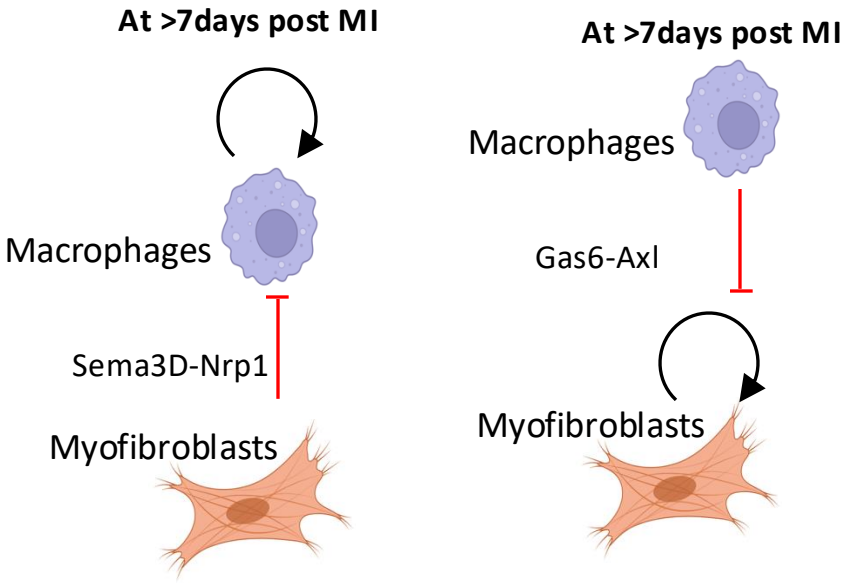
Macrophage proliferation assay

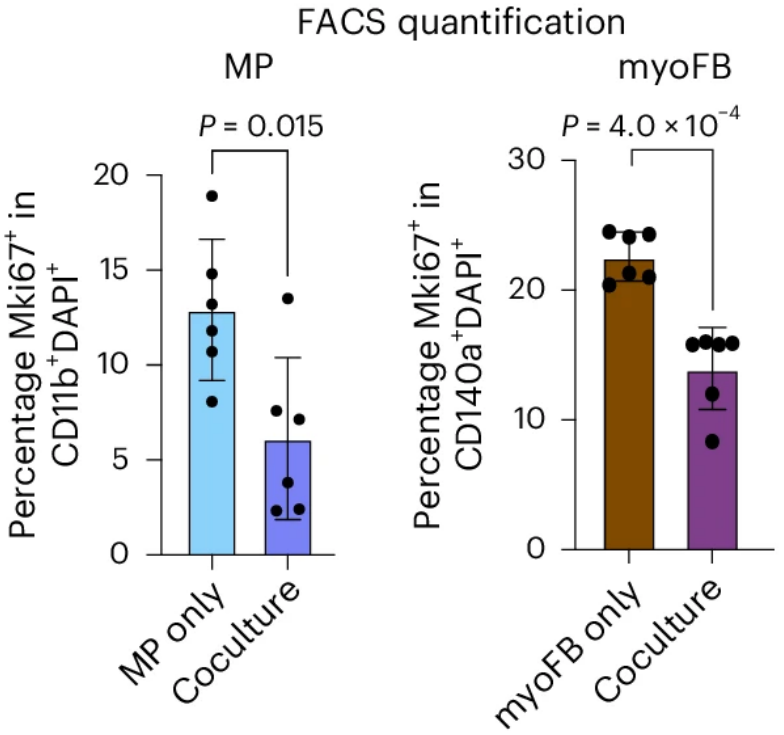




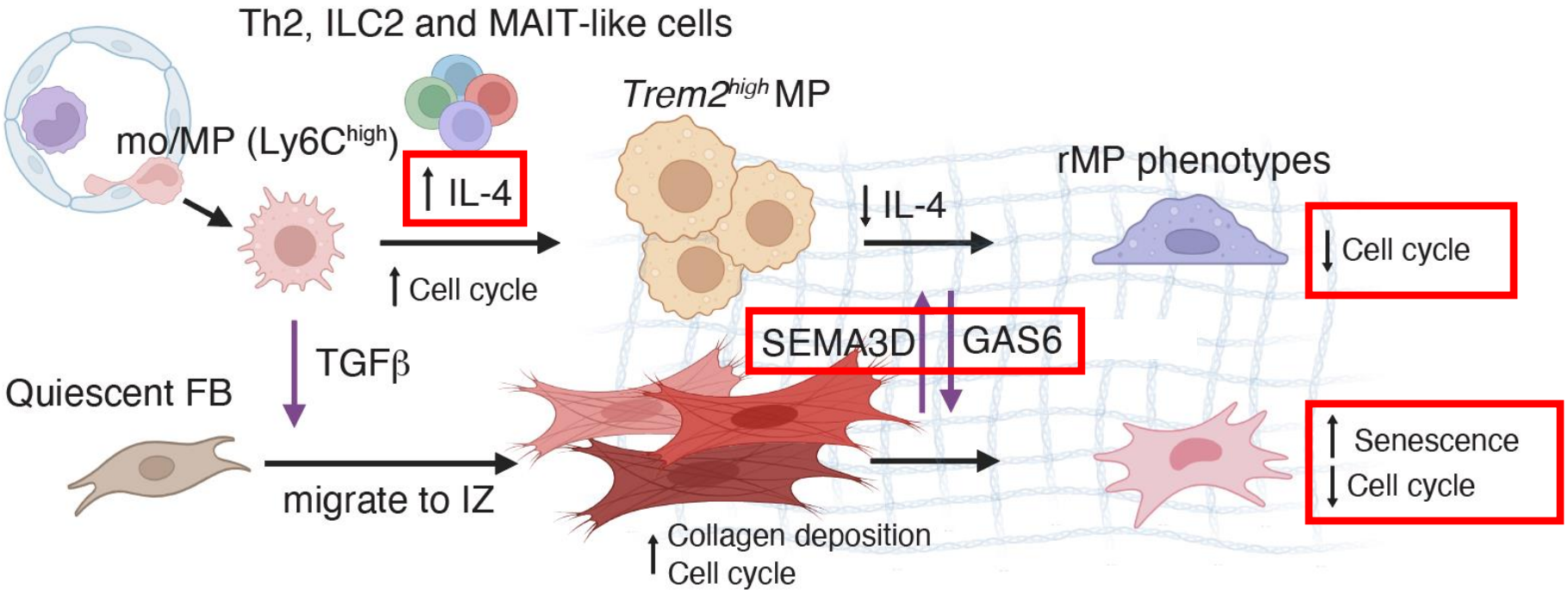


Mutual suppression of proliferation between Macrophages and Fibroblast



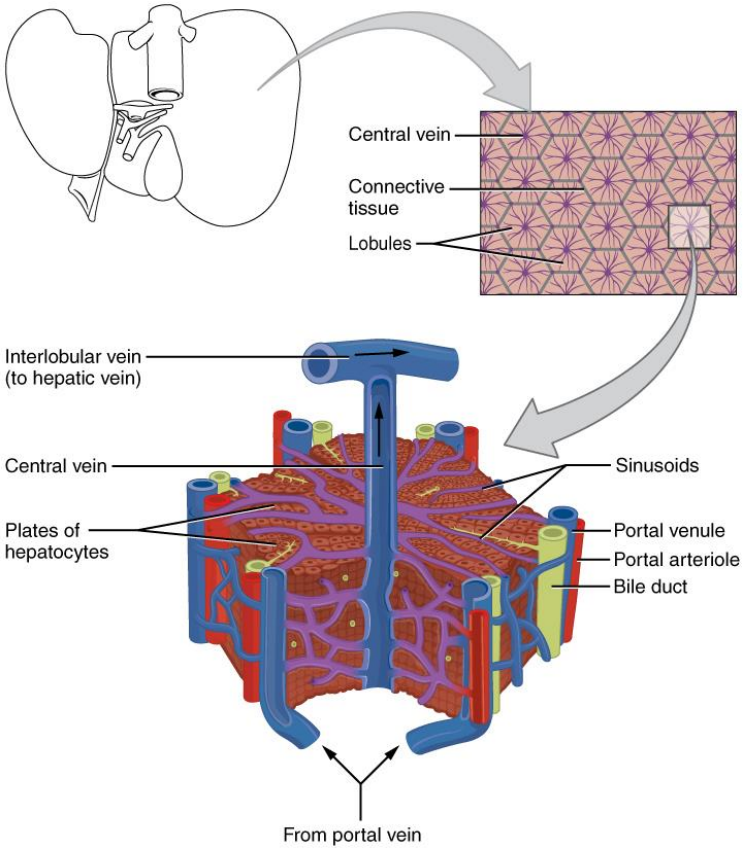


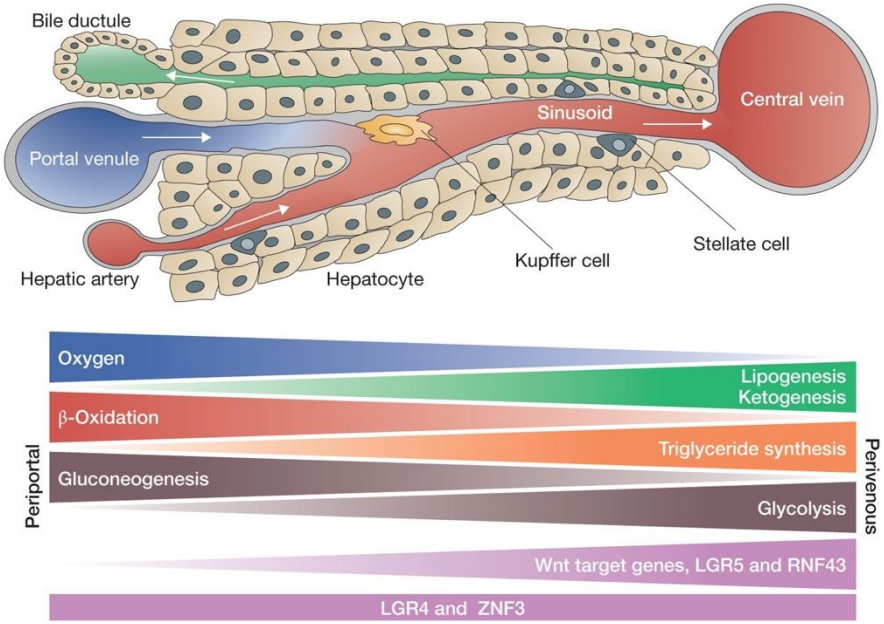
Mutual cell state control by MP-FB interactions in the fibrotic niche



NiCo on MERSCOPE data
Mouse Liver

The zonated liver niche

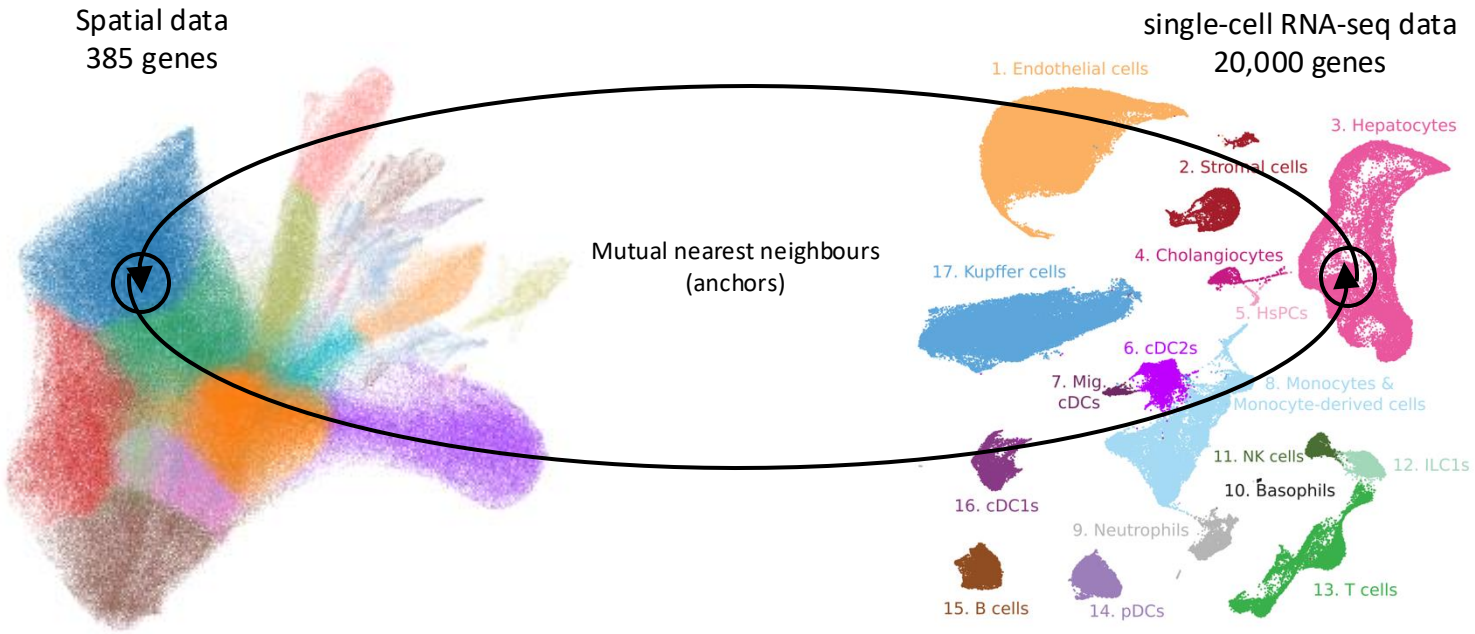




Birchmaier (2016) Nature Cell Biology

Genome-wide zonation of hepatocytes and Sinusoids:
Halpern et al. (2017) Nature
Halpern et al. (2018) Nature Biotechnology
Aizarani et al. (2019) Nature

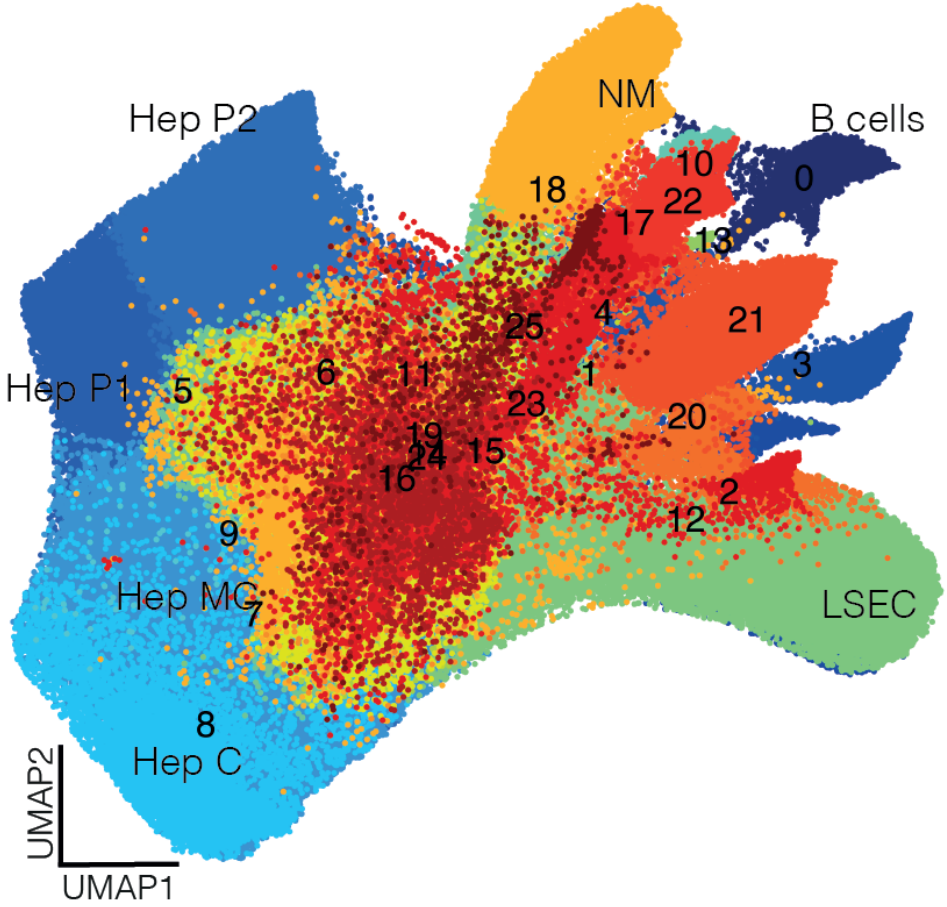
Cell type annotation by label transfer



Guilliams et al. (2022) Cell

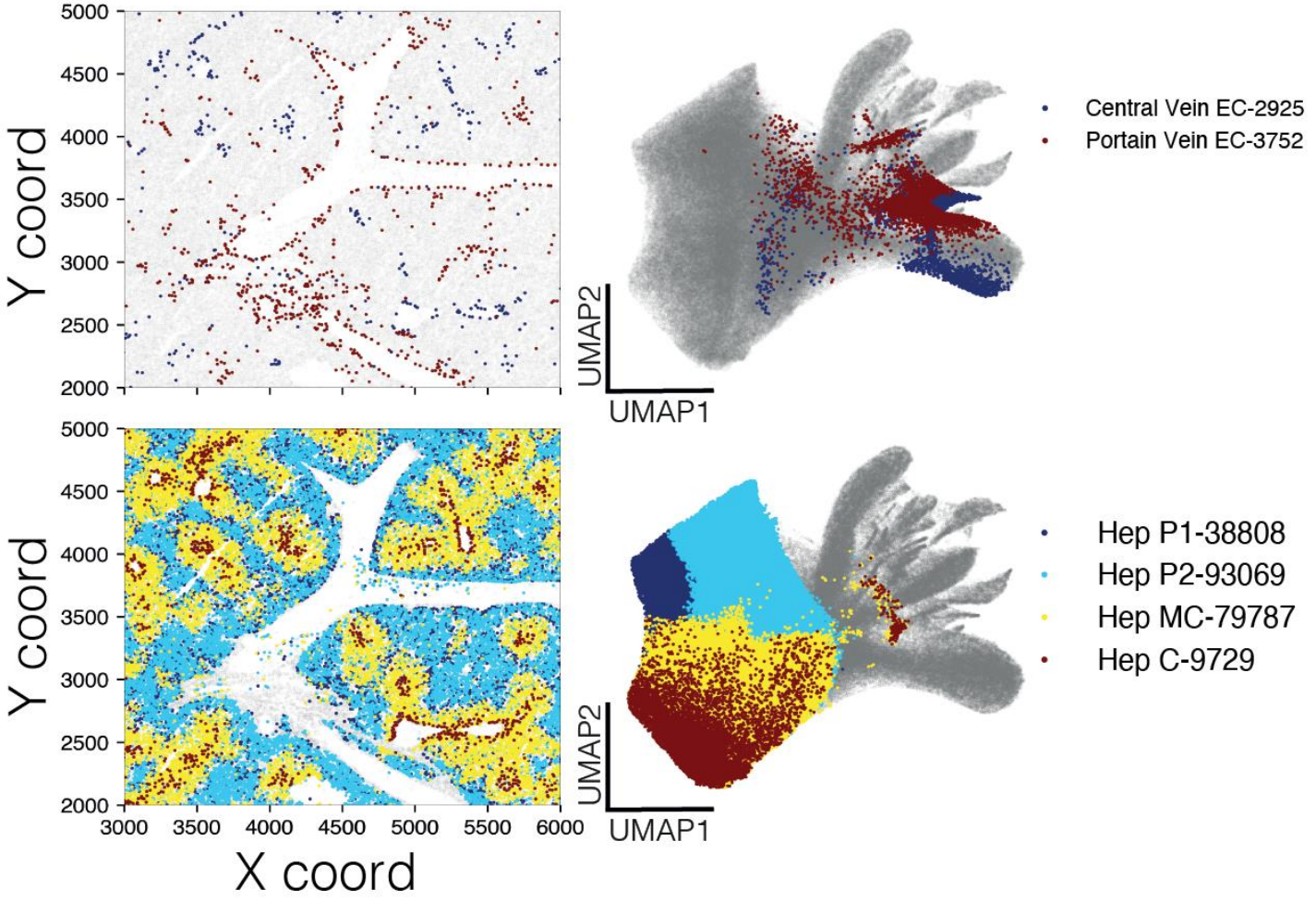
- Elimination of total count dependence by regression
- Mutual nearest-neighbourhoods: anchors
- Use spatial clusters for anchor pruning
- Iterative annotation of non-anchors based on neighbouring anchors

Cell type annotation by label transfer

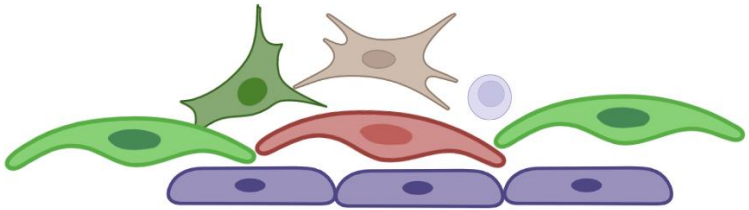


- 0-B cells-3419
- 1-Basophils-59
- 2-Central Vein EC-2925
- 3-Cholangiocytes-3125
- 4-Fibroblast-5894
- 5-Hep P1-38808
- 6-Hep P2-93069
- 7-Hep MC-79787
- 8-Hep C-9729
- 9-HsPCs-100
- 10-ILC1s-1230
- 11-KCs-38932
- 12-LSECs-55495
- 13-Lymphatic EC-280
- 14-Macro-2216
- 15-Mig. cDCs-388
- 16-Monocytes-9007
- 17-NK cells-96
- 18-NM-16518
- 19-Neutrophils-883
- 20-Portain Vein EC-3752
- 21-Stellate cells-11881
- 22-T cells-3352
- 23-cDC1s-7521
- 24-cDC2s-2228
- 25-pDCs-984

Cell type annotation by label transfer

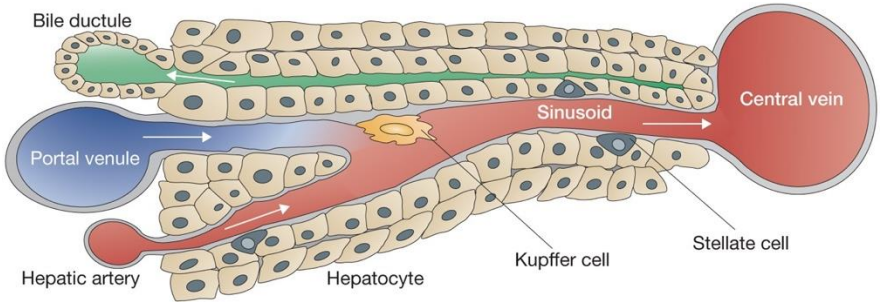


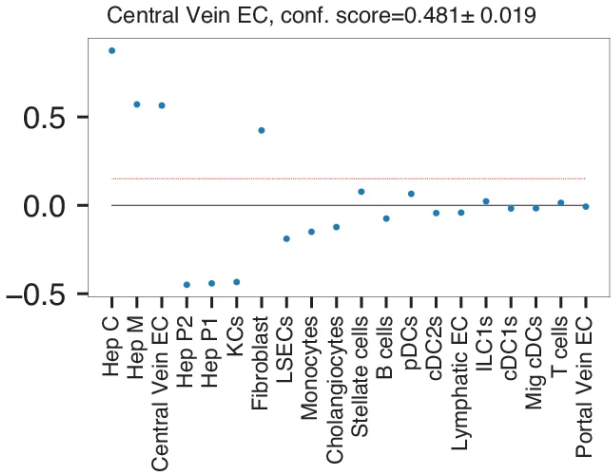
Inference of predictive neighbours by multinomial logistic regression

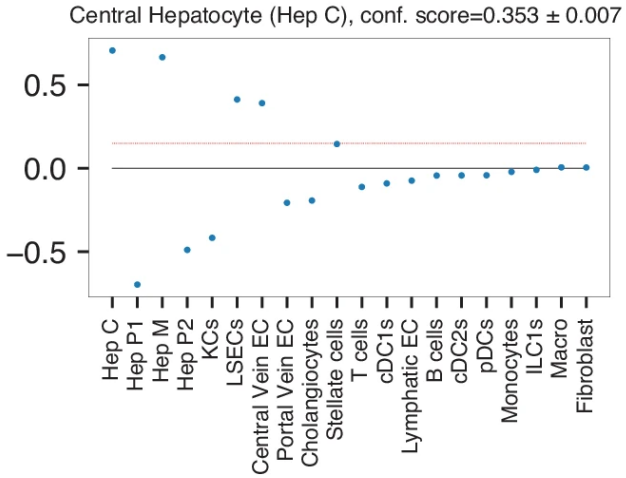


$$P(\text{cell}) \sim w_{\text{cell}} x_{\text{cell}} + w_{\text{cell}} x_{\text{cell}} + w_{\text{cell}} x_{\text{cell}} + w_{\text{cell}} x_{\text{cell}} + w_{\text{cell}} x_{\text{cell}}$$

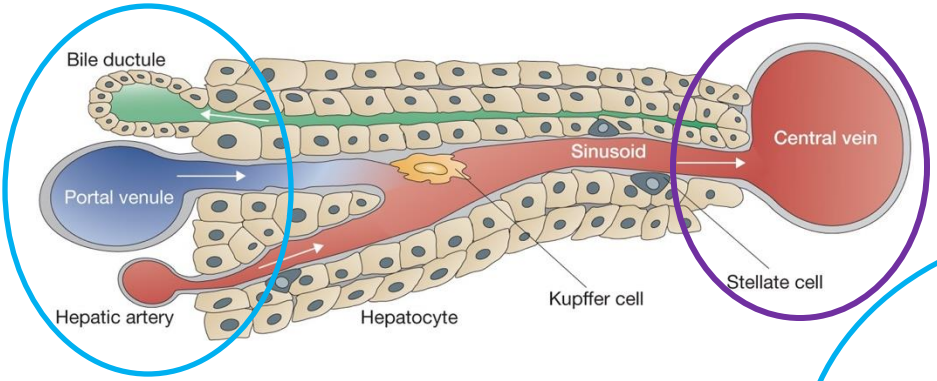
The liver niche interactions

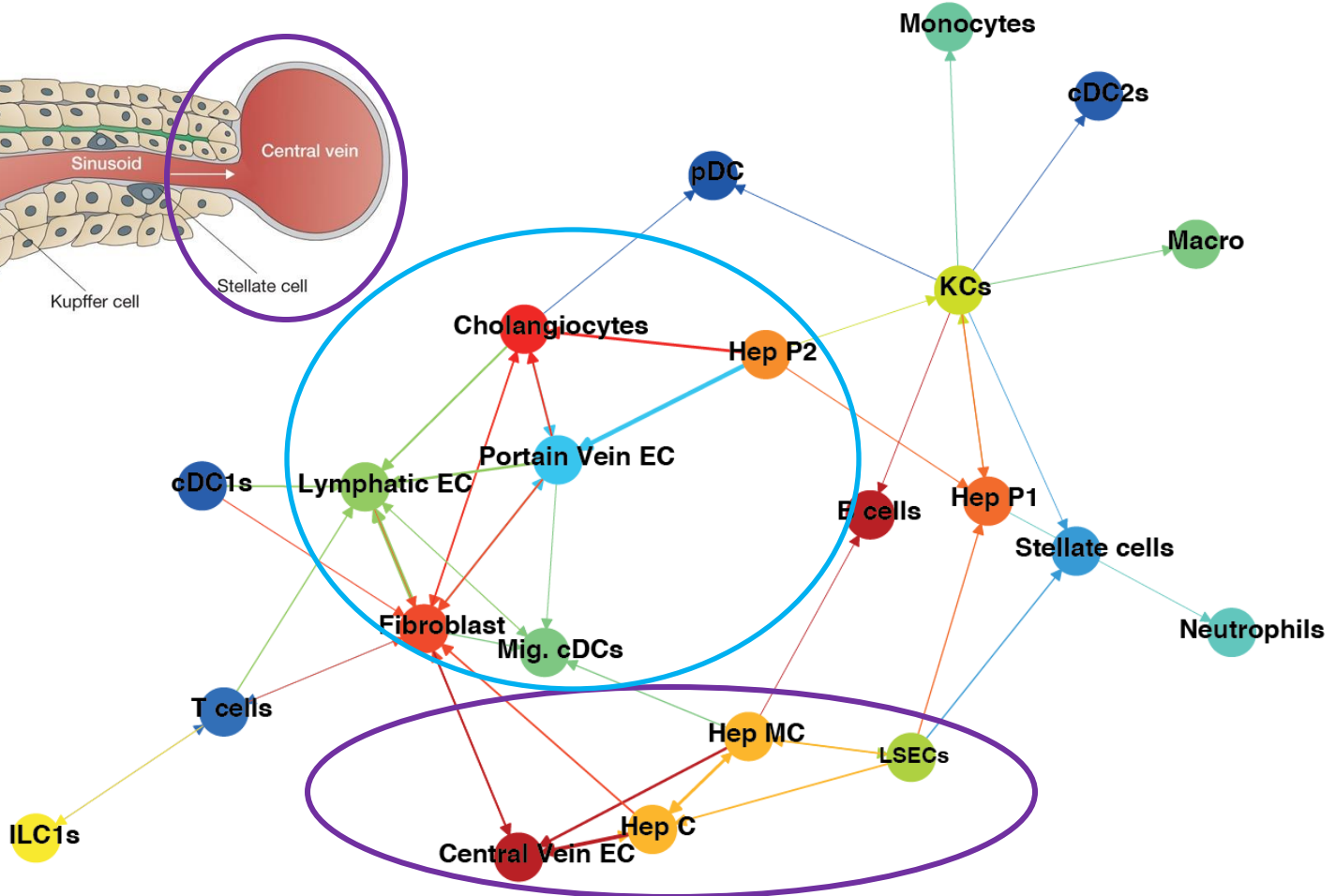






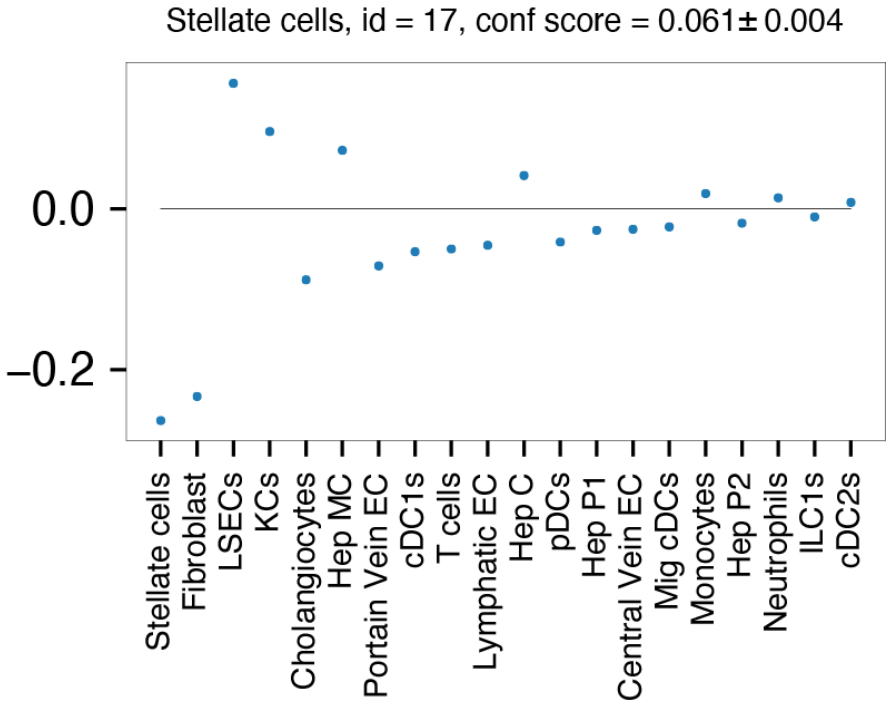
The liver niche interactions

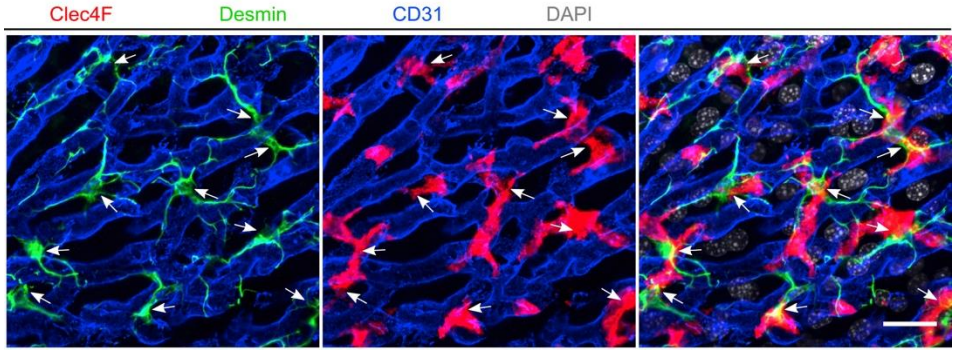




The stellate cell niche

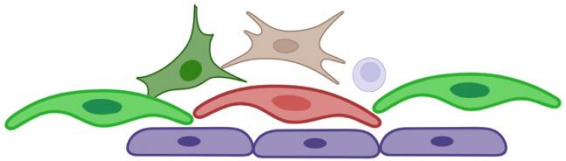
Stellate cells: extracellular matrix production upon liver damage – fibrosis, requires activation

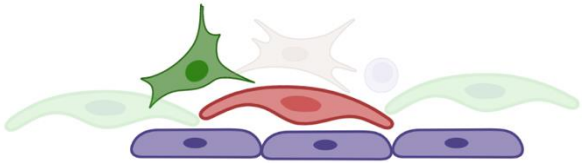


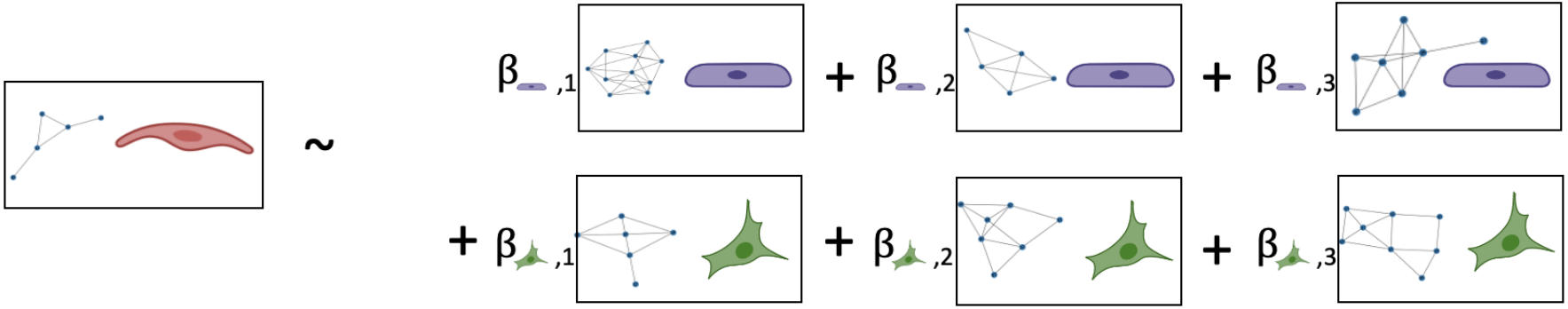


Bonnardel et al. (2019) Immunity

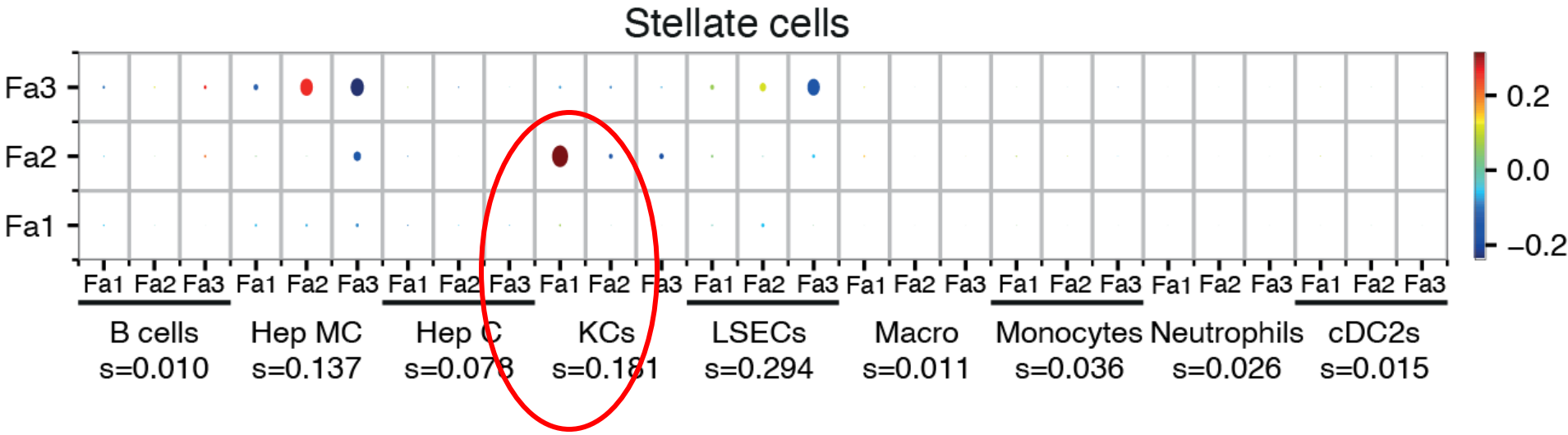
Inference of factor covariation within the niche







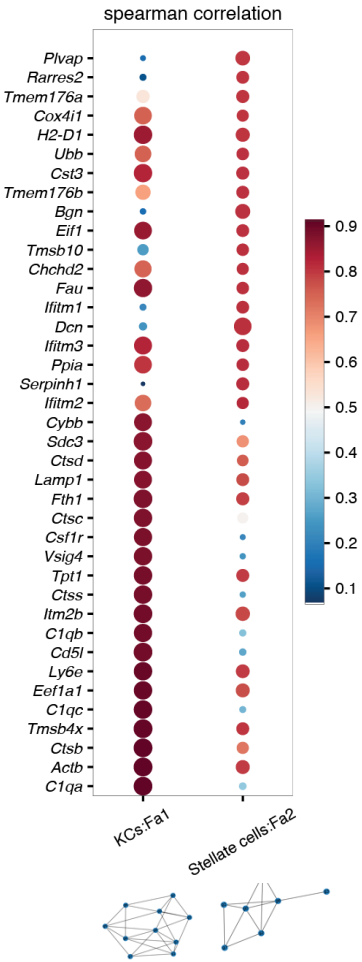
Covariation between Kupffer cells and Stellate cells



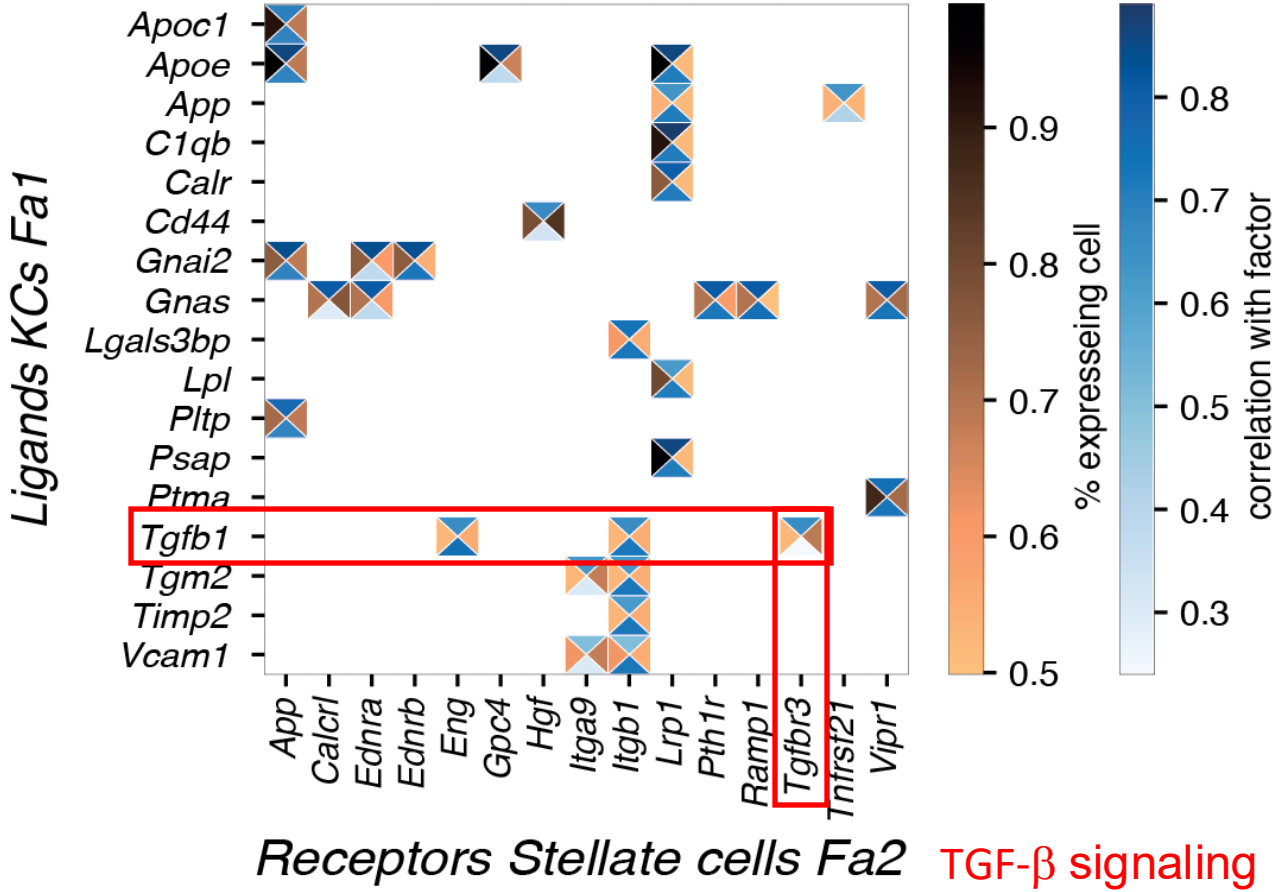
Stellate cells: activated upon liver damage to produce extracellular matrix - fibrosis

Kupffer cells: liver-resident macrophages - phagocytosis, innate immune response

Association of latent factors with full transcriptome



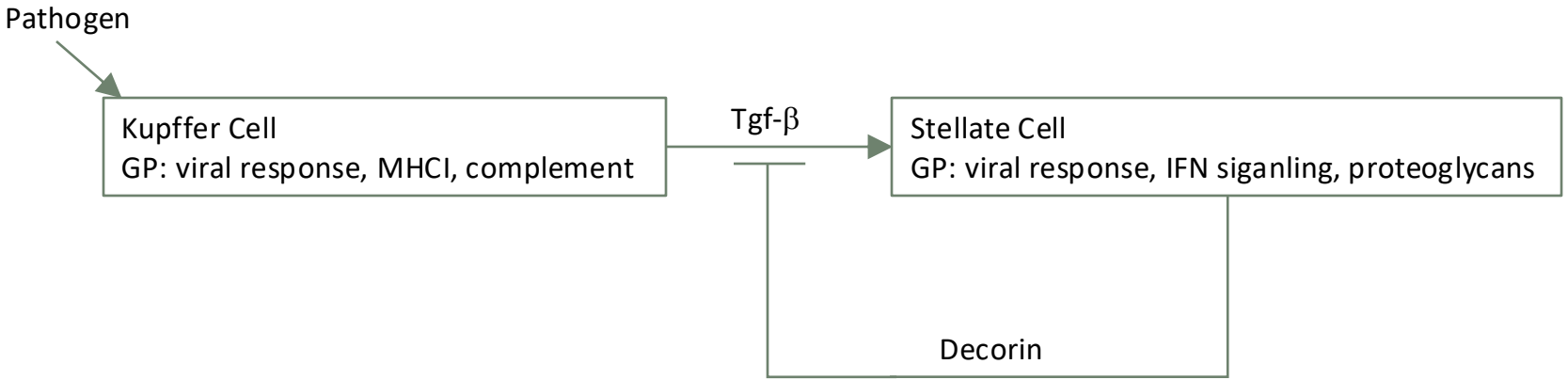
Signaling-mediatpor of niche interaction



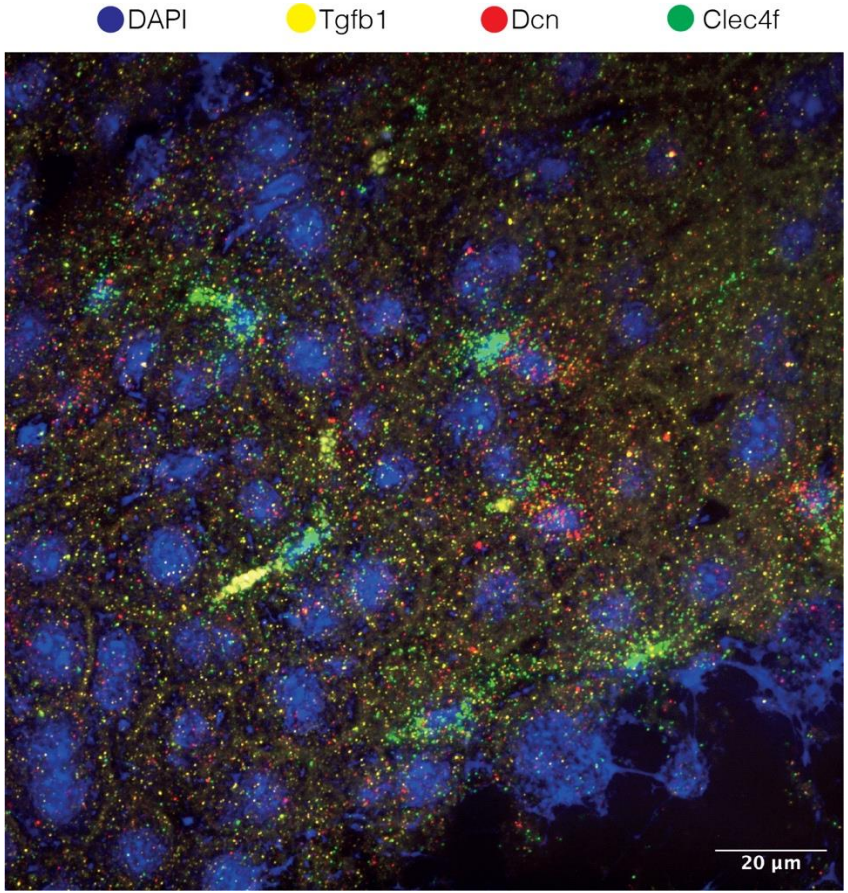
TGF-β signaling

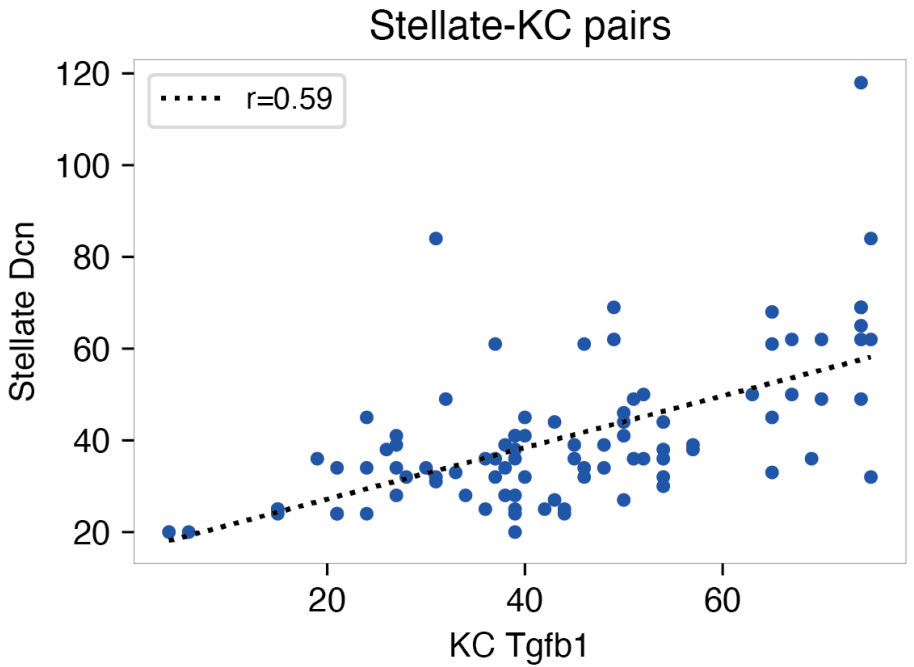
Hypothesis

- Phagocytic KCs sense low levels of viruses and other pathogenes
- Induction of pro-fibrotic TGF-β1 in KCs
- Stellate cells receive TGF-β1 singalling via Tgfbr3
- TGF-β1 signalling induces upregulation or proteoglycans (Dcn, Bgn)
- TGF-β1 binds to proteoglycans bound to the hepatic extracellular matrix or to the cell surface
- Binding may act as a signal to avoid the fibrotic response of stellate cells unless TGF-β1 exceeds a critical threshold



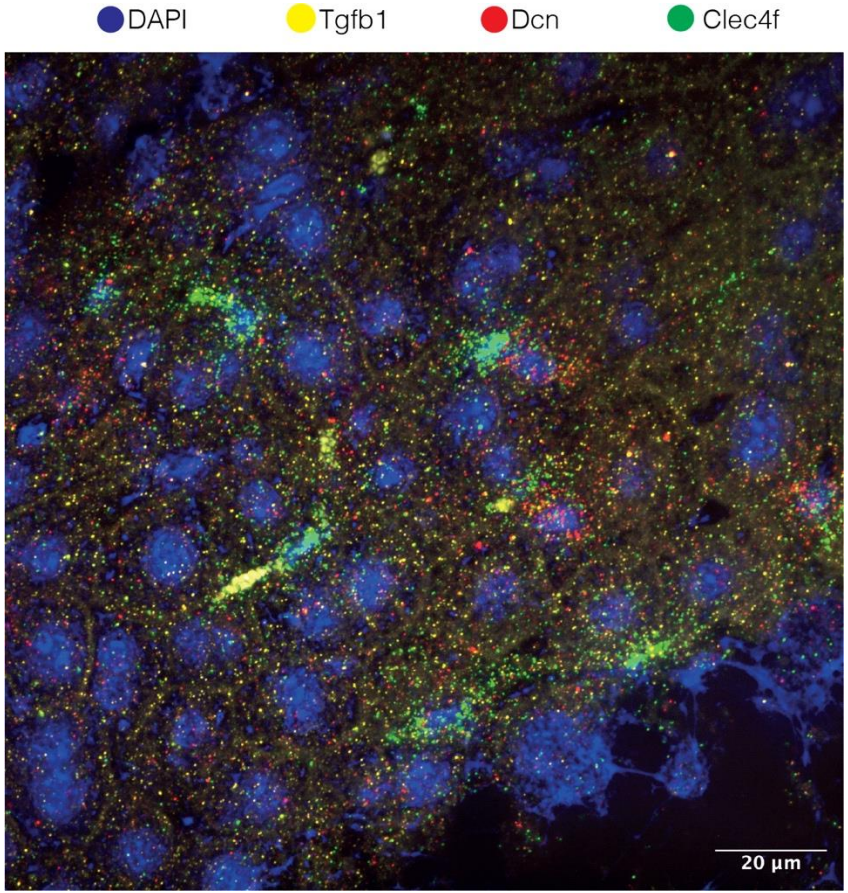
Validation by HCR-smFISH

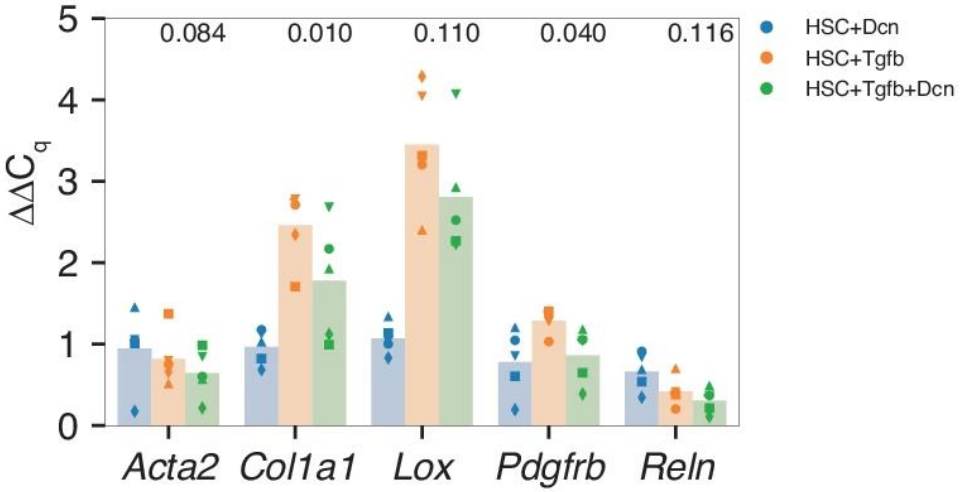




Stefan Thomann

In vitro validation

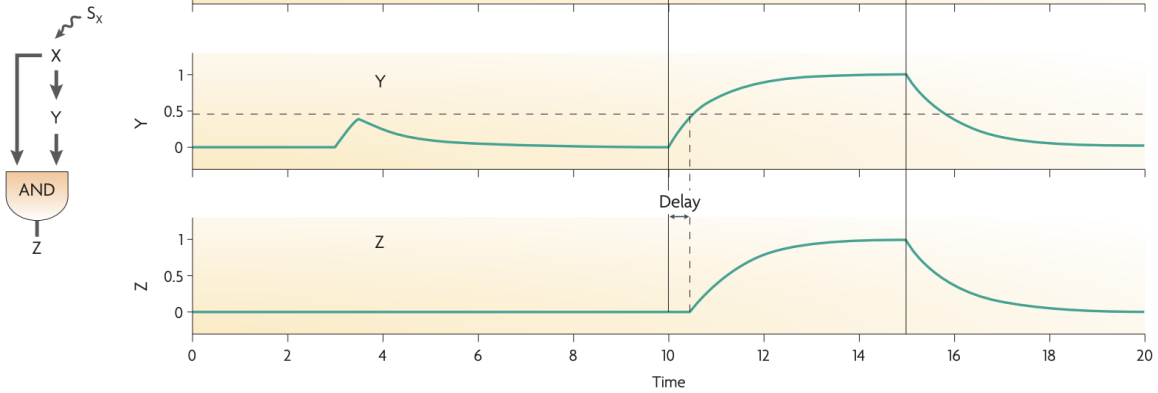




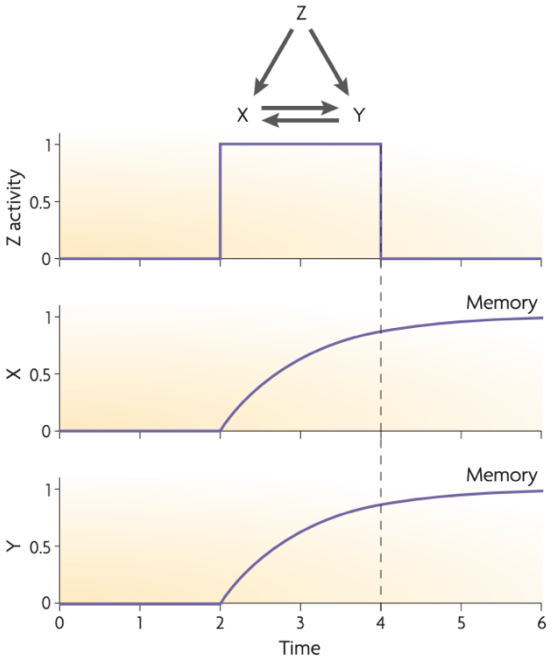
Modeling Tissues: From Gene Regulatory Networks to Cellular Networks

Gene Regulatory Networks are dominated by particular network motifs

Activation Filter

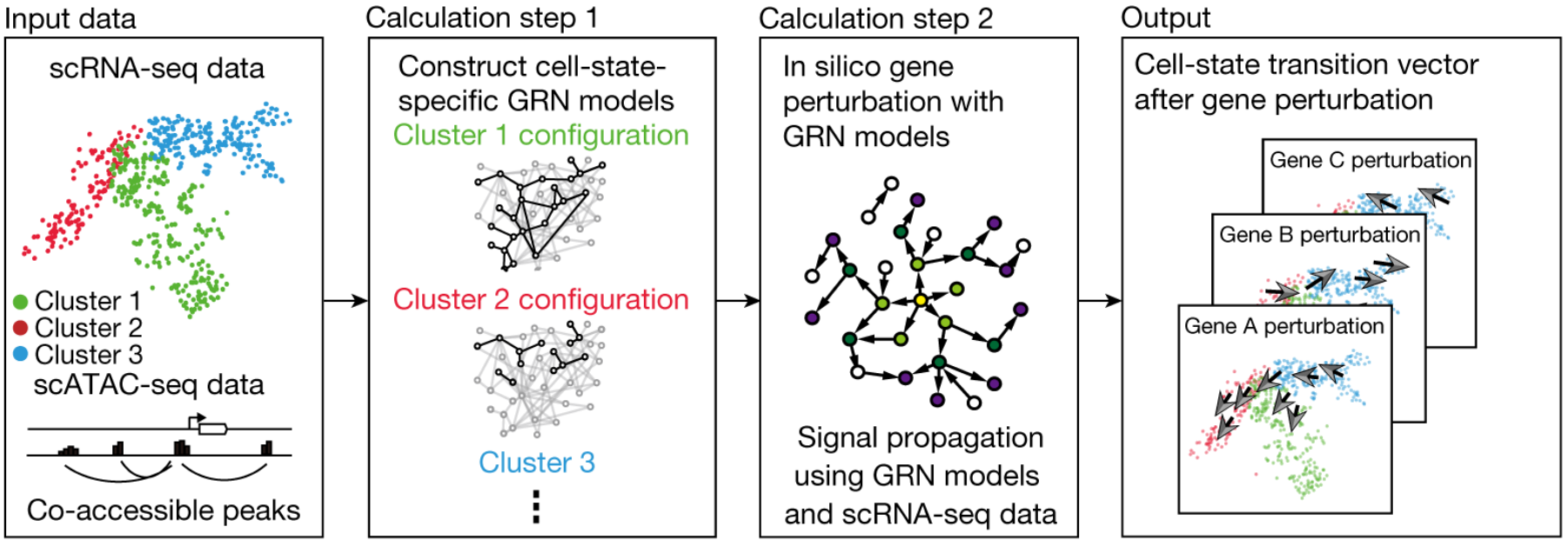


Memory switch



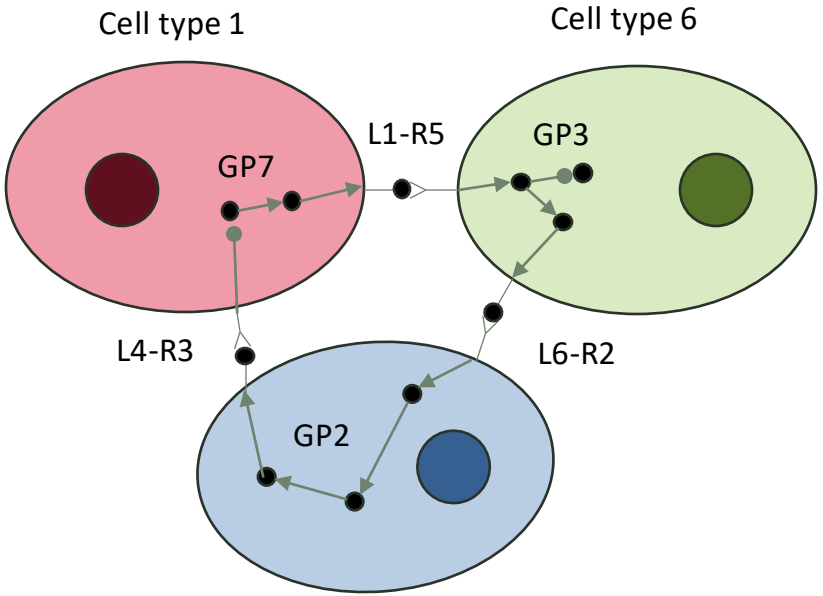
Alon (2007) *Nature Reviews Genetics*

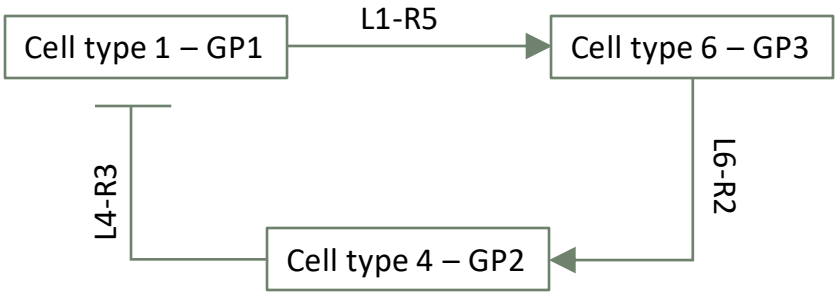
In Silico Perturbation of Gene Circuits: CellOracle

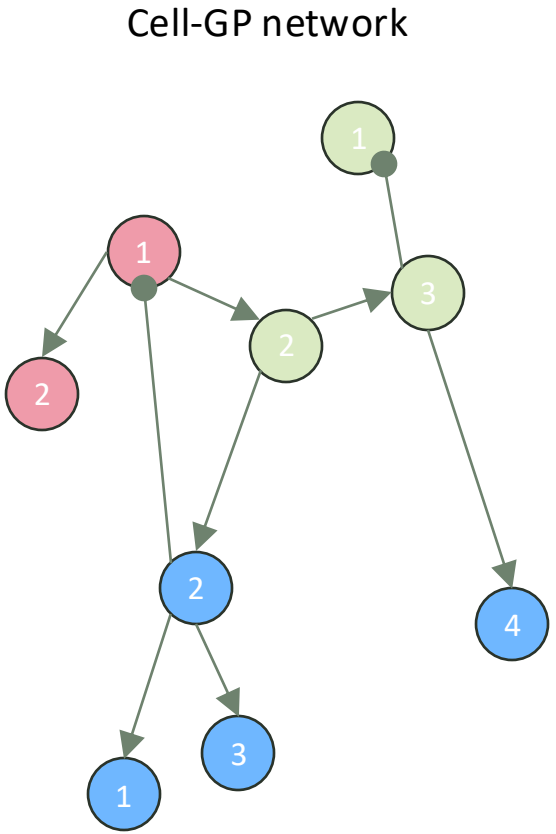


Kamimoto et al. (2023) *Nature*

Cellular Circuits







Summary of NiCo

- Improved **cell type annotation** of imaging-based spatial transcriptomics by MNN-based label transfer.
- Inference of **niche architecture** in heterogeneous tissues and identification of **predictive neighbours.**
- Prediction of **covarying latent factors** in neighbouring cell types capturing **intra-cell type variability** within local niches.
- Identification of **covarying gene programs** in co-localized cell types.
- Prediction of **ligand-receptors interactions** as mediators of covariation between neighbouring cells.
- NiCo predicts **suppression of myofibroblast proliferation by macrophages** 7 days post myocardial infarction in mouse, potentially **limiting scar formation**
- NiCo predicts **niche cell types** and **ligands** that synergistically induce **cardiomyocyte dedifferentiation** and proliferation

Next Steps

- What is the ”right” number of factors for each cell type (dimensionality)?
- Improve factor inference: interpretabillity and disentanglement (e.g. by using priors)
- Multi-sample analysis (cross-sample integration and differential analysis)
- Non-linear interactions between factors (saturation behaviour) and synergistic interactions
- Optimize probe set selection to capture intra-cell type variability
- Impute full transcriptome in space for direct modeling of interactions
- Higher order cell type-niche motifs

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